# Selection on Exit and Sticky Prices

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October 18, 2026

[MOST RECENT VERSION]

**Abstract**

This paper shows that the composition of entry and exit of products matters for the transmission and amplification of monetary shocks. Using retail scanner data, I show that low-revenue products exit more often and change prices less frequently. I develop a menu-cost model that matches the empirical evidence. In the model, price changes are costly, leading firms with higher exit risk to endogenously choose reset prices less frequently. This mechanism of selective product turnover is missing in standard models and amplifies the real effects of monetary policy: compared to a model with random turnover calibrated to the same moments, selective turnover raises the one-year cumulative output response to a monetary shock by 15%. Amplification reaches 35% near price stability and declines at higher inflation.

**JEL Codes**: E31, E50, L11.
**Keywords**: Selective exit, price stickiness, menu costs, monetary non-neutrality, firm dynamics, scanner data.

*n.zaccaria@tilburguniversity.edu I thank Kian Abbas Nejad, Quamrul Ashraf, Corina Boar, Jeff Campbell, Elena Casanovas, Gian Luca Clementi, Patricio Dalton, Shubhdeep Dep, Alberto Ferreira (discussant), Konstantin Gantert, Jonas Gathen, Bumsoo Kim, Paolo Martellini, Marta Morazzoni, Giuseppe Musillo, Guzmán Ourens, Greg Phelan, Florian Sniekers, Sigrid Suetens, Michael Tallent (discussant), Burak Uras, Ludo Visschers, Has van Vlokhoven, and Till Wicker for insightful comments. I also thank seminar participants at Tilburg University, NYU, UCL, as well as participants at the 3rd EUI PhD Conference, the 31st ENTER Jamboree UC3M, and the EEA-ESEM 2026 Dublin conference. I acknowledge financial support from CentER. Access to the data is provided by AiMark (https://aimark.org), which is a not-for-profit organization that supports and provides academic researchers with scanner data.

# 1 Introduction

Canonical menu-cost models generate small and short-lived real effects of nominal shocks because of selection into price adjustment: firms with the strongest incentives to reprice adjust, allowing relatively few price changes to absorb most of an aggregate shock (Golosov and Lucas, 2007). But product markets select on survival as well as on price adjustment. Turnover is substantial—about 8 percent of products are reallocated each quarter (Argente et al., 2018)—and the products that leave are not a random sample of price setters. The monetary consequences of this selection depend on how survival prospects affect incumbent pricing decisions and on which incumbents' prices are displaced by entry. This paper shows that selective turnover amplifies the real effects of nominal shocks, even when the aggregate turnover rate is held fixed.[1]

I use Dutch retail scanner data covering 2008–2020 to document a systematic relationship between product survival and price adjustment. The data follow individual products over time, allowing me to relate their sales, pricing decisions, and subsequent exit. Products in the bottom decile of revenue are about 50% more likely to exit within six months than products in the top decile. The same low-sales products also reprice less frequently: they change prices in about 7 percent of months, compared with 10 percent in the top decile.[2] Turnover therefore disproportionately replaces products belonging to groups with less frequent price adjustment. These findings establish a connection between product survival and pricing behavior, but do not determine whether turnover increases aggregate price flexibility: that depends both on how exit risk affects incumbent pricing decisions and on which prices are replaced. They motivate a model in which the economic conditions governing survival also determine price adjustment.

I build a quantitative menu-cost model with firm heterogeneity and aggregate risk, in which productivity jointly determines firms' sales, pricing incentives and survival.[3] A deterioration in productivity changes the price a firm would like to charge and lowers its continuation value, increasing its probability of exit. Firms therefore decide whether to pay the menu cost while accounting for the possibility that they will not survive to collect the future gains from repricing. When exit occurs, an entrant's newly chosen price replaces the incumbent's old price, changing the distribution of prices through which nominal shocks are absorbed. I compare selective turnover with random replacement at the same aggregate exit rate. This comparison isolates

[1] Nakamura and Steinsson (2010) incorporate product replacement through opportunities to reset prices independently of incumbents' adjustment incentives. Here, survival depends on economic conditions that also govern repricing.

[2] The pricing pattern is consistent with producer-price evidence: large firms change prices more frequently than small firms (Goldberg and Hellerstein, 2009), and firms with more products adjust more often (Bhattarai and Schoenle, 2014). Argente and Yeh (2022) document a different dimension of pricing heterogeneity, across product ages, and show how learning over the product life cycle affects monetary transmission.

[3] I use the terms 'firm' and 'product' interchangeably throughout the paper, and the model assumes one product per firm. Exit represents product discontinuation and need not imply that the producer goes out of business: Argente et al. (2018) document that most product creation and destruction occurs within surviving firms. I abstract from the producer's broader portfolio decisions and represent product discontinuation through a value-dependent exit hazard. The exit diagnostics in Appendix A.4 support this interpretation: revenue-weighted, 82 percent of measured exits involve disappearance from all chains at the national level.

how the composition of entry and exit changes monetary transmission, holding the amount of turnover fixed.

I calibrate the model to match pricing and product turnover moments from the scanner data, including the relative revenues of entering and exiting products. The model also reproduces the empirical relationships between product revenue, survival, and price adjustment, which are not directly targeted in the calibration. Following an unexpected increase in nominal aggregate demand, prices do not adjust immediately, generating a temporary increase in real output. Selective turnover amplifies the one-year cumulative output response by about 15 percent relative to random turnover, with the amplification arising from the composition of entry and exit rather than changes in menu costs or aggregate turnover.

Previous work has reconciled microeconomic price flexibility with larger real effects through changes in the pricing problem: sectoral heterogeneity and intermediate inputs (Nakamura and Steinsson, 2010), decreasing returns to scale (Burstein and Hellwig, 2007), multiproduct pricing (Midrigan, 2011), duopoly (Mongey, 2021), and variable demand elasticity and idiosyncratic demand shocks (Aruoba et al., 2024), among others. My results identify selective survival as an empirically grounded source of amplification: economies that look similar along the targeted pricing dimensions can respond differently to nominal shocks because survival risk changes pricing decisions and the allocation of replacement.[4]

The main source of amplification is firms' anticipation of exit. Low-value firms face poor survival prospects, shortening the horizon over which they expect to recover the cost of changing their price. Paying the menu cost becomes less attractive, and these firms become less willing to reprice in response to an aggregate shock. I call this the option-value channel.[5] Random exit shortens the expected horizon uniformly; selective exit concentrates this effect on low-value firms, widening their inaction regions and allowing their inherited prices to persist. The same amount of turnover therefore generates greater nominal rigidity because survival risk changes the pricing decisions of firms still in the market.[6]

Realized exit provides a second source of amplification by changing which incumbent price adjustments entrants replace. Holding pricing decisions fixed, random exit removes firms uniformly from the whole distribution, including products whose prices would otherwise remain unchanged. The entrant replacing such a product posts a newly chosen price, so random replacement adds aggregate price flexibility. Selective exit instead falls more heavily on products close to repricing, where replacement adds less flexibility because the incumbent's correction was already about to occur. These firms have been hit by a negative productivity shock that raises their marginal costs and lowers their profitability. They would otherwise raise their

[4]This comparison does not hold the entire distribution of price changes fixed. Higher-order moments can change, consistent with the importance of the full distribution of adjustments for monetary non-neutrality. See Alvarez et al. (2016).

[5]Price adjustment can be seen as exercising an option (Stokey, 2009).

[6]A related anticipation mechanism appears in Aruoba et al. (2024): firms expecting greater future demand dispersion postpone repricing because a price chosen today may soon become suboptimal, amplifying the impact real effects of nominal shocks. Here, firms postpone repricing because the product may disappear before they recover the adjustment cost.

prices in response to the higher costs, but exit before doing so. Replacing them with entrants adds little price flexibility, because their prices would have changed anyway. At the same total exit rate, replacing these firms means replacing fewer firms whose prices would otherwise have remained unchanged. I call this the censoring channel.[7] Censoring contributes less than anticipation in the benchmark, but its importance increases when productivity is less persistent. The two channels interact negatively because realized exit removes some of the firms whose delayed adjustments would otherwise contribute to amplification.

The real effects of monetary shocks depend on the inflation environment in which they occur. I examine how trend inflation affects the amplification generated by selective turnover and find that it is strongest when inflation is low. Near zero inflation, selective turnover raises the cumulative output response by about 35 percent relative to random turnover, compared with 15 percent in the benchmark calibration. The difference declines as inflation rises and becomes smaller at high inflation. A longstanding argument links the output–inflation trade-off to the greater frequency of price adjustment induced by higher inflation (Ball et al., 1988); recent work emphasizes the feedback between inflation and adjustment frequency (Blanco et al., 2025; Karadi et al., 2026). My results identify a further dimension of this relationship: inflation changes how strongly survival selection affects aggregate adjustment. Near price stability, monetary policy therefore has larger real effects in economies with selective turnover, even when their aggregate turnover rates and pricing moments are otherwise similar.

Trend inflation determines this amplification by changing the relative importance of idiosyncratic conditions and common price erosion in repricing decisions, a distinction central to understanding pricing across inflation environments (Gagnon, 2009). At low inflation, adjustment is driven largely by idiosyncratic conditions, leaving substantial room for high exit risk to widen the inaction regions of low-value firms. At high inflation, common price erosion pushes firms throughout the entire distribution alike to reprice. Poor survival prospects have less scope to delay adjustment, while random replacement increasingly removes prices that incumbents would soon have changed themselves. Both the anticipation and censoring differences between selective and random turnover therefore shrink. Accounting for survival selection is consequently most important when assessing monetary transmission near price stability, where frequent product replacement can coexist with substantial nominal rigidity.

The amplification depends quantitatively on firms' ability to choose when to reprice. Replacing the menu cost with a Calvo lottery, while leaving the other model blocks unchanged, preserves the positive effect of selective turnover but reduces its magnitude to about 2 percent relative to random turnover. With exogenous adjustment opportunities, firms can no longer postpone repricing in response to poor survival prospects, removing the timing margin through which anticipation generates most of the amplification in the menu-cost model. Appendix G examines the remaining effects through reset prices and the composition of surviving price gaps. The amplification survives moderately fat-tailed productivity shocks. With

[7] An analytical version of the model (see Section 6 and Section J of the Appendix) establishes a sufficient condition under which reallocating exit toward imminent adjusters raises the cumulative output response: the effect depends on which incumbent corrections replacement displaces.

a mixture of normals, or with Gaussian innovations that arrive in discrete jumps, the amplification results remain qualitatively the same (Appendix H).

**Related literature.** This paper contributes to the literature on how firms' adjustment decisions affect monetary non-neutrality. Golosov and Lucas (2007) show that selection into price adjustment can make the aggregate price level flexible even when individual prices are sticky, generating real effects that are too small compared to the ones estimated from the data. Subsequent work obtains larger real effects through production structure (Nakamura and Steinsson, 2010), multiproduct pricing (Midrigan, 2011), and strategic complementarities or variable demand elasticity (Mongey, 2021; Wang and Werning, 2022; Alvarez et al., 2023; Aruoba et al., 2024). Empirical work examines adjustment selection and its response to aggregate shocks (Carvalho and Kryvtsov, 2021; Karadi et al., 2024; Alvarez et al., 2025). I show that selection on entry and exit increases monetary non-neutrality. Poor survival probabilities discourage low-value firms from paying the menu cost, while selective replacement supplies less additional price adjustment than random replacement. Survival selection therefore changes both the rigidity of incumbent prices and the flexibility provided by turnover.

Firm dynamics research has long emphasized selection (Jovanovic, 1982; Hopenhayn, 1992), its consequences for aggregate productivity (Baily et al., 1992; Hopenhayn and Rogerson, 1993; Olley and Pakes, 1996; Foster et al., 2001), and its sway on the composition of turnover (Dunne et al., 1989; Ericson and Pakes, 1995; Campbell, 1998). Selective exit can remove less productive firms, although frictions may weaken or reverse this cleansing effect (Caballero and Hammour, 1994; Barlevy, 2002; Osotimehin and Pappadà, 2017; Foster et al., 2016), while selection at entry determines subsequent growth and survival (Sedláček and Sterk, 2017; Sterk et al., 2021). These dynamics can themselves amplify and propagate the effects of aggregate shocks. In Clementi and Palazzo (2016), a positive aggregate productivity shock expands entry, and the subsequent growth of surviving entrants extends the expansion after the aggregate shock recedes. I show that survival selection also amplifies nominal shocks by making aggregate prices less responsive. Whereas amplification in Clementi and Palazzo (2016) arises from the growth of surviving entrant cohorts, here it arises from the pricing decisions of firms exposed to exit and the allocation of replacement. Selection therefore strengthens monetary non-neutrality even without a response of aggregate entry and exit.

In monetary models, entry and exit transmit shocks through changes in product variety and productive capacity. Endogenous entry changes the range of available products (Bilbiie et al., 2008; Lewis and Poilly, 2012; Bilbiie et al., 2014); under the conditions established by Bilbiie (2021), this adjustment can substitute for price flexibility. Nominal rigidity can in turn amplify the entry and exit response to supply disruptions (Bilbiie and Melitz, 2020). In models with heterogeneous producers, Read (2020); Hamano and Zanetti (2022); Gamber (2026) examine how entry and exit following monetary shocks contribute to the responses of output and inflation.[8]

[8] In Read (2020); Gamber (2026), producer turnover occurs upstream of a separate sector with nominal price rigidity. Here, survival and costly price adjustment concern the same products.

I isolate a different margin by fixing both the number of products and the aggregate turnover rate. Firm dynamics thus affect monetary transmission through the adjustment of incumbent prices and through changes in the population of firms. This identifies a role for composition of entry and exit in monetary transmission even when aggregate turnover itself does not respond.

The distinction between selective and random turnover becomes particularly important for low values of inflation, where selection generates the largest amplification. At high inflation, firms reprice mostly because the price level erodes their relative price, reacting more to aggregate than to idiosyncratic conditions (Ball et al., 1988; Gagnon, 2009; Alvarez et al., 2019; Gautier et al., 2026). Recent quantitative work studies the resulting feedback between inflation and adjustment frequency (Blanco et al., 2024, 2025). When common inflationary pressure is the leading cause of repricing, different compositions of entry and exit have less scope to alter price adjustments, and selective and random turnover generate more similar monetary responses. Random replacement can therefore approximate the consequences of selective turnover more closely at high inflation, while severely understating monetary non-neutrality at low inflation near price stability. This matters for assessing monetary policy transmission across inflation environments: the sensitivity of output to nominal shocks depends on the interaction between inflation and market selection, even when the aggregate turnover rate is unchanged.

These results suggest an interaction between monetary stabilization and policies that affect selection. Existing work shows how creative destruction and productivity differences across product cohorts affect optimal inflation (Oikawa and Ueda, 2018; Adam and Weber, 2019, 2026), and how nominal growth changes entry selection and reallocates activity toward more innovative firms (Miyakawa et al., 2022). My analysis suggests that policies that weaken the selection of firms or products may therefore have (unintended) effects on the effectiveness of monetary policy, alongside effects on real allocations. To the extent that such interventions weaken selection, they could attenuate the amplification I identify here, affecting the output response to nominal shocks. This possibility connects policies about firm survival to the state-dependent stabilization trade-offs studied by Karadi et al. (2026): their consequences for monetary transmission may be particularly important near price stability, where selection has the greatest scope to affect transmission of aggregate shocks.

Section 2 documents the stylized facts in the scanner data. Section 3 lays out the model. Section 4 reports the calibration, the aggregate result, and the decomposition into channels. Section 5 varies trend inflation and the size of the shock. Section 6 derives the mechanism in closed form in a reduced model. Section 7 concludes.

## 2 Empirical evidence

This section documents three facts in product-level data, organized by product revenue: small products exit more often and change their prices less often, and revenue predicts both margins through different components.

**Table 1: Descriptive statistics.**

| Complete panel | Value |
|---|---|
| Sample period | 2008m1–2020m12 |
| Transactions (non-promotion) | 163,524,586 |
| Recorded purchase value (mn €) | 341 |
| Unique products (UPCs) | 999,062 |
| Brands | 20,404 |
| Retail chains | 231 |
| Product categories (groups) | 719 (86) |

*Notes*: The complete panel counts all non-promotion transactions, 2008–2020.

## 2.1 Data and coverage

I use Dutch retail scanner data for the years 2008-2020 provided by AiMark. The dataset covers fast-moving consumer goods (FMCG): food, beverages, cleaning products, beauty, and over-the-counter drugs. The panel captures a substantial share of supermarket expenditure: about €341 millions of sector revenue. This setting makes it a good case to study price stickiness and product dynamics: the Netherlands is a fairly homogeneous economy with little cross-province variation, and the long structure of the panel allows me to obtain enough variation in the product lifecycles. Products are sold across chains: each observed store belongs to a national retail chain. For each purchase I observe the purchase date and province, the retail chain, the UPC barcode, units, whether the sale was on promotion, brand and group and category assignments, from which I construct unit values. Product groups are larger groups of similar products (e.g. *cat food* or *soft cheese*); categories represent a more granular subdivision within a group (e.g. *canned cat food*, *foreign cheese*). In total, I have 999,062 unique UPCs that fall into 86 product groups and 719 product categories. The goods belong to 20,404 unique manufacturing brands, and are sold across 231 different retail chains.[9]

Table 1 describes the data. The raw panel records 163.5 million non-promotion transactions over 156 months. From these I build cells, a UPC in a chain in a province, collapse each cell's transactions within a month to a modal posted price, and keep the cell-months and product lifecycles that satisfy coverage filters. Appendix A.1 details each step of the construction: the unit-value price, the coverage filters, the removal of unflagged sales, and the classification of price changes, entry, and exit.

I summarize the size of a product by its lagged log revenue: the natural log of the previous month's revenue at the product-chain-province cell. Lagging by one month avoids the mechanical contemporaneous link between today's price change and today's revenue. Appendix A.3 repeats Facts 1 and 2 with the mean of log revenue over the previous twelve months, which captures persistent size at the cost of a slightly smaller sample; Section 4.2 uses the same measure to compare the model with the data.

[9] In Section A.2 in the Appendix I show that price changes of the same product are synchronized across chains and move in the same direction, so I read them as coming from the product rather than from the retailer.

**Table 2: Exit within $h$ months by decile of lagged log revenue.**

| | Unweighted | | | Revenue-weighted | | |
|---|---|---|---|---|---|---|
| Horizon | Bottom decile | Top decile | Ratio | Ratio | Slope | $N$ |
| 3 months | 5.5 | 3.6 | 1.52 | 1.54 | −0.5 (0.4) | 255,738 |
| 6 months | 10.1 | 6.9 | 1.46 | 1.45 | −0.7 (0.8) | 250,744 |
| 12 months | 19.4 | 13.4 | 1.45 | 1.44 | −1.4 (1.6) | 238,849 |
| 24 months | 35.0 | 22.1 | 1.58 | 1.57 | −3.4 (2.5) | 215,540 |
| Average | | | 1.50 | 1.50 | | |

*Notes*: Probabilities in percent. The ratio is the bottom-decile probability over the top-decile probability. Revenue-weighted ratios weight each product-month by its lagged revenue within deciles that hold equal numbers of product-months. The slope is a revenue-weighted OLS regression of the exit indicator on lagged log revenue, in percentage points per log point, with standard errors clustered by product category in parentheses.

## 2.2 Three facts by product size

Product turnover disproportionately replaces products belonging to groups with less frequent price adjustment. Low-revenue products are more likely to exit and change their prices less often, making revenue predictive of both survival and pricing behavior. In this section I present three simple stylized facts that relate revenues, exit, and pricing. The evidence motivates studying these margins jointly to understand how product turnover affects monetary transmission.

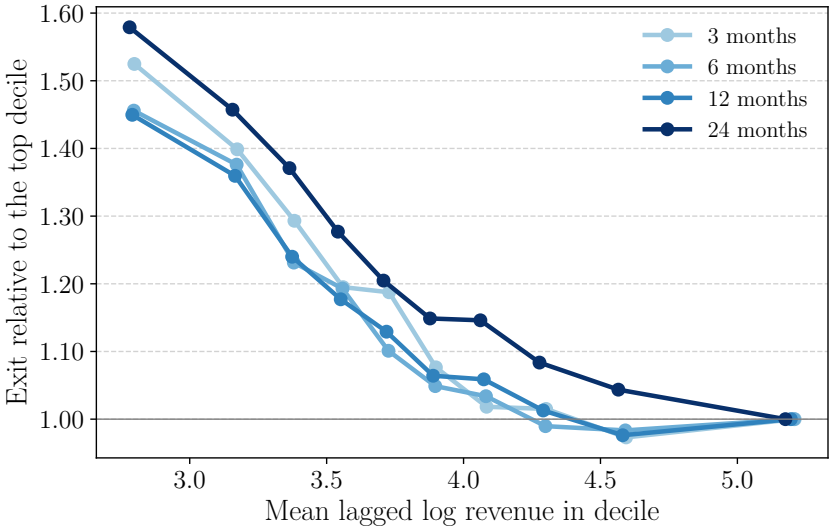
**Fact 1.** *Small products exit more often. At every horizon from three months to two years, products in the bottom decile of revenue exit about one and a half times as often as products in the top decile.*

Table 2 reports the share of product-months that exit within $h$ months by decile of lagged log revenue, and Figure 1 divides each profile by its top decile. The bottom decile exits between 1.45 and 1.58 times as often as the top decile, and most of the gradient sits in the bottom three deciles, which supply 36 percent of all six-month exits. Weighting product-months by revenue within each decile leaves the ratios unaffected. Using bins that contain a tenth of total revenue, instead of a tenth of product-months, just moves the six-month ratio to 1.35 (Appendix A.5), but the same pattern remains.

One possible concern about exit is that it might simply represent product substitutions, or retailers changing their shelf composition, and not a product genuinely leaving the market. It does not appear so. 82 percent of product exits, weighted by revenue, are followed by the product leaving the whole panel within six months, and 70 percent among products sold in two or more chains, where a delisting would be visible (Table A.3). In a similar fashion, relabeling accounts for little of measured exit. A same-brand product enters the same category around an exit only 12 percentage points more often than around a random month of the same brand. Besides, exiting products sell less than their competitors. In the twelve months before it exits, a product sells about a quarter less than products in the same category and month (Table A.4).[10]

[10]Selection on size sharpens when the economy turns down. For a one-standard-deviation rise in unemployment

**Figure 1: Exit within $h$ months relative to the top decile.**




*Notes*: For each decile of lagged log revenue and each horizon $h$, the figure plots the share of product-months for which the cell exits within the next $h$ months, divided by the same share in the top decile.

**Fact 2.** *Small products change their prices less often. Products in the top decile of revenue change their prices about 1.4 times as often as products in the bottom decile.*

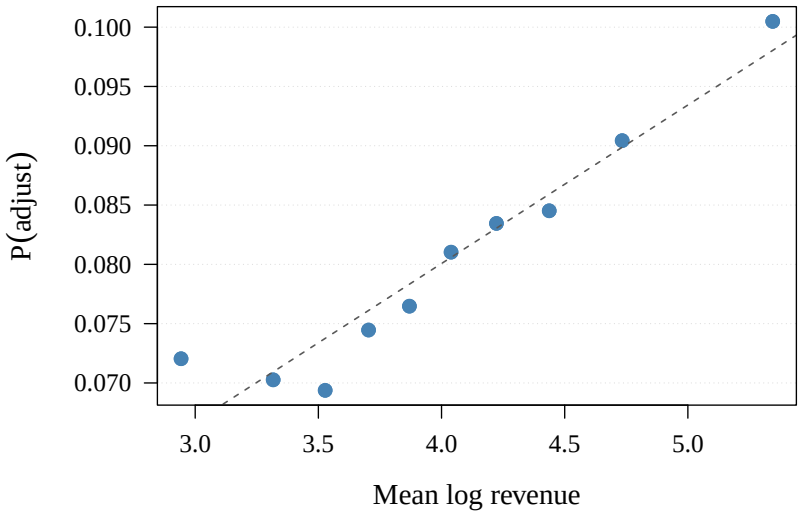
Products with lower revenue change their prices less often: the monthly frequency rises from 7.2 percent in the bottom revenue decile to 10.1 percent in the top (Figure 2). Across the full sample, this relationship mainly reflects differences between categories, with more frequent adjustment in higher-revenue categories. Among branded products, it also holds within categories, both in monthly observations and when comparing each barcode's average revenue with its lifetime frequency of price changes (Table 3).

Differences across categories matter for monetary transmission because aggregate adjustment depends on the distribution of price rigidity across the economy. Nakamura and Steinsson (2010) show that sectoral heterogeneity in adjustment frequency substantially increases monetary non-neutrality. Here, the evidence connects this heterogeneity to turnover: the low-revenue groups with less frequent price adjustment also experience more exit. Replacement therefore falls unevenly across groups with different pricing behavior, motivating a joint analysis of survival and price adjustment. The within-category relationship among branded products complements this aggregate pattern and is consistent with producer-price evidence that larger firms and firms selling more products adjust prices more frequently (Goldberg and Hellerstein, 2009; Bhattarai and Schoenle, 2014).

**Fact 3.** *Lower revenue predicts both a higher probability of exit and less frequent price adjustment. Lower quantity sold predicts exit, but does not predict the frequency of price changes.*

over twelve months, the size gradient of six-month exit steepens by 11 to 17 percent of its average (Table A.8). See Section A.6 in the Appendix for a discussion about entry and exit of products over the business cycle.

**Figure 2: Frequency of price changes by decile of lagged log revenue.**




*Notes*: For each decile of lagged log revenue, the figure plots the mean indicator that the cell changes its price in the current month. The dashed line is the OLS fit through the ten bin means; no fixed effects.

**Table 3: Frequency of price changes on lagged log revenue, all and branded products.**

| | Month FE | Category × month FE | Lifetime rate, category FE |
|---|---|---|---|
| All products | 1.30** (0.41) | −0.11 (0.17) | 0.19 (0.61) |
| Branded products | 1.97*** (0.49) | 0.97* (0.45) | 4.50** (1.54) |
| *N*, all / branded | 169,079 / 13,803 | 169,079 / 13,803 | 1,370 / 187 |

*Notes*: Coefficients in percentage points per log point. The first two columns regress the monthly indicator of a price change on lagged log revenue, on consecutive months within spells of at least twelve months. The last column regresses each barcode's lifetime share of months with a price change on its mean log revenue, one observation per barcode with at least twelve months, with category fixed effects. Standard errors clustered by category in parentheses. *, **, *** denote significance at the 5, 1 and 0.1 percent levels.

Revenue captures a relationship with both survival and price adjustment that quantity sold alone does not. Table 4 compares revenue with its price and quantity components as predictors of six-month exit and monthly price adjustment, controlling for month, chain and province. Products selling half as many units have a six-month exit probability about 1.5 percentage points higher, but no statistically detectable difference in adjustment frequency. Conversely, products with twice the price adjust about 1.0 percentage point more frequently each month, without a statistically detectable difference in exit probability. Revenue combines these two dimensions and is associated with both outcomes. This comparison motivates using revenue to summarize product size when studying survival and pricing jointly: sorting by quantity alone captures the exit gradient but misses the relationship with price adjustment (Appendix A.5). These comparisons run across categories; the within-category evidence is reported separately above.

The relationship between revenue and exit is not explained by differences across product categories. Within category and month, the estimated gradient is nearly three times as steep as

**Table 4: Exit and price changes on lagged revenue, quantity and price.**

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| A. Exit within six months, percent | | | | |
| Lagged log revenue | −1.59** | | | |
| | (0.54) | | | |
| Lagged log quantity | | −2.14*** | | −2.30*** |
| | | (0.43) | | (0.35) |
| Lagged log price | | | 0.15 | −0.60 |
| | | | (0.97) | (1.00) |
| Mean of outcome, % | | 7.93 | | |
| *N* | | 250,744 | | |
| B. Price change in the month, percent | | | | |
| Lagged log revenue | 1.33*** | | | |
| | (0.33) | | | |
| Lagged log quantity | | 0.58 | | 0.96 |
| | | (0.76) | | (0.67) |
| Lagged log price | | | 1.50** | 1.80*** |
| | | | (0.55) | (0.43) |
| Mean of outcome, % | | 8.03 | | |
| *N* | | 169,079 | | |

*Notes*: Linear probability models with month, chain and province fixed effects. Coefficients in percentage points per log point. Panel A: exit within six months. Panel B: a price change in the month, on consecutive months within spells of at least twelve months. Standard errors clustered by category in parentheses. **, *** denote significance at the 1 and 0.1 percent levels.

in the comparison across categories: −4.7 versus −1.6 percentage points per log point of revenue. The relationship holds for private labels and, with less precision, for branded products (Table A.6). It also persists when exit requires disappearance from every chain and province within six months, when the sample is split into two periods, and when comparing products within multi-product brands (Table A.2). Low revenue therefore predicts exit across several definitions of product disappearance and among otherwise more comparable products.

Together, these facts motivate a model in which the economic conditions governing survival also shape pricing decisions. In Section 3, productivity links a firm's revenue, its incentive to reprice and its continuation value. Low-productivity firms earn less revenue and face a higher probability of exit; the shorter expected horizon reduces the return to paying a menu cost and can discourage price adjustment. This mechanism provides an explanation for why low-revenue products both exit more often and reprice less frequently. Neither revenue gradient is targeted in the calibration, so their comparison with the data in Section 4.2 provides a validation check on the model's ability to connect survival and pricing behavior.

# 3 Model

The empirical evidence establishes a systematic relationship between product survival and price adjustment, but does not determine its implications for monetary transmission. To quantify these effects, I build a general equilibrium menu-cost model with selective firm turnover

and aggregate risk. The model allows me to isolate how selection through the entry and exit margins affects the real response to nominal shocks, and how this relationship changes across inflation environments. The pricing block follows the tradition of Golosov and Lucas (2007), Burstein and Hellwig (2007), Nakamura and Steinsson (2008), and Aruoba et al. (2024): a representative household, a competitive final-good producer, and a mass of intermediate producers that pay a menu cost to change their price. Incumbents face a state-dependent exit hazard that decreases in the firm's value, so low-value firms exit disproportionately. Selection operates on both the entry and exit margins, governed by two elasticities that determine its magnitude. Setting both elasticities to zero collapses the model to uniform turnover, which I use to isolate the effect of selection. The model is then mapped to the Dutch scanner data, targeting the frequency and average size of price changes and the exit rate; the two elasticities are calibrated to the relative size of exiting and entering products. The model reproduces, untargeted, the size gradients of exit and of the frequency of price changes of Section 2.2, and Section 4.2 compares them with the data.

### 3.1 Household

A representative household supplies labor to the intermediate producers, holds a complete set of state-contingent securities, and owns the firms. The representative household's problem is

$$\max_{C_t, h_t, \boldsymbol{B}_{t+1}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \chi h_t\right] \quad \text{s.t.} \quad P_t C_t + \boldsymbol{Q}_t \boldsymbol{B}_{t+1} \leq B_t + W_t h_t + \Pi_t, \tag{1}$$

where $C_t$ is consumption of the final good, $P_t$ its price, $W_t$ the nominal wage, $\boldsymbol{Q}_t$ the price vector of the state-contingent securities $\boldsymbol{B}_{t+1}$ purchased at $t$, $B_t$ the realized payoff of the securities bought at $t-1$, and $\Pi_t$ aggregate profits rebated from the intermediate producers, net of menu costs. First-order conditions deliver labor supply

$$\frac{W_t}{P_t} = \chi C_t,$$

and the stochastic discount factor $\Xi_{t,t+1} = \beta\, C_t / C_{t+1}$.

### 3.2 Aggregate nominal demand

Monetary policy targets nominal expenditure $S_t \equiv P_t C_t$, which follows a random walk with drift,

$$\log S_t = \bar{\pi} + \log S_{t-1} + \sigma_S\, \eta_t, \qquad \eta_t \sim N(0,1). \tag{2}$$

With log-linear preferences, the labor supply equation implies $W_t = \chi S_t$, so firms' marginal cost moves one-for-one with nominal demand.

Monetary policy in this economy sets a path for nominal expenditure, unlike standard New Keynesian models in which it targets the nominal interest rate.[11] As is standard in the litera-

[11] The two setups share the same object but differ in which instrument is used to achieve it: standard New

ture, since a one-time change in $\eta_t$ creates a permanent change in money balances, I interpret it as a monetary policy shock, which is the only source of aggregate risk in the model. This specification is the workhorse of the quantitative menu-cost literature (Golosov and Lucas, 2007; Nakamura and Steinsson, 2010; Midrigan, 2011; Aruoba et al., 2024).[12] Because (2) is common to every economy I solve, all of them face the same sequence of nominal demand shocks, and real outcomes differ only through the cross-section of firms. How selective exit interacts with a rule that responds to inflation and output is a question I leave for future research.

### 3.3 Final good producer

A perfectly competitive final-good producer combines a measure $\mu_t$ of intermediate varieties via a CES technology, taking input prices as given and earning zero profits. Demand for variety $i$ and the aggregate price are

$$y_{it} = Y_t \left( \frac{p_{it}}{P_t} \right)^{-\theta}, \qquad P_t = \left( \int p_{it}^{1-\theta} \, d\mu_t \right)^{\frac{1}{1-\theta}}, \tag{3}$$

where $\theta > 1$ is the elasticity of substitution and $\mu_t$ is the cross-sectional measure of operating firms.

### 3.4 Intermediate producers

There is a fixed mass of intermediate producers indexed by $i$, whose composition changes endogenously through exit and entry. Each operates a linear labor technology

$$y_{it} = z_{it} \, l_{it},$$

where $z_{it}$ is firm-specific productivity that follows an AR(1) process in logs with conditional cdf $F$,

$$\log z_{it} = \rho \, \log z_{i,t-1} + \sigma_\varepsilon \, \varepsilon_{it}, \qquad \varepsilon_{it} \sim N(0,1).$$

A firm's static real flow profit, given its inherited price $p_{i,t-1}$ and its choice of price $p_{it}$, is

$$\pi(p_{it}, z_{it}, \mathcal{S}_t) = \frac{p_{it}}{P_t} \, y_{it} - \frac{W_t}{P_t} \, l_{it} - \frac{W_t}{P_t} \, f \, \mathbb{1}\{p_{it} \neq p_{i,t-1}\},$$

where $f$ is a labor-denominated menu cost paid whenever the firm changes its price, and $\mathcal{S}_t = (P_t, Y_t, W_t, \mu_t)$ collects the aggregate state, including the cross-sectional measure of firms.

Keynesian models use a price (the nominal interest rate), while here I use a quantity (nominal spending). Both pin down the path of nominal aggregate demand, and the real response is whatever the pricing friction determines. In a cashless economy the central bank can set the nominal interest rate so as to deliver a target path for nominal aggregate demand (Woodford, 2003).

[12] The regime also has normative standing of its own: Bullard and Di Cecio (2025) characterize nominal GDP targeting as optimal under household heterogeneity and nominal contracting frictions.

**Incumbent's problem.** Let $s = (p_{-1}, z, \mathcal{S})$ denote the firm's full state. The incumbent's value function, conditional on having survived this period's exit draw, is

$$V(s) = \max\{V_N(s),\, V_A(s)\}, \tag{4}$$

where $V_N$ is the value of keeping the inherited price and $V_A$ is the value of paying the menu cost and resetting:

$$\begin{aligned} V_N(p_{-1}, z, \mathcal{S}) &= \pi(p_{-1}, z, \mathcal{S}) + \mathbb{E}[\Xi\,(1 - \lambda(s'))\,V(p_{-1}, z', \mathcal{S}')], \\ V_A(z, \mathcal{S}) &= \max_{p} \Big(\pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi\,(1 - \lambda(s'))\,V(p, z', \mathcal{S}')]\Big). \end{aligned}$$

Continuation values are discounted by the household's stochastic discount factor $\Xi$ and by the next-period survival probability $1 - \lambda(s')$. Because the exit hazard depends on the firm's own continuation value through equation (5), the firm internalizes the effect of its pricing decision on its own future survival when solving the Bellman equation: a price that delivers higher continuation value also lowers the firm's exit probability. As is standard in this literature, expectations over $\mathcal{S}'$ require firms to track the (infinite-dimensional) distribution of states. I follow Krusell and Smith (1998) and assume that firms forecast aggregates using a finite set of moments of the distribution, as detailed in Section 3.5.

**Exit hazard.** The exit hazard takes the form of an exponential tilt of the cross-sectional value distribution, normalized to deliver an aggregate exit rate of $\delta$:[13]

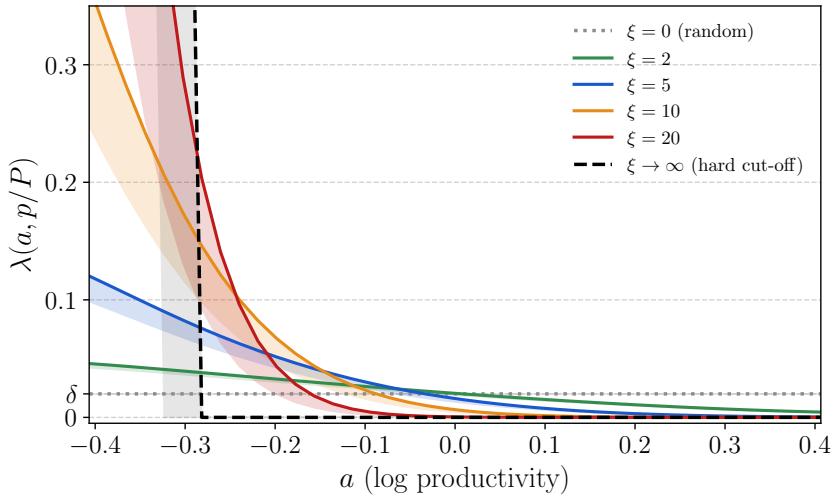
$$\lambda(s) \;=\; \delta \cdot \frac{\exp(-\xi\, V(s))}{\mathbb{E}_{\mu}[\exp(-\xi\, V)]}. \tag{5}$$

Two parameters control turnover. The first is $\delta$, the average exit rate: by construction, $\mathbb{E}_{\mu}[\lambda] = \delta$. The second is the exit elasticity $\xi \geq 0$, which controls selectivity. With $\xi = 0$ the hazard collapses to a constant $\delta$ and exit is iid random, independent of the firm's state. With $\xi > 0$ the hazard is decreasing in $V(s)$, and low-$V$ firms exit disproportionately; the tilt becomes steeper as $\xi$ grows. Selective exit is therefore a single-parameter generalization of pure-replacement turnover. Appendix B provides a microfoundation for (5) based on the problem of a retailer with a fixed shelf that sees the value of each product with noise and drops the products with the lowest signal. Two economies with the same $\delta$ but different $\xi$ have the same stationary exit flow and differ only in which firms leave. Figure 3 plots the calibrated hazard as a function of firm productivity.

To first order, exit depends on productivity: low-productivity firms exit at much higher rates than high-productivity ones, and the dependence on the inherited price is second order. In Figure 3, the solid line is the hazard for adjusters, who reset to the same flexible price regardless

[13] In the numerical solution the hazard is also capped below one and the residual mass is redistributed across the remaining states so that the aggregate constraint binds with equality.

**Figure 3: Exit hazard by firm productivity.**




*Notes*: The figure plots the exit hazard $\lambda(a, p/P)$, for different values of the exit elasticity $\xi$. For each $\xi$, the solid line reports $\lambda$ over the price-adjustment region conditional on $a$: since the hazard is flat in $p/P$ within that region, the line captures the exit rate of adjusting firms. The shaded band shows the [min, max] range of $\lambda$ within the no-adjustment region, where the hazard varies with $p/P$: low-productivity firms can effectively lower their exit probability by sticking to their old price, and save on the menu cost. Figure A.17 in the Appendix shows the exit hazard as a two-dimensional function of productivity and relative price.

of $p_{-1}$, and the shaded band shows the variation in the hazard across the firm's inaction region. The band widens with the exit elasticity $\xi$: by avoiding the menu cost, a low-productivity firm can lower its own exit probability to some extent, and this margin grows with the strength of selection.

**Entrants.** Entry is one-for-one with exit. Entrants arrive after the exit draw and draw productivity from the entry distribution

$$dG_e(z) \;\propto\; z^{-\tau}\, dG(z), \tag{6}$$

where $G$ is the ergodic distribution of incumbent productivity and $\tau \geq 0$ is the entry elasticity. At $\tau = 0$, $G_e = G$ and entrants are statistically identical to the incumbent pool. At $\tau > 0$, the entry distribution is tilted toward low-productivity types, a reduced-form representation of entrants, which comes from the data, as I observe new products starting smaller, with lower revenues. An entrant sets the reset price $p^\star(z, \mathcal{S})$, the price that attains $V_A(z, \mathcal{S})$, without paying the menu cost (as it has no old price to inherit), and produces in its first period. From the next period on it behaves as an incumbent: it carries $p^\star(z, \mathcal{S})$ as its inherited price, draws productivity from $F(\cdot \mid z)$, and faces the same exit and pricing decisions. The mass of entrants in each period is closed by replacement,

$$M = \int \lambda(s)\, d\mu(s),$$

which equals total exit.[14]

**Distribution of firms.** Combining the exit hazard, the incumbent transition kernel, and the entrant inflow yields the law of motion for the cross-sectional measure $\mu$ over the state space:

$$\mu'(s') \;=\; \int (1-\lambda(s))\, Q(s,s')\, d\mu(s) \;+\; M\, \nu_e(s'), \tag{7}$$

where $Q$ is the incumbent transition kernel induced by the price-adjustment policy, the productivity process, and the law of motion of the aggregate state, and $\nu_e$ is the distribution of this period's entrants over next period's states: productivity $z \sim G_e$, inherited price $p^\star(z, \mathcal{S})$, and next-period productivity drawn from $F(\cdot \mid z)$.

**Timing.** The within-period sequence of events is as follows:

1. The aggregate state $\mathcal{S}_t$ is realized. Incumbents enter the period with their inherited price $p_{i,t-1}$ and draw productivity $z_{it} \sim F(z_{it} \mid z_{i,t-1})$.
2. The exit hazard $\lambda(s)$ fires on incumbents. Exiting incumbents leave before producing. Entrants of mass $M$ arrive and draw productivity $z \sim G_e$ from (6).
3. Surviving incumbents decide whether to keep their inherited price or to pay the menu cost and reset. Entrants set $p^\star(z, \mathcal{S}_t)$ at no menu cost.
4. Production takes place and surviving incumbents and entrants collect flow profits.

From the following period on, an entrant carries an inherited price and draws productivity from $F$ like any other firm. Figure 4 summarizes the timing.

### 3.5 Equilibrium

Appendix C.1 defines the state space $\mathbf{S}$, the incumbent transition kernel $Q$, and the entrant distribution $\nu_e$.

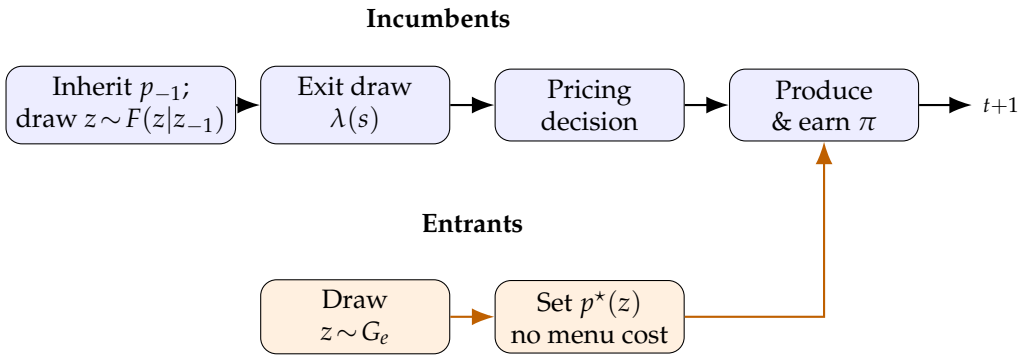
Heterogeneity, rational expectations, and aggregate risk together make the firm's problem depend on the cross-sectional distribution $\mu_t$, an infinite-dimensional object. Following Krusell and Smith (1998) and Nakamura and Steinsson (2010), I assume firms only care about a finite number of moments of the distribution, namely, the aggregate state $\mathcal{S} := \log(S/P_{-1})$, nominal demand deflated by last period's price index. Firms forecast the price level with a log-linear rule

$$\log \frac{P_t}{S_t} \;=\; \alpha_0 + \alpha_1 \log \frac{P_{t-1}}{S_t}. \tag{8}$$

Equivalently, firms expect log inflation $\log(P_t/P_{t-1}) = \Gamma(\mathcal{S}_t)$, with $\Gamma(\mathcal{S}) = \alpha_0 + (1-\alpha_1)\,\mathcal{S}$.

[14] A free entry condition would connect the entrant mass to the (expected) profits, creating an interaction across entry conditions, pricing decisions, and aggregate state. I instead fix the entrant mass through one-for-one replacement, since the focus of this paper is on the composition of the surviving cross-section under selective exit rather than on aggregate market size. Replacement shuts down the level of the extensive margin that Bilbiie (2021) and Bilbiie and Melitz (2020) study, allowing me to focus on the compositional margin.

**Figure 4:** Timing within a period.




*Notes*: Incumbents (top lane) inherit last period's price and draw productivity, face the exit hazard, choose whether to adjust, then produce. Entrants (bottom lane) arrive after the exit draw, draw productivity from $G_e$, set the flexible price at no menu cost, and produce. From $t+1$ they inherit that price, draw productivity from $F$, and face exit and pricing as incumbents.

**Definition.** *A recursive competitive equilibrium consists of value functions $V_N(s), V_A(s)$, a price-adjustment policy $p(s)$, an exit hazard $\lambda(s)$, an entrant distribution $\nu_e$ and entrant mass $M$, an invariant cross-sectional measure $\mu$, aggregate quantities $\{C, Y, W/P\}$, and a forecasting rule $\Gamma$ for the aggregate state, such that:*

1. *household optimization gives $C = S/P$ and $W/P = \chi C$;*
2. *given $\Gamma$, the wage $W/P$, and the hazard $\lambda$, the policy $p(s)$ solves the incumbent's Bellman equation (4);*
3. *the exit hazard satisfies (5);*
4. *the entrant mass equals total exit, $M = \int \lambda \, d\mu$;*
5. *the cross-sectional measure $\mu$ is invariant under the law of motion (7);*
6. *goods and labor markets clear at the implied $(P, Y)$;*
7. *the forecasting rule $\Gamma$ is consistent with the inflation path generated by firms.*

Appendix C.2 provides the algorithm in detail; Appendix D reports how well the rule tracks realized inflation and re-solves the model under a cubic alternative.

I use the rule for the stationary equilibrium and for the long simulations behind the unconditional variance of output in Section 4. The headline impulse responses come from a deterministic perfect-foresight transition that tracks the cross-section $\{\mu_t, V_t, \lambda_t, \pi_t\}$ period by period over the IRF horizon, described in Section 4.3 and in Appendix C.3.

# 4 Results

This section uses the model to study how selective exit affects monetary transmission. Section 4.1 describes the calibration. Section 4.3 reports the main amplification result: selective

exit raises the unconditional variance of output and the cumulative real response to a one-time monetary shock. Section 4.4 decomposes the CIR gap into an option-value channel and a censoring channel with counterfactual economies that activate one channel at a time, and Section 4.5 describes each channel.

### 4.1 Calibration

The model period is one month. Five parameters are internally calibrated to match five empirical moments from Dutch retail scanner data (Section 2); the remaining structural parameters are externally calibrated. The internally calibrated parameters and targets are (i) the menu cost $f$, targeting the monthly frequency of price changes among incumbents; (ii) the average exit rate $\delta$, targeting the monthly UPC exit rate; (iii) the exit elasticity $\xi$, targeting the relative size of exiting products (computed as the average revenue of exiting products over the average revenue of incumbents); (iv) the entry elasticity $\tau$, targeting the relative size of entrants (computed as the average revenue of new products over the average revenue of incumbents); (v) the productivity volatility $\sigma_\varepsilon$, targeting the mean log price change. The exit and entry rate moments together pin down $\delta$ and the entrant mass.[15] Externally, I follow Nakamura and Steinsson (2010) for the trend nominal-demand drift $\bar{\pi}$ and the volatility of the nominal-demand innovation $\sigma_S$, take the productivity autocorrelation $\rho = 0.83$ from the autocovariances of prices in the same panel (Appendix A.7), and fix the standard preference and demand parameters at $\beta = 0.96$, $\theta = 4$, and $\chi = 1$. Table 5 reports the full parameter set together with model and data values for the targeted moments and several untargeted moments.

The model matches the targeted moments closely: the calibrated economy has frequency 0.09, exit rate 0.02, relative size of exiters 0.64 (data: 0.66) and of entrants 0.69 (data: 0.67). Higher moments of the price-change distribution are not targeted: the mean absolute price change in the model is about twice as large as in the data, and the standard deviation is similarly inflated.[16] The fraction of upward changes is matched without targeting.

### 4.2 Model validation

The model reproduces two empirical patterns that the calibration does not directly target: low-revenue products exit more often and change prices less frequently. The quantitative fit is close for exit over a two-year horizon and for the probability of repricing within a year. These comparisons assess whether the model captures the relationship between product size, survival, and price adjustment beyond the targeted moments.

[15] The number of products in the data is not strictly stationary: the entry rate is slightly above exit, meaning that over time the sample of products grows. I solve the model with an invariant distribution of firms, where entry and exit are balanced.

[16] This is a standard feature of menu-cost models with a single Gaussian productivity shock and no leptokurtic component (Midrigan, 2011; Aruoba et al., 2024): with one shock, the model cannot generate both small adjustments (from firms close to the inaction band) and large adjustments (from firms in the tails). Since the goal of this paper is to isolate the role of selective exit, I do not target these moments. Appendix H re-solves the model with leptokurtic innovations; the fit of these moments does not improve, and the amplification survives.

**Table 5: Benchmark calibration.**

| Parameter | Description | Value | Target | Data | Model |
|---|---|---|---|---|---|
| $f$ | Menu cost | 0.155 | Frequency of price changes | 0.09 | 0.09 |
| $\delta$ | Average exit rate | 0.020 | Monthly exit rate | 0.02 | 0.02 |
| $\xi$ | Exit elasticity | 20 | Rel. size of exiting firms | 0.66 | 0.64 |
| $\tau$ | Entry elasticity | 5 | Rel. size of entrants | 0.67 | 0.69 |
| $\sigma_\varepsilon$ | St. dev. productivity | 0.086 | Mean $\Delta \log p$ | 0.04 | 0.04 |
| *Untargeted moments* | | | | | |
| — | — | — | Mean $\lvert\Delta \log p\rvert$ | 0.10 | 0.21 |
| — | — | — | St. dev. $\Delta \log p$ | 0.11 | 0.21 |
| — | — | — | Fraction upward changes | 0.63 | 0.62 |
| *Externally calibrated* | | | | | |
| $\bar{\pi}$ | Trend nominal demand | 0.0028 | Nakamura and Steinsson (2010) | | |
| $\sigma_S$ | St. dev. nominal demand | 0.0065 | Nakamura and Steinsson (2010) | | |
| $\rho$ | Productivity autocorrelation | 0.83 | Estimated, Appendix A.7 | | |
| $\beta$ | Discount factor | 0.96 | Externally calibrated | | |
| $\theta$ | Demand elasticity | 4 | Externally calibrated | | |
| $\chi$ | Labor disutility | 1 | Externally calibrated | | |

*Notes*: The table reports calibrated parameter values, targeted moments (data vs. model), and selected untargeted moments. Five parameters are internally calibrated jointly to match the five targeted moments. The model matches the targeted first-moment pricing and turnover moments closely. The mean absolute price change and the standard deviation of price changes are not targeted and are larger in the model than in the data, a known feature of single-shock menu-cost models that abstract from leptokurtic shocks or random adjustment opportunities. The fraction of upward changes is essentially matched.

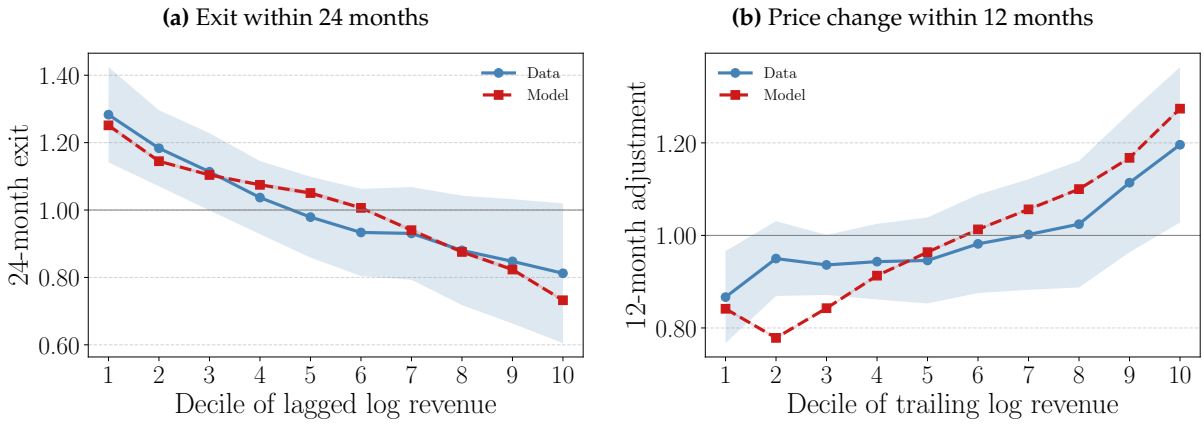
I construct the model statistics using the same definitions as in the data. I simulate 50,000 firm slots for 300 months after a burn-in of 300 months, starting from the stationary benchmark equilibrium without aggregate shocks. Log revenue equals $(1-\theta)$ times the log relative price, up to a common constant. Exit indicators and revenue deciles follow Section 2.2. For the pricing comparisons, I sort products by their mean log revenue over the previous twelve months in both the model and the data, reducing the influence of short-term price movements on the ranking. I normalize each profile by its sample mean to compare variation across revenue deciles rather than aggregate levels.[17]

Figure 5a compares exit probabilities across revenue deciles over a two-year horizon. Products in the bottom decile exit 1.71 times as often as those in the top decile in the model, compared with 1.58 in the data. The model profile falls within the empirical 95 percent confidence intervals in all ten deciles. Although the calibration targets the relative size of exiting products, it does not target this exit profile. The comparison therefore provides additional support for the model's account of selection across product sizes.

The model also captures the increase in price adjustment with product size. The probability of at least one price change within a year rises from 0.84 to 1.27 times the sample mean across revenue deciles in the model, compared with 0.87 to 1.20 in the data (Figure 5b). The model falls within the empirical confidence intervals in eight of the ten deciles. Together, these results

[17] Data confidence intervals use standard errors clustered by category. Model confidence intervals use 50 batches of 1,000 firm slots and are too narrow to be visible in the figures.

**Figure 5: Exit and price changes by revenue decile at long horizons, model and data.**




*Notes*: Panel (a): for each decile of lagged log revenue, the share of product-months followed by exit within 24 months. Panel (b): for each decile of mean log revenue over the previous twelve months, the probability of at least one price change within the next twelve months. Deciles run from smallest (1) to largest (10). Both quantities are divided by their sample mean. Data: scanner data. Model: simulated benchmark panel. Shaded: 95 percent intervals.

show that the model captures both the higher exit risk and the less frequent repricing of low-revenue products. The fit is worse at shorter horizons, where exit in the model concentrates more among low-revenue products than in the data (Appendix F.1).

### 4.3 Aggregate effect of selective exit

I compare two economies, both matched to the same moments (Appendix E):

1. **Benchmark.** The fully calibrated economy: selective exit ($\xi = 20$) and selective entry ($\tau = 5$), at $\delta = 0.02$.

2. **No selection.** Same average exit rate ($\delta = 0.02$) but no selectivity ($\xi = 0$, $\tau = 0$). Total exit equals the benchmark, but turnover is i.i.d. random across firms and entrants are statistically identical to the incumbent pool.

I summarize aggregate dynamics with two objects. The first is the one-year cumulative impulse response (CIR) of output to a one-time nominal shock, computed along a perfect-foresight transition path that tracks $\{V_t, \mu_t, \lambda_t, \pi_t\}_{t \geq 0}$ period by period (Appendix C.3). The second is the unconditional variance of output, $\mathrm{Var}(Y)$, over a 25,000-period simulation in which nominal demand follows (2) and firms apply the forecasting rule $\Gamma$. Table 6 reports both. The Benchmark's CIR is about fifteen percent above No selection's, and its output variance is higher by a similar amount. I also report an economy without entry or exit (No churn.): no exit at all ($\delta = 0$), and consequently no entry, whose CIR increases only by half a percent of No selection's, showing how the composition of entry and exit matters, rather than the amount.

**Table 6: Cumulative impulse response and variance of output.**

| Model | CIR (1 year) | vs. No selection | $\text{Var}(Y) \times 10^5$ | vs. No selection |
|---|---|---|---|---|
| Benchmark | 0.178 | +14.6% | 2.36 | +15.1% |
| No selection | 0.155 | – | 2.05 | – |

*Notes*: One-year cumulative impulse response CIR $= \frac{1}{12}\sum_{s\leq 12}\psi_s$, in annualized units, where $\psi_t = (\log Y_t - \log Y_t^0)/\Delta$ is the output response to a one-time log-nominal innovation $\Delta$ at $t = 0$, computed along the perfect-foresight transition path and differenced against the contemporaneous no-shock path $Y_t^0$ (Appendix C.3). Firms set their date-0 prices before observing the shock, so $\psi_0 = 1$ in all economies. $\text{Var}(Y) \times 10^5$ is the variance of real value-added output across a 25,000-period stationary simulation of the model with nominal aggregate shocks. The percent differences are relative to No selection, the economy with the Benchmark's exit flow and no selectivity.

### 4.4 Decomposing the CIR

The exit hazard works through two channels in the model. The first one is anticipation: survival discounts every continuation value, so a selective hazard affects pricing incentives before any firm has actually exited: this is where the option-value channel operates. The second one is realization. The hazard determines which firms are deleted from the cross-section, and this is where the censoring channel operates. In equilibrium a single hazard carries both channels at once. To quantify each channel in units of monetary non-neutrality, I separate the two mechanisms: firms solve their pricing problem under a perceived hazard $\lambda_B$, while the cross-section evolves under a realized hazard $\lambda_F$. Each hazard comes from (5) with its own exit elasticity and with the Benchmark's value function in the weights,

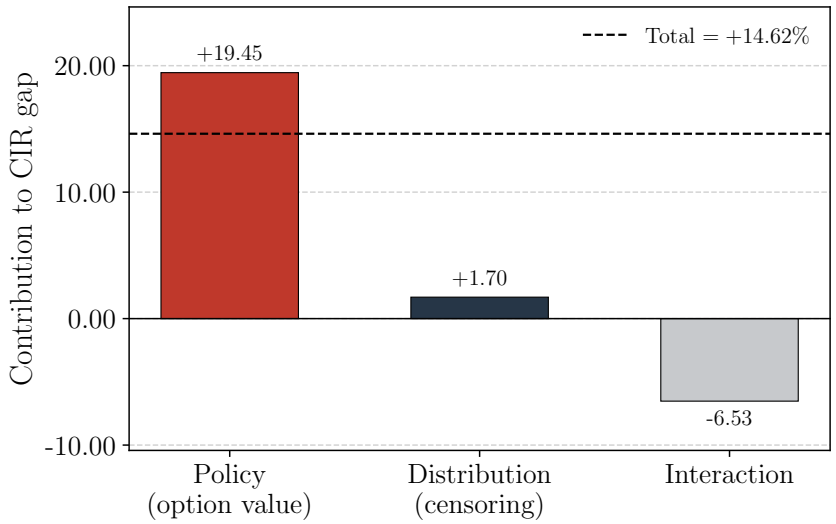
$$\lambda_j(s) = \delta \cdot \frac{\exp(-\xi_j V^b(s))}{\mathbb{E}_\mu[\exp(-\xi_j V^b)]}, \qquad j \in \{B, F\},$$

where $V^b$ is the stationary value function of the Benchmark and $\mu$ is the cross-section of the counterfactual economy. The weights $\exp(-\xi_j V^b)$ stay fixed across economies and dates, and the normalizer integrates them over each economy's own cross-section, so both hazards carry the same exit flow $\delta$ and all economies below share the same total exit rate. Each switch therefore turns on the Benchmark's ranking of which firms exit, perceived in one economy and realized in the other, at the Benchmark's amount of exit. This counterfactual exercise provides the same kind of comparison that defines the Benchmark against No selection: same turnover, different composition of entry and exit.

The entry elasticity $\tau$ of (6) enters the model only through the entrant distribution $\nu_e$ in the law of motion, so it is assigned to the realization side. A configuration $(\xi_B, \xi_F, \tau)$ then defines a complete economy: I solve its stationary equilibrium, hit it with the same one-time nominal shock, and compute the one-year CIR under the procedure of Section 4.3.[18] The mixed configurations embody deliberately inconsistent beliefs, since firms perceive an exit process different from the realized one, which is the point of the exercise: switching off one transmission channel at a time.

[18]The computation is identical to Appendix C.3, with $\lambda_B$ entering the backward pass and $\lambda_F$ the forward pass.

**Figure 6: CIR decomposition: No selection to Benchmark.**




*Notes*: Contribution of each channel to the one-year CIR gap between the No-selection economy $(\xi_B, \xi_F, \tau) = (0, 0, 0)$ and the Benchmark $(20, 20, 5)$, in percent of the No-selection CIR. The policy bar is the CIR of the economy $(20, 0, 0)$, the distribution bar that of $(0, 20, 5)$, each in deviation from the base; the interaction is the residual to the total gap (dashed line). All CIRs are computed along the perfect-foresight transition path of Section 4.3.

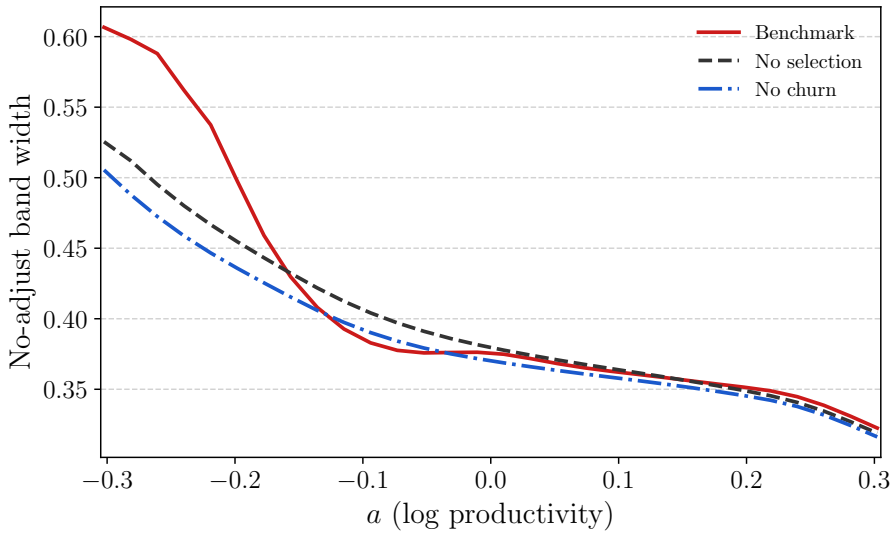
Turning on the selection only in the Bellman equation quantifies the option-value channel: firms price as if exit were selective, but actual exit falls at random. Turning it on only in the law of motion prices the censoring channel: exit falls selectively, but firms price as if it were random. Each counterfactual CIR, in deviation from the economy with no selection anywhere, gives one channel's standalone contribution; the two contributions sum to the total gap up to an interaction term. Figure 6 reports the decomposition. Anticipation alone raises the CIR by more than the whole gap. Censoring alone adds under two percent.[19] The interaction is negative: the two channels act on the same firms.

## 4.5 Mechanisms

The real effect of a nominal shock is governed by the amount of misaligned prices in the economy, and by how fast these prices catch up with the shock. This is pinned down by two features of the model: which firms find it worthwhile to pay the menu cost in any given period, and which firms are present in the cross-section to do so. Selective exit affects both. It changes the pricing policy by lowering the option value of paying the menu cost for the firms most exposed to exit (the option-value channel), and it changes the composition of the cross-section by removing a non-random subset of firms (the censoring channel). Amplification depends on how the removal differs between selective and random exit.

[19] Selection on exit and selection on entry work in opposite directions: on its own, selective exit raises the CIR by 18.5 percent relative to No selection, and selective entry by 0.3 percent. Appendix F.3 explains why adding selective entry to selective exit lowers the amplification.

**Figure 7: Inaction region by productivity, across models.**




*Notes*: The figure shows the width of the inaction region in $\log(p_{-1}/P)$ as a function of log productivity, at the stationary aggregate state, for the three economies. Band edges are interpolated between grid points. Inside the band, firms keep their inherited price; outside, they pay the menu cost and reset. The band is wider for low-$z$ firms in the Benchmark relative to the no-selection and no-churn economies, reflecting the option-value effect of a higher exit hazard at low $V$.

Higher exit hazards lower the present value of paying the menu cost, especially for low-productivity firms whose hazard is largest. The result is that the inaction region $V_N(s) \geq V_A(s)$ widens for those firms relative to an economy with random exit. Figure 7 plots the implied $Ss$ band (the no-adjustment region of $\log(p_{-1}/P)$ as a function of productivity) across the three economies.

The inaction band is widest at low productivity in all three economies: less productive firms have lower flow profits, and the gain from a price reset is small relative to the menu cost. Selective exit amplifies this pattern. In the Benchmark, low-productivity firms face a disproportionately higher exit hazard, which acts as an additional discount on the future returns to any current price reset, and they therefore find it less worthwhile to pay the menu cost today. For more productive firms the hazard is closer to the cross-sectional average, the discount effect is small, and the band is essentially unchanged across economies. The selective hazard therefore reshapes the pricing policy at low productivity before any firm has actually exited. In a similar fashion, the No churn economy has the smallest inaction band as this additional discounting due to exit is absent.

The censoring channel works through the firms removed by exit. The key difference is that the firms removed by exit in the Benchmark economy are firms that received a sequence of bad productivity draws: they face a very high marginal cost, their inherited price is too low, and hence they are about to raise it.[20] These would-be adjusters are replaced by fresh entrants

[20]Followed forward under the No-selection dynamics, the cohort that exits in the Benchmark would have changed its price in 31 percent of cases in the month of exit, against 9 percent for a random draw of firms. Appendix F.2 reports this counterfactual exercise and the distributions of price changes by productivity.

that set their optimal price on arrival, so exit simply swaps one price change for another. In this sense, selection is harmless. In the No selection economy, instead, exit removes firms at random, and the average firm in this economy has not re-optimized in a while: it is most likely mid-spell, with a stale price. Exit removes it and replaces it with a fresh entrant, hence de facto subsidizing aggregate flexibility. Selection takes this subsidy away, and the real effects of the nominal shock grow.[21]

## 5 Selection and monetary transmission across inflation regimes

The relationship between trend inflation and the real effects of monetary shocks is not necessarily linear, implying that monetary transmission may differ substantially across inflation regimes. Changes in trend inflation can alter how firms adjust prices and, consequently, how nominal shocks are transmitted to the real economy. Understanding how monetary transmission varies across inflation regimes is therefore important, particularly when moving between periods of price stability and high inflation. This section examines how selective turnover contributes to this relationship, and whether its role in amplifying monetary shocks changes across inflation environments.

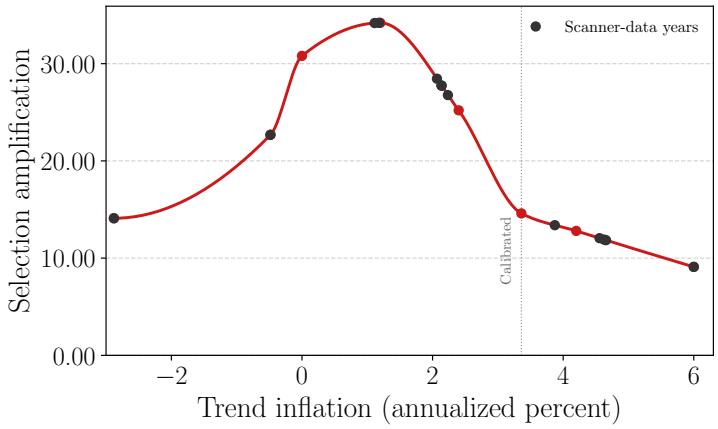
To investigate this interaction, I vary trend inflation while holding the average exit rate $\delta$ and all other parameters at their calibrated values, and compare the one-year cumulative output responses under selective and random exit. The amplification from selection is strongest near price stability, reaching approximately 35% around zero inflation, including mild deflation. It then declines as inflation increases, falling markedly above six percent and approaching zero at high inflation. For extremely high inflation rates, selection may even dampen the real effects of monetary shocks. Figure 8 illustrates this relationship.

Trend inflation acts on both channels: it affects the width of the inaction bands, and it changes which firms selection censors. Both channels are stronger when there is virtually no general increase in prices. In general, firms can reprice for aggregate reasons (i.e., trend inflation) or for idiosyncratic reasons. Since the amplification comes from differences across firms, low trend inflation leaves much more room for idiosyncratic motives to matter, and hence accentuates the differences between selective and random exit. When trend inflation is instead the main reason for repricing, all firms are more likely to reprice alike, leaving little room for individual idiosyncrasies to move prices.

Anticipation is stronger near zero inflation because of how long a wider inaction band can keep a price stale. Near zero inflation, only idiosyncratic shocks move a firm toward the edge of its band, so a wider band can delay its price change for a long time. At high trend inflation, instead, every relative price erodes by the trend each month, so the drift forces the price change

[21] Section 6 makes this argument formally in a more general case. Monetary non-neutrality equals the dispersion of price gaps in the cross-section, so any exit rule that interacts with price gaps changes it, depending on where in the gap distribution exit lands. Holding the exit flow fixed, Theorem 1 gives a sufficient condition for selection to amplify, proved in Appendix J.5.

**Figure 8: Selection amplification across trend inflation.**




*Notes*: The amplification is the percent increase in the one-year CIR of output under selective relative to random exit, at a common total exit rate. The amplification across trend inflation including deflation, with each Dutch scanner-data year (2008–2020) placed at its realized annual price drift; the dotted line marks the calibrated rate.

within a few months whatever the width of the band. Likewise, censoring is stronger for low values of inflation. At higher trend inflation every price erodes faster, so much more of the cross-section has already queued up a price increase, and a randomly placed exit increasingly lands on a firm that was going to reprice anyway. Random exit buys less flexibility, the two rules converge, and the amplification decreases.

To see this, I compare what an exiting firm is about to do with what the average firm is about to do. For a firm in state $s = (p_{-1}, z, \mathcal{S})$, let $p^\star(z, \mathcal{S})$ be the reset price, the maximizer in $V_A(z, \mathcal{S})$, and let

$$g(s) \;=\; \mathbb{1}\{V_A(s) > V_N(s)\}\,[\log p^\star(z, \mathcal{S}) - \log p_{-1}] \tag{9}$$

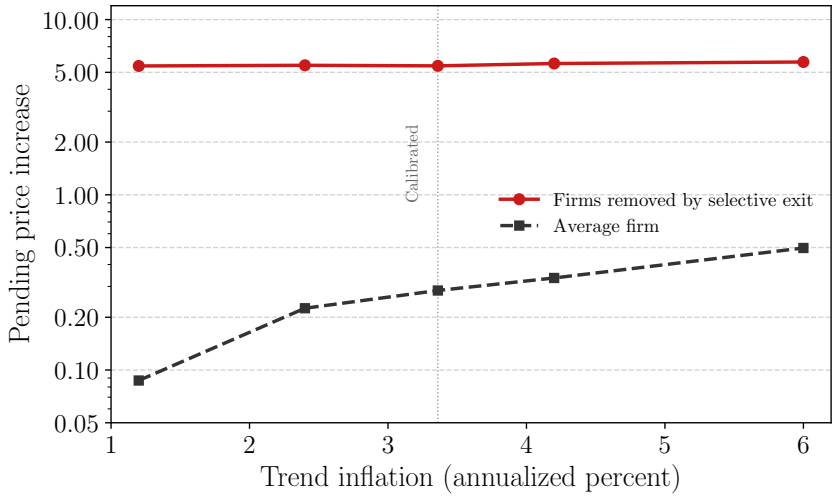
be the pending price change the firm is about to post: zero if it keeps its price, the distance from its inherited price to its reset if it adjusts. Then I can compare two different averages of $g$: one that runs over the firms removed by exit, the other one over all firms of the No-selection economy,

$$\mathbb{E}_\lambda[g] \;=\; \frac{1}{\delta}\int \lambda(s)\, g(s)\, d\mu_b(s), \qquad \mathbb{E}[g] \;=\; \int g(s)\, d\mu_{ns}(s). \tag{10}$$

$\mathbb{E}_\lambda[g]$ weights each firm by its chance of dying, which (5) normalizes to average $\delta$, and it represents the pending price increase of the firm removed by selective exit, with $\mu_b$ the cross-section of the Benchmark. $\mathbb{E}[g]$ instead weights every firm of the No-selection economy alike, with $\mu_{ns}$ its cross-section, and gives the pending increase of a firm drawn at random, i.e. the firm removed by random exit. In a stationary economy prices rise with the trend, so $\mathbb{E}[g]$ is close to trend inflation. These are the two lines Figure 9 plots against trend inflation: the pending increase of an exiting firm, and the pending increase of the average firm. The exiting firm is about to raise its price by roughly 5.5 percent at every trend, while the average firm goes from 0.09 to 0.50 percent a month. The ratio between the two falls, so the differences between the two specifications increasingly shrink.

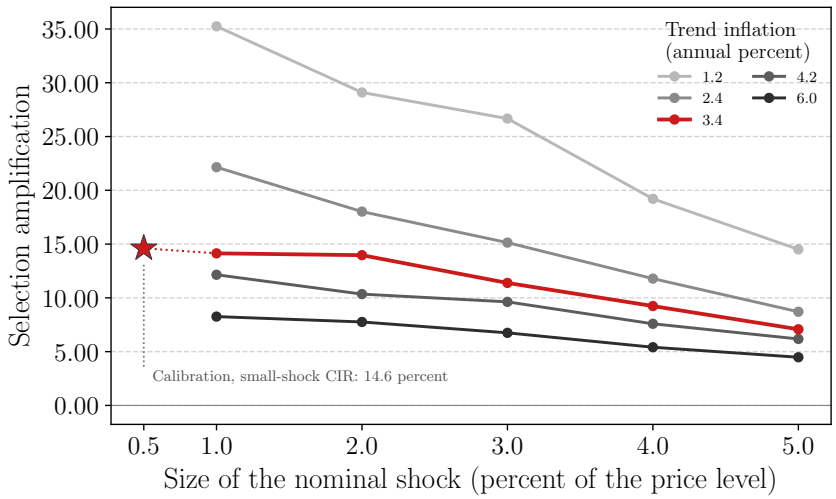
The size of the shock acts on the amplification the same way trend inflation does. I hold

**Figure 9: Pending price increase of the firms removed by selective exit and of the average firm.**




*Notes*: The monthly log price increase a firm records if it survives to the pricing stage, $g(s)$ of equation (9), in percent per month, averaged over the firms exit removes with weight $\lambda(s)\, d\mu_b(s)/\delta$ and over the whole cross-section with weight $d\mu_{ns}(s)$. $g$ and the adjustment policy come from the No-selection economy, the hazard and $\mu_b$ from the Benchmark, and $\mu_{ns}$ from the No-selection economy. The vertical axis is logarithmic, so the gap between the two series reads as the ratio $\mathbb{E}_\lambda[g]/\mathbb{E}[g]$. The dotted line marks the calibrated rate.

**Figure 10: Selection amplification across shock sizes.**




*Notes*: Selection amplification of the one-year CIR (the percent increase under selective relative to random exit, at a common total exit rate) against the size of the one-time nominal shock, one line per annualized trend-inflation level. Shocks are expressed as a percent of the price level. The star marks the calibrated economy at the small-shock limit, the headline amplification of Section 4.3. Zero trend inflation is omitted: its net repricing flow is near zero, so the ratio is numerically unstable.

every parameter at its calibrated value, vary the size of the one-time nominal shock from half a percent to five percent of the price level, and recompute the one-year CIR under both exit rules at each trend inflation level. Trend inflation and shock size both raise the repricing that happens for reasons other than exit. The trend does it every month; a large shock does it once, by pushing firms out of their inaction band on impact. A large shock pushes firms out of their inaction band on impact whatever its width, so, as with a high trend, the wider bands of firms facing a high exit hazard delay less.

Figure 10 reports the result. Amplification falls with the size of the shock at every trend inflation level, and the ranking across levels holds at every size. A five percent shock cuts the amplification by half or more: from 14.6 to 7.1 percent at the calibrated trend, from 34.2 to 14.5 at a 1.2 percent trend, from 9.1 to 4.5 at six percent. Shock size and trend inflation substitute: a five percent shock at 1.2 percent trend inflation gives 14.5 percent amplification, the same amplification a small shock gives at the calibrated 3.36 percent trend. The fall comes from the selective side. At the calibrated trend the No-selection CIR moves from 0.155 to 0.151 as the shock grows from half a percent to five percent, while the Benchmark CIR falls from 0.178 to 0.161.

# 6 Understanding the results

To gain additional insights on the mechanisms, I develop a reduced model that clarifies how survival prospects affect pricing and when selective replacement amplifies nominal shocks.[22] I study the two channels separately: first, firms choose their inaction band under a constant exit hazard; second, the band is fixed and exit varies with the price gap. This separates the incentive to postpone adjustment from the flexibility supplied by replacement. All derivations and proofs are in Appendix J.

## 6.1 The reduced model

The only state of a firm is its price gap $x$: the log difference between the price it charges and the price it would set if adjustment were free. Trend inflation erodes every gap downward, and idiosyncratic shocks move the target price around, so between adjustments the gap follows $dx = -\bar{\pi}\,dt + \sigma\,dW$, with $\sigma$ the standard deviation of gap innovations. A firm whose gap reaches the edge of the inaction band $(-a, a)$ pays the menu cost and resets its gap to zero. Exit arrives at a hazard $\lambda(x)$ that may depend on the gap, and the exiting firm is replaced by an entrant born at gap zero: the entrant prices optimally on entry, so it starts correctly

[22] I use continuous time to follow the analytical menu-cost literature: with the gap following a diffusion, expected spell values and the stationary density solve ordinary differential equations, which is what makes the CIR computable in closed form. I use tools coming from the literature on lumpy adjustments (Baley and Blanco, 2021). The aggregation result below extends formulas in Alvarez and Lippi (2014), Alvarez et al. (2016), and Alvarez et al. (2022) to an economy with entry and state-dependent exit, while the exit layer, the pricing of exit placement and the amplification theorem, is new to the best of my knowledge.

priced. Adjustment and exit therefore both place a correct price at that slot of the economy; what distinguishes them is which firms they hit. The monetary shock is a small once-and-for-all increase in nominal demand, which lowers every gap by $\Delta$ on impact, and the object of interest is the cumulative impulse response of output per unit of shock, the continuous time counterpart of the CIR of Section 4.3. Every comparison holds the exit flow fixed: the average hazard over the cross-section equals $\delta$ in each economy, mirroring the normalization of the structural hazard (5). The no-selection benchmark is the uniform hazard $\lambda_u := \delta$; a selective economy is any hazard $\lambda_s(x)$ with the same flow. Appendix J.1 discusses the limitations of the reduced model.

**Proposition 1** (Non-neutrality is gap dispersion)**.** *In the reduced model with zero trend inflation, for any exit hazard and any inaction band,*

$$\text{CIR} = \frac{\text{Var}_f(x)}{\sigma^2},$$

*where $f$ is the stationary distribution of price gaps. Trend inflation adds a correction term of order $\bar{\pi}$.*

Money is non-neutral because, at any moment, the economy carries a spread of wrong prices, and the shock survives in that spread: $\text{Var}_f/\sigma^2$ measures how much standing mispricing the economy tolerates per unit of new mispricing injected by idiosyncratic shocks each period. The formula is the price-gap version of the sufficient statistics of Alvarez et al. (2016) and Baley and Blanco (2021): the aggregation logic survives entry and exit exactly, for any hazard and any placement of exit. Every reset, whether an adjustment or an exit and replacement, moves one firm from wherever it stands to a correct price, so what exit does to money depends only on where in the cross-section it lands. Exit that removes a firm sitting in the middle of the distribution deletes mispricing that would otherwise have persisted, compresses the spread, and makes money more neutral. Exit that removes a firm about to adjust deletes nothing, because that firm's price was about to be corrected anyway.

With a fixed inaction band, a higher uniform exit rate lowers the cumulative output response, while a wider band raises it. Random replacement removes prices that would otherwise persist; allowing firms to tolerate larger price gaps works in the opposite direction. These comparative statics distinguish the flexibility supplied by realized replacement from the rigidity induced by anticipated exit.[23]

### 6.2 Exit risk and price adjustment

The option-value experiment gives one firm the pricing problem of Section 3 in quadratic form. The flow loss from a gap $x$ is $Bx^2$, the second-order expansion of profits around the frictionless

[23] A related limiting case appears in Bilbiie (2021): when every firm is replaced each period and entrants set prices when they begin producing, no inherited prices remain and money is neutral. Here, increasing uniform replacement likewise reduces monetary non-neutrality. Selective replacement changes how much flexibility a given turnover rate supplies.

price. The firm discounts at rate $r$, pays a menu cost $\kappa$ per adjustment, and faces an exit hazard $\lambda$ that is constant along its spell. Exit ends the stream of losses and of future menu costs, so survival acts as a higher discount rate, $\hat{\rho} := r + \lambda$. At zero trend inflation the optimal policy is a symmetric band $\pm\bar{y}$ with reset to zero (Stokey, 2009).

**Proposition 2** (Exit risk widens the band)**.** *The optimal half width $\bar{y}$ is strictly increasing in the effective discount rate $\hat{\rho} = r + \lambda$, and so in the exit hazard.*

This is the mechanism of the option-value channel of Section 4.5, firm by firm. The proposition prices the level of one firm's hazard, not how the hazard is spread across firms. Selection holds the average hazard at $\delta$, so relative to uniform exit the firms with a hazard above $\delta$ widen their bands and those with a hazard below $\delta$ narrow them. The net effect on the CIR then depends on which firms the hazard reaches and where they sit in the cross-section, which the decomposition of Section 4.4 measures in the full model.

### 6.3 Exit allocation and amplification

The censoring experiment fixes the band, as the censoring margin of Section 4.4 fixes pricing policies, and lets the hazard depend on the gap. In the full model the hazard depends on value, and value on productivity; the results below use only where exit lands in gap space, which Figure 11 measures. It plots the mean exit hazard conditional on the price gap, $\mathbb{E}[\lambda(s) \mid \text{gap} = x]$, over the stationary cross-section of the Benchmark. The hazard is flat and negligible over the interior of the gap distribution and rises steeply at negative gaps. A firm that has just drawn low productivity has a target price above its posted price, a negative gap, and a low value, so the firms exit removes are the firms about to raise their price. Productivity drives exit and predicts the gap at once: in gap space, selection on value is selection on imminent repricing.

Random exit lands mostly in the interior of the band, where it removes standing mispricing. At the other extreme, i.e. exit landing outside of the inaction band, nothing changes: the exit hazard only decides whether the event is recorded as a price adjustment by the incumbent or as an exit followed by a correctly priced entrant (Proposition A.3). Between random exit and exit at the margin, the effect of a unit of exit depends on the gap where it lands.

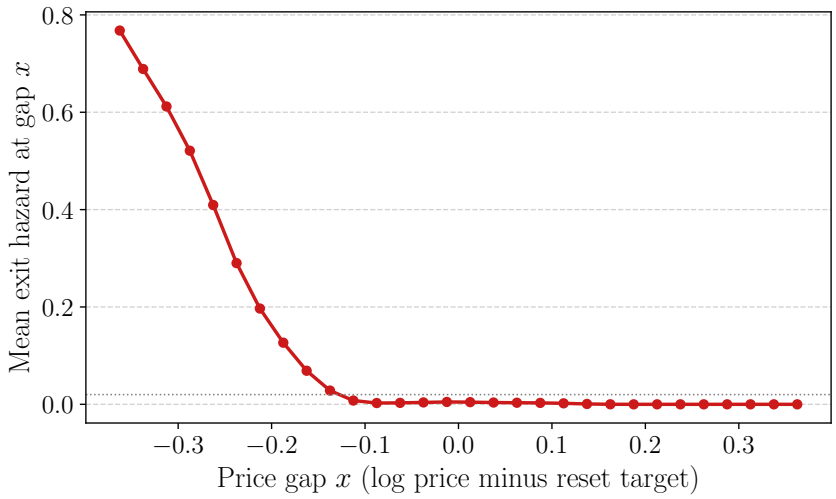
**Proposition 3** (The price of exit placement)**.** *At zero trend inflation, let $\lambda$ be any baseline hazard with stationary distribution $f$ and mean gap $\bar{x}$. Adding exit in any direction $h$, $\lambda \to \lambda + \theta h$, changes the CIR at rate*

$$\frac{d\,\mathrm{CIR}}{d\theta}\Big|_{\theta=0} = \int_{-a}^{a} \omega(x)\, h(x)\, dx, \qquad \omega(x) = \frac{f(x)}{\sigma^2}\, \mathbb{E}_x\left[\int_0^{\tau} \left(\mathrm{Var}_f - (x_t - \bar{x})^2\right) dt\right], \tag{11}$$

*where $\mathbb{E}_x$ follows one firm from gap $x$ until its next reset at time $\tau$, by adjustment or exit.*

The kernel measures the effect of additional exit by comparing a firm's expected remaining mispricing with average dispersion in the economy. It accumulates the difference between $\mathrm{Var}_f$ and the firm's squared deviation from the mean until its next adjustment or exit. Additional

**Figure 11: The exit hazard in the space of price gaps.**




*Notes*: Mean monthly exit hazard conditional on the price gap, $\mathbb{E}[\lambda(s) \mid \text{gap} = x]$, over the stationary cross-section of the Benchmark economy. The dotted line marks the average exit rate $\delta = 0.02$. The gap is the log distance between a firm's price and its reset target, so the left side collects underpriced firms who would raise their price but are most likely to exit.

replacement raises monetary non-neutrality when this accumulated difference is positive and lowers it when it is negative. Both the extent of mispricing and its expected duration therefore matter.

Comparing selective and random turnover requires evaluating where exit is added and where it is withdrawn. At the no-exit baseline, the kernel $\omega_0$ is negative away from the reset point and adjustment margins, where it is zero (Figure A.18). Replacement thus removes persistent mispricing in the interior but adds no correction at the margin, where the incumbent would adjust anyway. Redirecting exit toward imminent adjusters can therefore leave more mispricing in place than random replacement. The following theorem states the weighted condition under which this reallocation amplifies nominal shocks, comparing exit flows that agree to first order.

**Theorem 1** (Selection amplifies)**.** *At zero trend inflation and a fixed inaction band, let $h(x) \geq 0$ be a bounded hazard profile and let $c = \mathbb{E}_{f_0}[h]$, where $f_0$ is the stationary density without exit. Consider selective exit $\lambda_s(x) = \theta h(x)$ and uniform exit $\lambda_u = \theta c$, whose aggregate exit flows agree to first order in $\theta$. If*

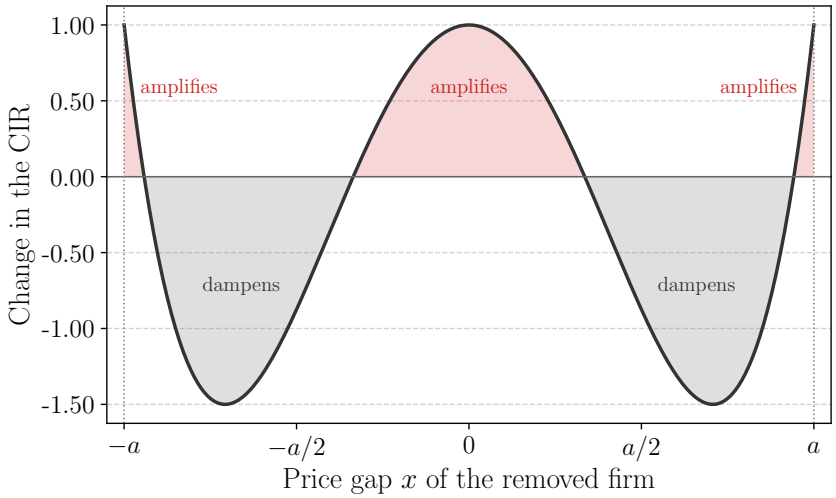
$$A(h) := \int_{-a}^{a} \omega_0(x)(h(x) - c)\, dx > 0,$$

*then*

$$\mathrm{CIR}_s - \mathrm{CIR}_u = \theta A(h) + O(\theta^2) > 0$$

*for all sufficiently small $\theta > 0$: selective replacement generates larger cumulative real effects than random replacement.*

**Figure 12: The effect of reallocating exit on monetary non-neutrality.**




*Notes*: Change in the CIR from moving one exit from a randomly drawn firm to a firm at gap $x$, in units of the effect of a random removal, at the no-exit baseline with zero trend inflation. The effect of removing a firm at $x$ is the kernel per firm, $\omega_0(x)/f_0(x)$, and the effect of a random removal is its average over the cross-section. Positive values amplify relative to random exit, negative values dampen.

The condition compares the flexibility supplied by two allocations of the same exit flow.[24] Selection amplifies when it redirects replacement toward firms whose removal reduces monetary non-neutrality less than random replacement would. Concentrating exit near the adjustment margins can satisfy this condition: replacement adds little flexibility there because the incumbent's correction is imminent, while the exit withdrawn from the interior leaves more persistent mispricing in place. Figure 12 illustrates which reallocations amplify and which dampen.

The reduced model isolates a broader economic point: turnover changes price flexibility through both the prospect of replacement and the price corrections replacement supplies. A shorter expected life reduces the return to paying an adjustment cost, while replacing a price adds little flexibility if that price would soon have changed anyway. The closed-form results rely on a simplified pricing environment, but these considerations apply whether replacement follows firm exit, product discontinuation, or the introduction of a new variety. What matters is how replacement relates to the pricing decisions that would otherwise occur. Treating turnover as random therefore makes an economically consequential assumption about both firms' incentives to adjust and which inherited prices disappear. The quantitative model establishes the magnitude of these effects; the reduced model clarifies why the same amount of turnover can support different degrees of monetary non-neutrality.

[24] The uniform level $c$ matches the two exit flows to first order in $\theta$, which is the sense in which the comparison holds the exit flow fixed.

# 7 Conclusion

Selective turnover amplifies the real effects of nominal shocks because survival and price adjustment are connected. Dutch scanner data show that low-revenue products exit more often and change prices less frequently. In the quantitative model, poor survival prospects discourage firms from paying to reprice, while selective replacement supplies less additional flexibility than random replacement. Anticipation accounts for most of the amplification. At the same turnover rate and matching the same targeted pricing moments, selective turnover raises the one-year cumulative output response by about 15 percent relative to random turnover. This difference reaches about 35 percent near price stability and shrinks as inflation or the size of the shock increases. The composition of turnover is therefore particularly important when common nominal pressures push weakly on prices.

These findings suggest that policies affecting firm survival may also affect monetary transmission. Support for vulnerable firms or products could weaken survival selection and attenuate the amplification identified here, changing the output response to monetary policy alongside its effects on productive allocation. Such interactions may matter most near price stability, where differences in survival prospects have greater scope to influence pricing. The results do not determine whether weakening selection is desirable; they identify a monetary consequence that should enter the assessment of policies shaping entry and exit.

A natural next step is to let aggregate entry and exit respond to economic conditions and examine how their composition interacts with a monetary policy rule. This would connect the mechanism studied here to the joint determination of turnover, inflation and output over the business cycle. Another direction is to study pricing and product discontinuation within multiproduct firms, where adjustment costs and survival decisions may be shared across products. Evidence on these joint decisions would help establish how broadly the connection between survival prospects and price rigidity applies.

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# A Data

This appendix describes the data cleaning and construction of the estimation samples used in Section 2. Section A.1 shows how to go from the raw transaction records to the sample of Table 1. Section A.2 shows that a chain posts one price per product nationwide, and that price changes of the same product are synchronized across chains. Section A.3 collects the robustness of the facts of Section 2.2, and Section A.4 collects the exit diagnostics. Section A.7 estimates the persistence of idiosyncratic productivity that the calibration of Section 4.1 uses.

## A.1 Sample construction

The raw data are transaction records: a household buys a barcode in a store of a retail chain in a province, on a date, in some units, for some value. The price of a transaction is the unit value, transaction value over units. I keep non-promotion transactions (12.8% of the records carry a promotion flag), drop the transactions with a missing province, round prices to the euro cent, and drop transactions with a price below €0.20 or above €25. The upper cut removes 0.05% of the remaining transactions, misplaced decimals and bulk-entry records. The lower cut removes 3.2% of transactions but 1.6% of value: items sold per piece for a few cents, half of them small bread rolls, where a one-cent move on the rounding grid is already a five percent price change.

The unit of observation is the cell, a UPC in a chain in a province. Within cell $i$ and month $t$, let $n_{it}(p)$ count transactions at price $p$. Since I do not observe the physical store location, but only the first two digits of a zip code, in principle there might be multiple stores belonging to the same chain in the same province posting different prices. To make sure I capture genuine price changes, and not some store-idiosyncratic temporary pricing policy, I use the following definition of posted price as the mode

$$p_{it} = \arg\max_{p} \, n_{it}(p), \qquad s_{it} = \frac{n_{it}(p_{it})}{\sum_{p} n_{it}(p)}, \tag{12}$$

where $s_{it}$ is the modal share. The mode returns a price that was actually posted, while a mean or a median of unit values moves with basket composition even when no posted price changes. A cell-month is retained when the mode rests on enough transactions and describes the month: $\sum_{p} n_{it}(p) \geq \underline{n}$ and $s_{it} \geq \underline{s}$.

The retained months of a cell form its lifecycle. Writing $\underline{t}_i$ and $\bar{t}_i$ for the first and last retained month, the span is $\bar{t}_i - \underline{t}_i + 1$, the density is the number of retained months over the span, and the gap is the longest run of consecutive missing months inside the span. A lifecycle is valid when its span is at least $\underline{T}$, its density at least $\underline{d}$, and its longest gap at most $\bar{g}$. Lifecycles that begin in the first sample month are left-censored and lifecycles that end in the last are right-censored: their start and end are not observed events, and they are excluded from the entry and exit calculations respectively.

I use the following thresholds: $\underline{n} = 20$, $\underline{s} = 0.50$, $\underline{T} = 12$, $\underline{d} = 1/2$, $\bar{g} = 6$. Two further steps separate a genuine price change from residual noise. The first removes temporary sales that the data provider failed to flag: the price drops for a month or two and returns exactly to

its pre-sale level. I replace these V-shaped drops with the surrounding price before classifying changes: whenever

$$p_{i,t+\ell} = p_{i,t-1} \text{ for some } \ell \leq 2, \qquad p_{i\tau} \neq p_{i,t-1} \text{ for some } \tau \in \{t, \ldots, t+\ell-1\}, \tag{13}$$

the filtered price is $\tilde{p}_{i\tau} = p_{i,t-1}$ for $\tau \in \{t, \ldots, t+\ell-1\}$, and $\tilde{p}_{i\tau} = p_{i\tau}$ otherwise. A sale that ends with a return to the old price is then no longer two price changes.

The second step restricts where a change can be observed at all. A month-to-month comparison is meaningful only when both months are retained, so I classify changes inside uninterrupted runs of consecutive retained months, and only in runs of at least twelve months. Within such a run, month $t$ is an adjustment when

$$0.001 \;\leq\; |\,\Delta \log \tilde{p}_{it}\,| \;\leq\; \log(10/3), \tag{14}$$

where the lower bound removes rounding artifacts and the upper bound removes barcode reuse and misplaced decimal points, and both follow from the literature, e.g. Alvarez et al. (2016). The frequency of price changes is the count of adjustments over the count of consecutive month pairs inside such runs.

Entry in month $t$ is a valid, non-left-censored lifecycle with $\underline{t}_i = t$; exit is a valid, non-right-censored lifecycle with $\bar{t}_i = t$. The monthly exit rate averages, over the interior of the sample,

$$x_t = \frac{\#\{i : \bar{t}_i = t,\ \text{not right-censored}\}}{\#\{i : \underline{t}_i \leq t \leq \bar{t}_i\}}, \tag{15}$$

where the interior drops the first and last twelve months, in which censoring mechanically depresses measured turnover; the entry rate is symmetric. The relative revenue of exiters is the average revenue of the products exiting in month $t$ over the average revenue of incumbents in the same month, averaged over interior months, and likewise for entrants.

## A.2 Geography of price setting

A chain sets one price per product for the whole country. Comparing the modal posted price across the provinces in which a chain sells a product, in 98% of product-chain-months the price is the same in every one of them, and the standard deviation of the log price across provinces averages 0.001 (Table A.1). The same statistic computed across chains rather than provinces is 0.028, twenty-eight times larger, so prices do differ, but along the chain dimension and not the geographic one. The province is a sampling unit, not a market: I keep it in the cell definition because the panel records it and because it lets me hold the local market fixed in the tests below, but it carries no separate decision. A product that stops appearing in a chain has left that chain's national assortment.

When one chain changes the price of a product, a different chain selling the same product changes its price within one month in 63% of cases (Table A.1). Under independent timing the estimated rate is 14%: shifting each cell's price-change months by a random offset of three to

twenty-four months and recomputing the index yields 0.14. The matched changes also agree in direction: among cross-chain pairs of price changes within one month of each other, 94% have the same sign. Price changes of the same product are timed together across chains, four times as often as chance, and move the same way. Common movements in the product's cost generate this pattern; chain-specific pricing policies do not. I therefore read an observed price change as belonging to the product, which is what the competitive final-good sector of the model assumes.

The test matters because the data report retail prices and contain no wholesale costs, so an observed price change could in principle reflect a repricing decision by the chain rather than a change in the product's underlying cost; DellaVigna and Gentzkow (2019) document that chains in the United States set prices uniformly across their stores, so chain-level pricing policies are a real possibility. Synchronization across chains discriminates between the two: a chain policy moves many products within one chain, while a producer-side shock moves one product across many chains.

A cross-chain comparison requires a product sold in at least two chains; these are branded products by construction, since private labels are sold in a single chain. For product $u$ in month $t$, an opportunity is an ordered pair of cells $(i, j)$ in different chains such that cell $i$ changes the price in $t$ and cell $j$ has a defined change indicator in $t$; a match occurs if cell $j$ changes the price in $t-1$, $t$, or $t+1$. The synchronization index is the share of matches over opportunities, pooled across products and months:

$$\text{Sync} = \frac{\sum_{u,t} \sum_{(i,j)\in\mathcal{O}_{u,t}} \mathbb{1}\left\{ \text{chg}_{j,t-1} + \text{chg}_{j,t} + \text{chg}_{j,t+1} > 0 \right\}}{\sum_{u,t} |\mathcal{O}_{u,t}|}, \tag{16}$$

where $\mathcal{O}_{u,t}$ is the set of opportunities and $\text{chg}_{j,t}$ indicates a price change of cell $j$ in month $t$. Table A.1 reports two variants that tighten the comparison: restricting the pair to cells in the same province, which holds the local market fixed, and aggregating cells to the product-chain-month before pairing, so that a chain selling the product in many provinces counts once. Both leave the conclusion unchanged. The same-month rate, with no window, is 0.46, six times the unconditional monthly frequency of price changes of 0.08 in this sample, so the co-movement is not an artifact of the one-month tolerance.

**Table A.1: Geography of price setting.**

| | Rate | Observations |
|---|---|---|
| Same price in every province, within chain | 0.98 | 51,907 |
| Synchronization, ±1 month | 0.63 | 4,778 |
| Same province only | 0.62 | 799 |
| Chain level | 0.55 | 2,067 |
| Same month only | 0.46 | 4,778 |
| Placebo: random time shifts | 0.14 | 4,778 |
| Same sign among matched changes | 0.94 | 3,446 |

*Notes*: The first row covers the 51,907 product-chain-months of the pricing sample observed in two or more provinces, and reports the share in which the modal posted price is identical in every one of them; the mean standard deviation of the log price across provinces is 0.001, against 0.028 across chains in the same product-month (4,242 product-months). The remaining rows use the 185 products of the pricing sample sold in two or more chains (22,165 cell-months with a defined change indicator). An opportunity is a pair (product, month) in which one chain changes the product's price and a different chain sells the product with a defined change indicator in that month; a match occurs if the second chain changes the price within one month. Same province restricts the pair to cells in the same province. Chain level aggregates cells to the product-chain-month before pairing. The placebo shifts each cell's change months by a random offset of between three and 24 months (20 draws, mean reported). The last row is the share of matched cross-chain change pairs in which the two changes have the same sign.

## A.3 Robustness of the facts

Table A.2 tests Facts 1 and 2 along three margins. First, the exit definition: within category, the gradient of Fact 1 survives restricting exit to national exits, in which the product leaves every chain and province within six months, and further to exits after which the brand leaves the category entirely. Second, the sample period: the exit gradient holds in both halves of the sample, and so does the pooled frequency gradient of Fact 2. Third, brand structure: the exit gradient holds within multi-product brands and among each brand's highest-revenue product, so small brands carrying small products do not drive it.

**Table A.2: Robustness of the exit and frequency gradients.**

| | Outcome and sample | Pooled | Cat. + month FE | $N$ |
|---|---|---|---|---|
| Exit definition | Cell exit within 6 months (baseline) | −1.40 (0.51) | −4.49 (0.55) | 250,744 |
| | National product exit | −0.45 (0.49) | −2.10 (0.42) | 250,744 |
| | National exit, brand leaves category | −0.20 (0.11) | −0.63 (0.12) | 250,744 |
| Subperiods | Exit, 2008–2013 | −1.39 (0.50) | −3.57 (0.46) | 98,207 |
| | Exit, 2014–2020 | −1.75 (0.72) | −4.98 (0.77) | 152,537 |
| | Frequency, 2008–2013 | +1.67 (0.73) | −0.02 (0.23) | 65,450 |
| | Frequency, 2014–2020 | +1.08 (0.32) | −0.54 (0.21) | 103,629 |
| Brand structure | Exit, multi-product brands | −1.58 (0.58) | −4.53 (0.60) | 244,968 |
| | Exit, star products | −3.08 (1.14) | −3.83 (1.26) | 17,838 |

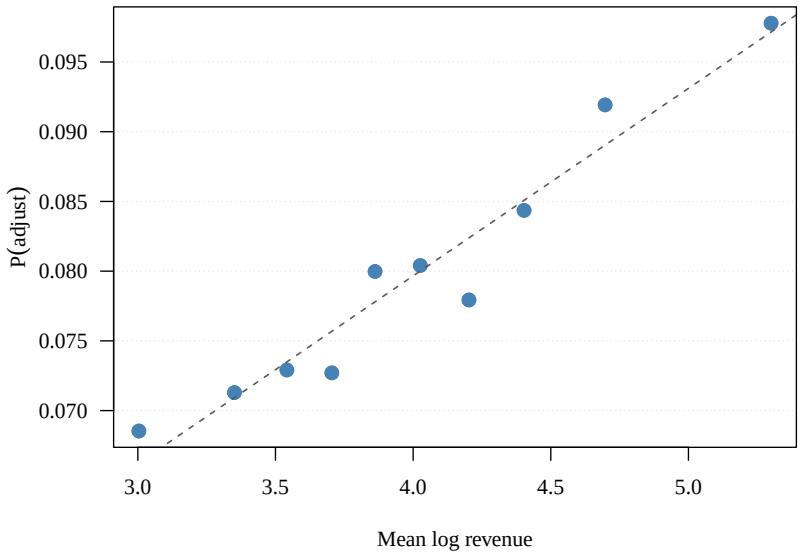
*Notes*: Each row reports the slope from a linear regression of the stated outcome on lagged log revenue, in percentage points per log point, with standard errors clustered by category in parentheses. The fixed-effects column includes additive category and month fixed effects. National product exit restricts the exit indicator to cell exits in which the UPC leaves the entire panel within six months; brand-leaves-category further requires that the brand has no other product active in the category afterwards. Star products are each brand's highest-revenue UPC.

Facts 1 and 2 use the previous month's log revenue at the product-chain-province cell as the size proxy. As an additional robustness check, I substitute the lagged trailing twelve-month

mean of log revenue. This proxy is dominated by persistent product size and is less affected by single-month sales spikes; the cost is a slightly smaller sample, since it requires at least twelve months of revenue history per cell.

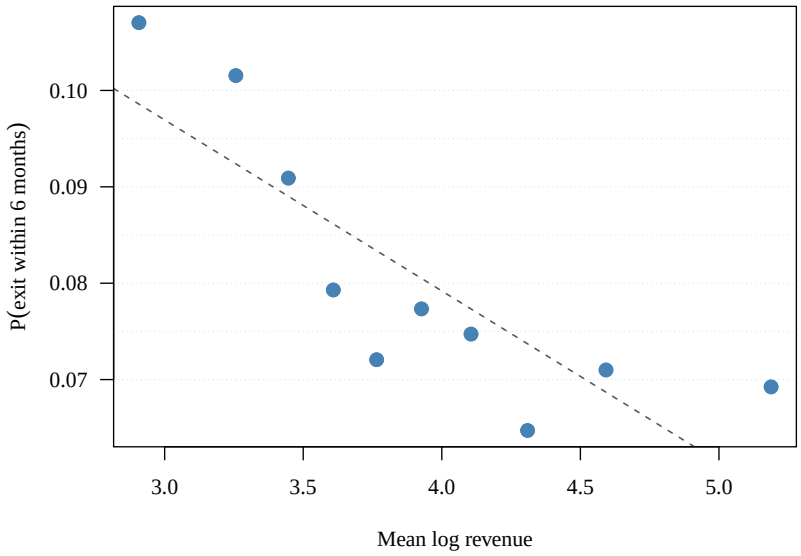
Figures A.1 and A.2 repeat the two bin scatters with this measure: the frequency of price changes rises with it, and six-month exit falls with it.

**Figure A.1: Adjustment frequency by trailing twelve-month size decile.**




*Notes*: For each decile of the trailing twelve-month log-revenue mean, the figure plots the mean indicator that the cell adjusted its price in the current month. Each decile contains 14,153 product-months. The dashed line is the OLS fit through the ten bin means; no fixed effects.

**Figure A.2: Six-month exit probability by trailing twelve-month size decile.**




*Notes*: For each decile of the trailing twelve-month log-revenue mean, the figure plots the share of product-months for which the cell exits within the next six months. Each decile contains 18,789 product-months. The dashed line is the OLS fit through the ten bin means; no fixed effects.

## A.4 Exit diagnostics

Section 2.2 reads measured exit as the exit of a product from the market. The first diagnostic follows every exiting product across all chains and provinces: an exit is national when the product disappears from the entire panel within six months of the cell's end. Weighted by revenue, 0.82 of all exit is national, and 0.70 among products sold in two or more chains, for which a within-panel substitution would be visible (Table A.3). Chain-level delisting, in which the product survives elsewhere, accounts for the remainder.

The second diagnostic studies barcode relabeling: a brand that changes a product's flavor or pack size retires one barcode and issues another, and the data would record an exit and an entry where no product actually leaves the market. For each national exit I look for a same-brand entrant in the same category within a window from three months before to six months after the exit; one appears in 0.62 of cases. However, the benchmark is not zero: these are large, active brands that launch products continuously. At a random interior month of the same brand and category, the same window contains an entry in 0.50 of cases, so the excess replacement tied to the exit itself is twelve percentage points. The pack sizes of the closest same-brand entrants split about evenly between smaller, identical, and larger than the exiting product, which is not the signature of systematic relabeling or shrinkflation.

**Table A.3: What barcode exit measures.**

| | Share |
|---|---|
| National product exit within 6 months, revenue-weighted | 0.82 |
| products sold in 2+ chains | 0.70 |
| Same-brand entrant in same category, $[-3, +6]$ months | 0.62 |
| at a random month of the same brand-category (placebo) | 0.50 |
| excess over placebo | 0.12 |
| closest entrant: same / smaller / larger pack | 0.34 / 0.30 / 0.34 |

*Notes*: Sample: valid lifecycles of the turnover sample (Table 1), interior months; 96,915 cell exits, of which 27,263 national product exits with an identified manufacturer brand. A national product exit occurs when the UPC leaves every chain and province in the panel within six months of the cell exit; shares are weighted by lifecycle revenue. The replacement rows report the share of national product exits followed by a new barcode of the same brand in the same category within three months before to six months after the exit; the placebo evaluates the same window at a random interior month for the same brand-category, averaged over ten draws. Pack size compares volume per unit of the exiting product and its closest replacement, when both share a measurement unit.

Exiting products are small for a long time before they leave. Table A.4 compares them with products in the same category and month over their last twelve months: they sell about a quarter less, and almost the whole gap is in units sold.

## A.5 Additional evidence on the three facts

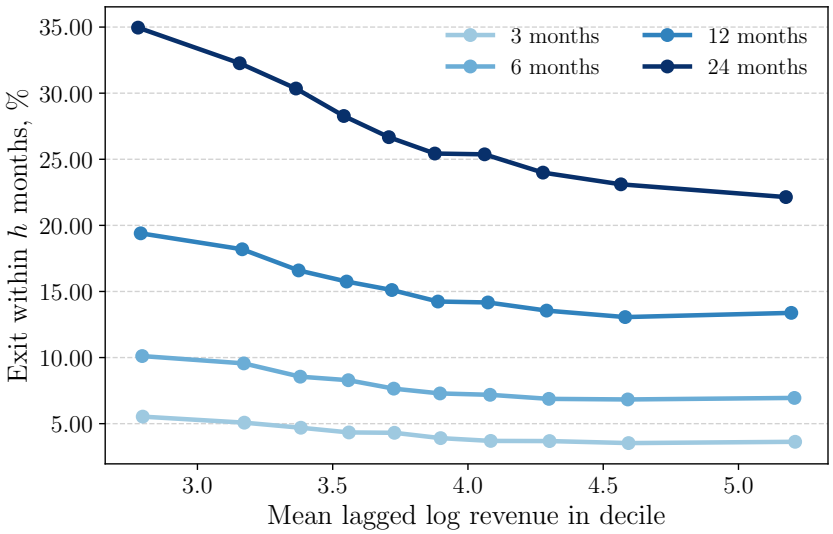
Fact 1 does not depend on how the bins are drawn. Figure A.3 plots the levels behind Figure 1. When every bin holds a tenth of total revenue, the bottom bin exits within six months 1.35 times as often as the top bin (Figure A.4). Seen from the side of the exits, the bottom decile supplies 12.8 percent of all six-month exits and the top decile 8.8 percent (Figure A.5).

**Table A.4: Revenue, quantity and price of exiting cells in the twelve months before exit.**

| | Months 1 to 12 before exit |
|---|---|
| Log revenue | $-0.29^{***}$ (0.06) |
| Log quantity | $-0.26^{***}$ (0.04) |
| Log price | $-0.03$ (0.03) |
| Exiting cells | 3,539 |
| Cell-months | 264,779 |

*Notes*: Each row is one regression of the stated log outcome on a dummy for the twelve months before the exit month and a dummy for the other months of exiting cells, including the exit month, with category × month fixed effects. The reference group is all months of cells that do not exit. Standard errors clustered by category in parentheses. *** denotes significance at the 0.1 percent level.

**Figure A.3: Exit within $h$ months by decile of lagged log revenue, levels.**



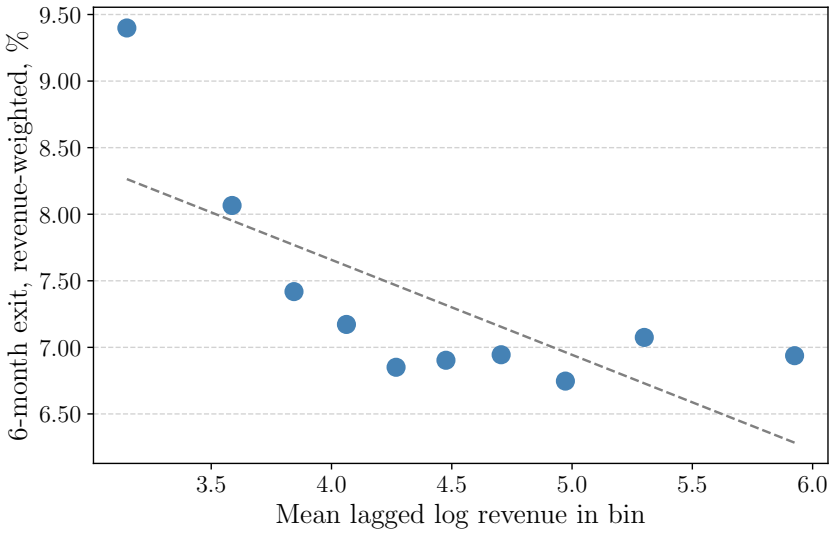

*Notes*: For each decile of lagged log revenue and each horizon $h$, the figure plots the share of product-months for which the cell exits within the next $h$ months.

Table A.5 divides each predictor by its standard deviation within category and month. Adding age and the growth of revenue over the past year shrinks by a fifth the revenue coefficient behind Fact 1. Private labels exit less often than branded products of the same size and age.

Sorting on quantity instead of revenue separates Facts 1 and 2, as Fact 3 implies. Six-month exit falls from 11.5 percent in the bottom decile of lagged log quantity to 5.7 percent in the top (Figure A.6), a steeper gradient than by revenue. The frequency of price changes stays between 7.6 and 8.2 percent in the first nine deciles and reaches 9.1 percent only in the top one (Figure A.7).

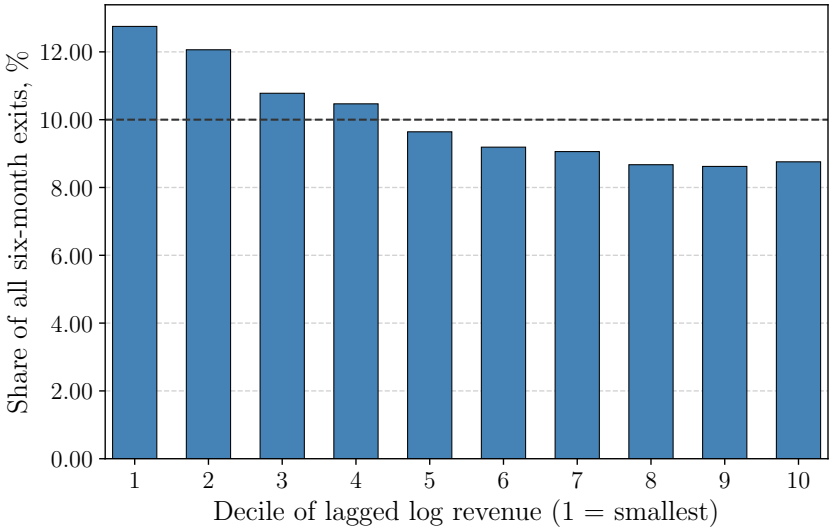
Within category and month, Fact 1 holds for branded products and for private labels (Table A.6). The branded slope is imprecise in the first three columns, which contain 27,446 branded product-months.

**Figure A.4: Six-month exit by bins of equal revenue.**




*Notes*: Each bin of lagged log revenue holds a tenth of total lagged revenue. The figure plots the revenue-weighted share of product-months for which the cell exits within the next six months.

**Figure A.5: Share of six-month exits by decile of lagged log revenue.**




*Notes*: The figure plots the share of all product-months followed by exit within six months that falls in each decile of lagged log revenue. The dashed line marks 10 percent, the share under exit independent of size.

## A.6 Entry and exit over the business cycle

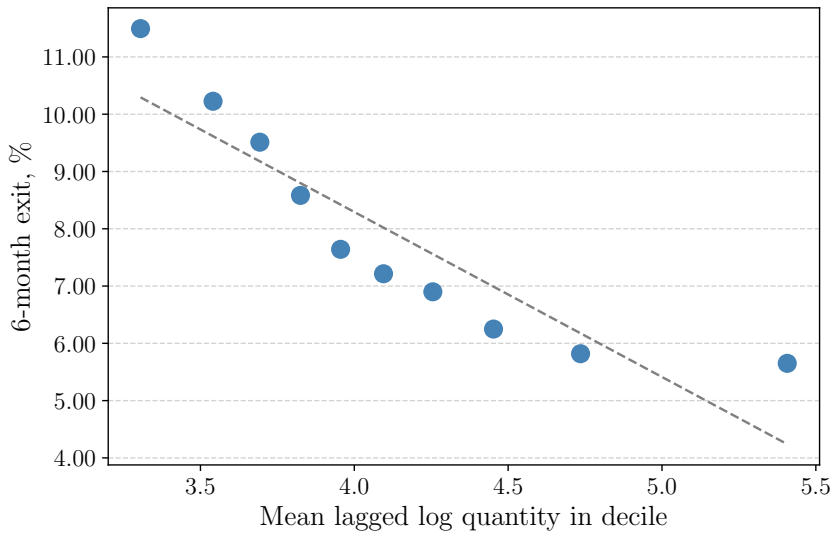
Exit rises and becomes more selective when unemployment rises. Whether downturns sharpen selection is an old question for firms. In Caballero and Hammour (1994) recessions cleanse: the least productive units leave first, so a downturn raises exit and makes it more selective. Barlevy (2002) shows that recessions can also sully the allocation, Osotimehin and Pappadà (2017) find that credit frictions weaken the selection of exiting firms without removing the cleansing effect, and Foster et al. (2016) find that reallocation in the Great Recession was less productivity-enhancing than in earlier US recessions. For products the evidence concerns the

**Table A.5: Six-month exit on predictors scaled by their within-category-month standard deviation.**

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Lagged log revenue | −1.50*** | | | | −1.17*** | | −1.05*** |
| | (0.31) | | | | (0.29) | | (0.30) |
| Lagged log quantity | | −1.52*** | | | | −1.09*** | |
| | | (0.20) | | | | (0.18) | |
| Lagged log price | | −0.64 | | | | −0.76 | |
| | | (0.53) | | | | (0.51) | |
| Log age (cells born in sample) | | | −5.32*** | | −5.04*** | −5.07*** | −4.36** |
| | | | (1.48) | | (1.49) | (1.48) | (1.40) |
| Born before 2008 | | | −17.33*** | | −16.38*** | −16.52*** | −14.05*** |
| | | | (3.96) | | (4.01) | (4.02) | (3.72) |
| 12-month change in log revenue | | | | −0.37 | −0.22 | −0.23 | −0.26 |
| | | | | (0.19) | (0.20) | (0.21) | (0.20) |
| Private label | | | | | | | −8.20*** |
| | | | | | | | (1.41) |
| Within $R^2$, % | 0.5 | 0.5 | 1.1 | 0.0 | 1.4 | 1.4 | 2.2 |
| $N$ | | | | 129,353 | | | |

*Notes*: Linear probability models of exit within six months, in percent, with category × month fixed effects. Within-category-month standard deviations: 0.48 (revenue), 0.47 (quantity), 0.25 (price), 1.54 (log age). Log age is set to zero for cells born before 2008, which carry their own dummy. Standard errors clustered by category in parentheses. **, *** denote significance at the 1 and 0.1 percent levels.

**Figure A.6: Six-month exit by decile of lagged log quantity.**

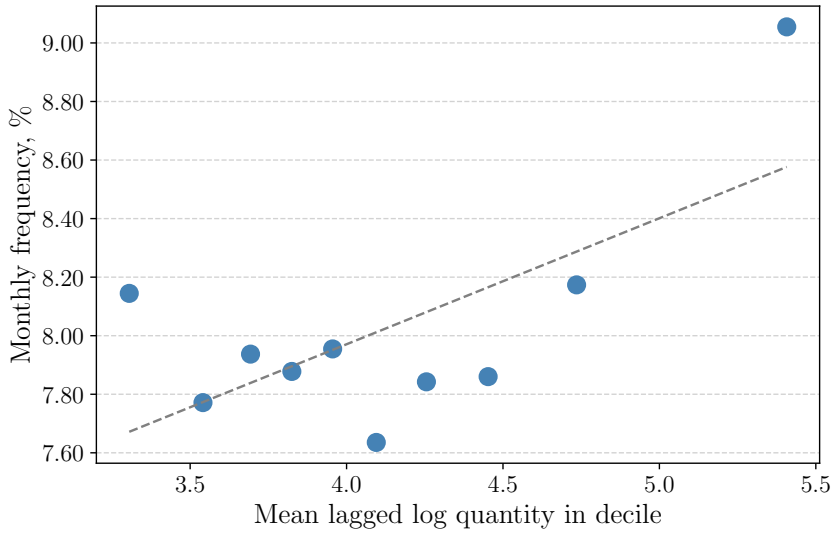



*Notes*: For each decile of lagged log quantity, the figure plots the share of product-months for which the cell exits within the next six months.

level of exit, and it does not agree. Argente et al. (2018) find that the quarterly product exit rate in the United States fell from 4.2 to 3.0 percent between 2007 and 2010, and report that Broda and Weinstein (2010) find product destruction mildly countercyclical around the 2001 recession. I know of no product-level estimate of how selective exit is over the cycle, so I check both the level and the selectivity in my data.

I use four indicators for the Netherlands from FRED: the unemployment rate, its change over the previous twelve months, the year-on-year growth of real GDP, and consumer confi-

**Figure A.7: Frequency of price changes by decile of lagged log quantity.**




*Notes*: For each decile of lagged log quantity, the figure plots the mean indicator that the cell changes its price in the current month.

**Table A.6: Six-month exit on lagged log revenue, by product type and sample.**

| | Pricing sample | | | Turnover sample | | |
|---|---|---|---|---|---|---|
| | Slope | Mean | $N$ | Slope | Mean | $N$ |
| All | $-4.7^{***}$ (0.5) | 7.9 | 250,744 | $-4.9^{***}$ (0.2) | 11.5 | 3,749,058 |
| Branded | $-1.8$ (1.2) | 14.7 | 27,446 | $-4.1^{***}$ (0.3) | 17.2 | 969,652 |
| Private label | $-4.6^{***}$ (0.6) | 7.1 | 223,298 | $-4.7^{***}$ (0.2) | 9.5 | 2,779,406 |

*Notes*: Linear probability models of exit within six months with category × month fixed effects. Slopes in percentage points per log point; means in percent. Standard errors clustered by category in parentheses. $^{***}$ denotes significance at the 0.1 percent level.

dence. Unemployment is the main one, because it is monthly, like the data. Its level rose from 5.4 percent in 2009 to 8.3 percent in 2014 and fell to 4.4 percent in 2019, so the level partly moves with the trend in the number of products; the twelve-month change measures whether the economy is entering or leaving a downturn. GDP is quarterly, and I assign each quarter's growth to its three months. Consumer confidence is monthly and captures expectations rather than realized activity.

Table A.7 regresses four monthly series on one indicator at a time: the exit rate, the entry rate, and the relative size of exiters and of entrants, defined as in the calibration. Each regression includes month-of-year fixed effects for seasonality and a linear trend, because the number of products grows over the sample. The sample runs from 2009 to 2019, which drops the first and last year, where censoring depresses measured entry and exit. I measure the series on the 29 chains present for the whole sample. On these chains a one-point rise in unemployment over twelve months raises the monthly exit rate by 0.24 percentage points, on a mean of 1.64 percent, and exiters are smaller relative to incumbents when unemployment is high. Entry does not respond to any indicator.

**Table A.7: Monthly entry and exit on cyclical indicators, stable chains.**

| | Exit rate | Entry rate | Relative size of exiters | Relative size of entrants |
|---|---|---|---|---|
| Unemployment rate | 0.056 (0.039) | −0.099 (0.088) | −0.024*** (0.006) | −0.014* (0.006) |
| 12-month change in unemployment | 0.244*** (0.065) | −0.067 (0.213) | −0.039* (0.018) | −0.018 (0.011) |
| GDP growth, year on year | −0.062* (0.030) | 0.139 (0.075) | 0.012 (0.009) | −0.008 (0.008) |
| Consumer confidence | −0.081 (0.042) | −0.139 (0.083) | 0.025** (0.009) | 0.014* (0.006) |
| Mean | 1.64 | 2.18 | 0.66 | 0.66 |

*Notes*: 29 chains present in every month. One regression per cell, with month-of-year fixed effects and a linear trend, 2009–2019. Rates in percent; relative sizes are the average revenue of exiters or entrants over that of incumbents. Newey–West standard errors with 12 lags in parentheses. Unemployment: FRED series LRHUTTTTNLM156S; GDP: CLVMNACSCAB1GQNL; consumer confidence: CSCICP03NLM665S. *, **, *** denote significance at the 5, 1 and 0.1 percent levels.

**Table A.8: Six-month exit on lagged log revenue interacted with the cycle.**

| | Turnover sample, stable chains | | Pricing sample | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Lagged log revenue | −4.20*** (0.22) | −4.18*** (0.22) | −4.77*** (0.53) | −4.79*** (0.51) |
| × unemployment rate | 0.05 (0.08) | | −0.65 (0.45) | |
| × 12-month change in unemployment | | −0.55*** (0.12) | | −0.95* (0.43) |
| $N$ | 3,193,751 | 3,193,751 | 239,671 | 239,671 |

*Notes*: Linear probability models of exit within six months, in percent, with category × month fixed effects, which absorb the level of exit in each category-month. The cycle variables are demeaned. Standard errors clustered by category in parentheses. *, *** denote significance at the 5 and 0.1 percent levels.

Table A.8 asks whether the gradient of Fact 1 changes with the cycle. It interacts lagged log revenue with the demeaned level and twelve-month change of unemployment in the product-level regression of six-month exit, with category × month fixed effects, which absorb the level of exit in each category and month. The level of unemployment does not move the gradient. The twelve-month change does: a one-standard-deviation rise in unemployment, 0.86 points, steepens the gradient by 11 to 17 percent of its average.

## A.7 Persistence of idiosyncratic productivity

I estimate the persistence of idiosyncratic productivity from the autocovariances of prices at lags of one to three months rather than from a regression on the first lag, because a monthly price computed from transactions carries measurement error, and the error moves the variance but not the autocovariances.

In the model the desired price of a firm is a constant markup over $W_t / z_{it}$, so $\log z_{it}$ enters the log relative price one for one with the opposite sign and the persistence of the relative price is the persistence of productivity. A relative price in the data also moves with the product's own demand, which $z$ does not represent. I therefore strip out the part of the price that moves with the product's market share and keep the rest. Within a category and month, the log relative price is the log unit value minus the median log unit value and the share is the product's revenue over category revenue; I regress the first on the second with product and month fixed effects and call the residual $x_{it}$. The residual is the cost state only up to the demand movements that leave a trace in the share, which is the assumption this step rests on.

The unit of observation is the product-month, pooling chains and provinces, since Section A.2 shows that a price change is determined mainly by the product, and not by the chain. I keep months with at least three transactions and a unit value between €0.20 and €25, and products observed for at least twelve consecutive months, splitting a product into separate spells at each interruption. This leaves 3,550,549 pairs of consecutive months.

A monthly price is an average over the transactions of the month, so $x_{it}$ measures the state with error. Write $x_{it} = x^*_{it} + u_{it}$, with $x^*$ an AR(1) of persistence $\rho$ and $u$ uncorrelated over time and with $x^*$. The error then sits in the variance alone,

$$C_0 = \mathrm{Var}(x^*) + \mathrm{Var}(u), \qquad C_k = \rho^k \,\mathrm{Var}(x^*) \quad (k \geq 1), \tag{17}$$

so any two successive autocovariances recover the persistence and the variance of the state, whatever the size of the error:

$$\rho = \frac{C_2}{C_1} = \frac{C_3}{C_2}, \qquad \mathrm{Var}(x^*) = \frac{C_1^2}{C_2}. \tag{18}$$

I compute the $C_k$ after demeaning by product and by month, and treat a lag as missing whenever the two months are not exactly $k$ apart on the calendar, so a hole in a product's panel never enters as an adjacent pair.

The two ratios in (18) give 0.830 and 0.833, and I set $\rho = 0.83$. Regressing on the first lag would have given 0.512: the variance of the state is 0.008 against a measured variance of 0.014, so 38 percent of the observed movement in relative prices is error, and the regression reads that error as mean reversion.

# B A microfoundation for the exit hazard

The hazard (5) is the delisting rule of a retailer owned by the household that has a fixed shelf and sees the value of each product with noise. Each period new products take a mass $\delta$ of the shelf, and the retailer drops the products whose signal of value is lowest. The exit elasticity $\xi$ is the precision of the retailer's signal, and the normalization in (5) is the signal cutoff that clears the shelf. The same constraint delivers the fixed mass of firms and one-for-one replacement.

**The retailer's problem.** A retailer owned by the household decides which products are listed on its shelf, which holds a unit mass of products. It does not buy or sell goods: listed products sell directly to the final-good producer, as in Section 3. It uses no inputs, earns no profits, and, like the final-good producer, takes prices and the cross-section as given. Each period a mass $\delta$ of new products is listed, so a mass $\delta$ of incumbents must be delisted. The retailer acts for the household, and with complete markets the household is best off when the value of its claims on the firms is as high as possible. A claim to each firm is equal to the firm's value. A delisted product loses this period's profit and all future profits, so delisting product $i$ costs the household $V(s_i)$. The retailer therefore maximizes the value of the products it keeps,

$\int (1-d_i)V(s_i)\,di$. The retailer observes the signal

$$v_i = V(s_i) - \frac{e_i}{\xi},$$

where $e_i$ is drawn after the product's state is realized, independently across products and over time, from a unit exponential distribution. Let $d_i \in \{0,1\}$ equal one if the retailer delists product $i$ this period and zero if it keeps it. The retailer chooses which products to drop to solve

$$\max_{\{d_i\}} \mathbb{E}\Big[\int (1-d_i)\,V(s_i)\,di \;\Big|\; \{v_i\}\Big] \qquad \text{s.t.} \qquad \int d_i\,di = \delta, \quad d_i \in \{0,1\}.$$

**Solution.** The retailer keeps the products with the highest expected value given their signal. The exponential density is log-concave, so a higher signal is good news about $V$ (Milgrom, 1981): the expected value rises with the signal, and the retailer drops the products whose signal falls below a cutoff $c$. The probability of being dropped is

$$\lambda(s) = \Pr\Big[V(s) - \frac{e}{\xi} < c\Big] = \min\Big\{1,\, \exp\big(-\xi\,(V(s)-c)\big)\Big\},$$

equal to one for a product with $V(s) < c$, since $e \geq 0$. The constraint $\int \lambda\,d\mu = \delta$ pins the cutoff. When $c$ lies below every value in the cross-section the minimum never binds, the constraint reads

$$\exp(\xi c) = \frac{\delta}{\mathbb{E}_\mu[\exp(-\xi V)]},$$

and substituting gives (5). When the cap binds, the hazard of the uncapped products is still proportional to $\exp(-\xi V)$, as in (5), but the constant now solves the capped constraint and is larger,

$$\exp(\xi c) > \frac{\delta}{\mathbb{E}_\mu[\exp(-\xi V)]},$$

since the capped products supply less exit than (5) assigns them and the rest make up the flow. At the calibration the cap binds: two percent of the cross-section supplies 97 percent of the exit flow. The numerical solution caps the hazard at 0.95 rather than at one and rescales the uncapped hazards so that the flow constraint holds (footnote 13). When the cross-section is worth more, the cutoff rises with it. Being delisted is worth zero to the firm, and it takes the cutoff $c$ as given because it is one of a continuum of products.

As $\xi \to 0$ the signal carries no information, every product is equally likely to go, and $\lambda = \delta$: No selection. As $\xi \to \infty$ the retailer sees values exactly and drops the products with the lowest $V$, below the cutoff that leaves a mass $\delta$ underneath it.

# C Equilibrium and computational details

This appendix defines the equilibrium of Section 3.5 (Section C.1) and collects the two solution algorithms used in the paper: the Krusell–Smith stationary solver (Section C.2) and the perfect-foresight transition path used to compute the impulse responses (Section C.3).

## C.1 State space and law of motion

**Aggregate state.** Nominal demand has a unit root, so I write the economy in real terms. A firm enters period $t$ with productivity $z$ and an inherited price that matters only relative to last period's price index, $\tilde{p} := \log(p_{-1}/P_{t-1})$. The economy enters period $t$ with nominal demand relative to the same index, $\mathcal{S}_t = \log(S_t/P_{t-1})$, and with the cross-section $\mu_t$ of firms over $(\tilde{p}, z)$. Given $(\mathcal{S}_t, \mu_t)$, the price policies and the CES aggregator (3) determine inflation $P_t/P_{t-1}$, and with it output $Y_t = S_t/P_t$ and the real wage $W_t/P_t = \chi Y_t$. Every aggregate variable in the firm's problem of Section 3 is therefore a function of $(\mathcal{S}_t, \mu_t)$. The cross-section $\mu_t$ is infinite-dimensional, so firms perceive inflation as a function of $\mathcal{S}_t$ alone, through the rule $\Gamma$ of Section 3.5. Under that perception $\mathcal{S}$ is the only aggregate state in the firm's problem.

**State space.** The firm's state is $s = (\tilde{p}, z, \mathcal{S}) \in \mathbf{S} \equiv \mathbf{P} \times \mathbf{Z} \times \mathbf{G}$, where $\mathbf{P} \subset \mathbb{R}$ is the set of inherited log real prices, $\mathbf{Z} \subset \mathbb{R}_{++}$ the set of productivities, and $\mathbf{G} \subset \mathbb{R}$ the set of aggregate states. I equip $\mathbf{S}$ with its Borel $\sigma$-algebra $\Sigma_{\mathbf{S}}$. The value functions, the exit hazard, and the price policy are functions on $\mathbf{S}$. Prices charged in period $t$ are in logs relative to $P_t$: $p^\star(z, \mathcal{S})$ is the reset price that attains $V_A(z, \mathcal{S})$, and the price policy is

$$p(s) = \begin{cases} \tilde{p} - \Gamma(\mathcal{S}) & \text{if } V_N(s) \geq V_A(z, \mathcal{S}), \\ p^\star(z, \mathcal{S}) & \text{otherwise.} \end{cases}$$

The measure $\mu$ of the equilibrium definition is a probability measure on $(\mathbf{S}, \Sigma_{\mathbf{S}})$. Its marginal on $\mathbf{G}$ is the ergodic distribution of the aggregate state, and its distribution conditional on $\mathcal{S}$ is the cross-section of firms at $\mathcal{S}$. The expectation $\mathbb{E}_\mu$ in the exit hazard (5) is taken under this joint measure, so the exit rate averages to $\delta$ over aggregate states and can differ from $\delta$ at a given $\mathcal{S}$. Along the perfect-foresight transition of Appendix C.3 the expectation is taken under the date-$t$ cross-section, and the exit rate equals $\delta$ in every period.

**Incumbent transition kernel.** By (2), the aggregate state moves with nominal demand and with perceived inflation,

$$\mathcal{S}' = \mathcal{S} - \Gamma(\mathcal{S}) + \bar{\pi} + \sigma_S \eta', \qquad \eta' \sim N(0,1).$$

Write $\Phi(\cdot \mid \mathcal{S})$ for the distribution of $\mathcal{S}'$ given $\mathcal{S}$. A surviving incumbent carries the price it charged in period $t$ into period $t+1$, draws $z'$ from $F(\cdot \mid z)$ independently of $\eta'$, and faces $\mathcal{S}'$. For Borel sets $A_P \subset \mathbf{P}$, $A_Z \subset \mathbf{Z}$, and $A_G \subset \mathbf{G}$, the kernel $Q\colon \mathbf{S} \times \Sigma_{\mathbf{S}} \to [0,1]$ is

$$Q(s, A_P \times A_Z \times A_G) = \mathbb{1}\{p(s) \in A_P\} \int_{A_Z} dF(z' \mid z)\, \Phi(A_G \mid \mathcal{S}).$$

Products of Borel sets generate $\Sigma_{\mathbf{S}}$, so this expression determines $Q(s, \cdot)$ on all of $\Sigma_{\mathbf{S}}$. The kernel describes survivors only; the law of motion (7) applies exit through $1 - \lambda(s)$.

**Entrant distribution.** An entrant that arrives at aggregate state $\mathcal{S}$ draws $z$ from $G_e$ in (6), charges $p^\star(z,\mathcal{S})$, and enters the next period with state $(p^\star(z,\mathcal{S}), z', \mathcal{S}')$. Its distribution over next period's states is

$$\nu_e(A_P \times A_Z \times A_G \mid \mathcal{S}) = \int \mathbb{1}\{p^\star(z,\mathcal{S}) \in A_P\} \int_{A_Z} dF(z' \mid z)\, dG_e(z)\; \Phi(A_G \mid \mathcal{S}).$$

Entrants arrive across aggregate states in proportion to the surviving incumbents, so the entrant distribution in (7) is

$$\nu_e(A) = \int_{\mathbf{G}} \nu_e(A \mid \mathcal{S})\, \omega(d\mathcal{S}), \qquad \omega(A_G) = \frac{\int \mathbb{1}\{\mathcal{S} \in A_G\}\,(1-\lambda(s))\, d\mu(s)}{\int (1-\lambda(s))\, d\mu(s)}. \tag{19}$$

With $M = \int \lambda\, d\mu$, integrating (7) over $\mathbf{S}$ gives $\mu'(\mathbf{S}) = (1-M) + M = 1$: replacement keeps the mass of firms at one.

## C.2 Krusell–Smith solution algorithm

Firms forecast log inflation with the rule $\Gamma(\mathcal{S}) = \alpha_0 + (1-\alpha_1)\,\mathcal{S}$ implied by (8). In the solver I write the same line centered on trend inflation,

$$\Gamma(\mathcal{S}) = \bar{\pi} + b_1\,(\mathcal{S} - s^\star), \qquad b_1 = 1-\alpha_1, \quad s^\star = \frac{\bar{\pi}-\alpha_0}{1-\alpha_1},$$

in terms of its slope $b_1$ and of $s^\star$, the aggregate state at which the rule returns trend inflation. The algorithm runs on a grid $\{\mathcal{S}_k\}_{k=1}^K$ for the aggregate state. Write $\hat{\mu}_k$ for the cross-section at $\mathcal{S}_k$ after one period of pricing, productivity shocks, and the aggregate shock; $p(s)$ for the log real price the policy assigns to state $s$; and $s-\Gamma$ for the state with its relative price and its aggregate state both deflated by $\Gamma$. The CES aggregator (3) then returns

$$\Gamma_{\text{res}}(\mathcal{S}_k) \;=\; \frac{1}{1-\theta}\,\log \int \exp[(1-\theta)\,p(s-\Gamma(\mathcal{S}_k))]\, d\hat{\mu}_k(s),$$

the log price index built from prices the rule has already deflated. A correct rule leaves that index at one and $\Gamma_{\text{res}}$ at zero. A positive $\Gamma_{\text{res}}(\mathcal{S}_k)$ says the announced deflation was too small: prices at that state aggregate above the price level the rule promised, so realized inflation runs above $\Gamma(\mathcal{S}_k)$. This is the model-implied inflation the forecasting rule has to match, and it is computed by integrating over the cross-section at each grid point rather than by simulating a path.

The loop is then the following.

1. **Guess.** Start from $(s^\star, b_1)$.
2. **Policies.** Iterate the firm Bellman (4) at $\Gamma$ and at the wage and discount factor it implies. After each pass I recompute the exit hazard from the contemporaneous $(V,\mu)$ through (5), so that value, distribution, and hazard are mutually consistent.
3. **Distribution.** Iterate the law of motion (7) at $\Gamma$, given the price policy, the hazard, and the

entry block, to its invariant measure $\mu$.

4. **Implied inflation.** Push $\mu$ one period forward through pricing, the productivity transition, and the aggregate-state transition. Evaluate $\Gamma_{\text{res}}(\mathcal{S}_k)$ at every grid point and record $w_k$, the mass the forward-pushed cross-section leaves at $\mathcal{S}_k$. Model-implied inflation is $\Gamma_{\text{target}}(\mathcal{S}_k) = \Gamma(\mathcal{S}_k) + \Gamma_{\text{res}}(\mathcal{S}_k)$.

5. **Project.** Fit
$$\Gamma_{\text{target}} \;\approx\; b_0 + b_1\,(\mathcal{S} - \bar{\mathcal{S}})$$
by least squares with weights $w_k$, and read off the fixed point the fit implies, $s^\star_{\text{new}} = \bar{\mathcal{S}} + (\bar{\pi} - b_0)/b_1$.

6. **Update.** Move $(s^\star, b_1)$ to $(s^\star_{\text{new}}, b_{1,\text{new}})$ and return to step 2, until $\|\Gamma_{\text{new}} - \Gamma\|_\infty < \epsilon$.

The projection in step 5 is the population counterpart of the Krusell and Smith (1998) simulation regression: the invariant distribution of step 3 supplies the cross-section at every aggregate-state grid point, so the weighted fit computes what a regression on a simulated path estimates, with the ergodic mass as weights and no sampling error (Nakamura and Steinsson, 2010). Appendix D tests a cubic alternative.

### C.3 Perfect-foresight transition path

The IRFs in Section 4.3 are computed along a deterministic transition of length $T$ following a one-off nominal innovation $\Delta$ at $t = 0$, similar to Clementi and Palazzo (2016). Write $y_t := \log Y_t$ and $\pi_t := \log(P_{t+1}/P_t)$, the inflation between the price indices of dates $t$ and $t+1$. Since $Y_t = S_t/P_t$ and nominal demand grows at $\bar{\pi}$ along the path, $y_{t+1} = y_t + \bar{\pi} - \pi_t$, deterministic given the inflation path $\{\pi_t\}_{t=0}^{T-1}$. The firm Bellman is solved on a time-varying $Y_t$:

$$\begin{aligned}
V_t(s) &= \max\{V_t^{\text{stay}}(s),\, V_t^{\text{adj}}(s)\},\\
V_t^{\text{stay}}(s) &= \pi(s; Y_t) + \beta\,(1 - \lambda_t(s))\,\rho_t\,\mathbb{E}_{z'}[V_{t+1}(s')],\\
V_t^{\text{adj}}(s) &= -K(Y_t) + \max_p \Big[\pi(p, z; Y_t) + \beta\,(1 - \lambda_t(p, z, \cdot))\,\rho_t\,\mathbb{E}_{z'}[V_{t+1}(p, z', \cdot)]\Big],
\end{aligned}$$

where $\rho_t = \exp(\gamma(\pi_t - \bar{\pi}))$ is the real part of the stochastic discount factor along the path, $(C_t/C_{t+1})^\gamma$ with $\gamma$ the coefficient of relative risk aversion, equal to one under the log preferences of Section 3, so that $\Xi_{t,t+1} = \beta\rho_t$; $K(Y_t)$ is the menu cost expressed at the date-$t$ wage, and the terminal condition $V_T = V^{ss}$ is the deterministic stationary value. The exit hazard is recomputed each period from the contemporaneous $(V_t, \mu_t)$ via (5).

The cross-section evolves under a deterministic operator

$$\mu_{t+1} \;=\; \mathcal{T}_t[\mu_t;\, \lambda_t, M_t, p_t, \pi_t],$$

which sequentially applies (i) exit at $\lambda_t$; (ii) injection of entrants of mass $M_t = \int \lambda_t\, d\mu_t$ from $\nu_e$ at the prevailing $Y_t$; (iii) the date-$t$ price policy $p_t$; (iv) the productivity transition $F(z' \mid z)$; and

(v) deflation of relative prices by $\pi_t$. The CES condition pins down $\pi_t$ at every $t$:

$$1 \;=\; \mathbb{E}_{\mu_t^{\text{post-}p,F}}[\exp((1-\theta)\, p_{t+1}(z', \tilde{p} - \pi_t, \mathcal{S}'))],$$

with initial conditions $\mu_0 = \mu^{ss}$ and $y_0 = y^{ss} + \Delta$.

The unknown is the inflation path $\{\pi_t\}_{t=0}^{T-1}$, with $T$ chosen large enough that the IRF returns to zero. The algorithm iterates as follows:

1. **Guess** a path $\{\pi_t\}_{t=0}^{T-1}$ and compute the implied $\{y_t\}_{t=0}^{T}$ from $y_{t+1} = y_t + \bar{\pi} - \pi_t$.
2. **Backward pass.** Set $V_T = V^{ss}$. For $t = T-1, \ldots, 0$, solve the Bellman to obtain $(V_t, p_t)$ at $(Y_t, \pi_t, \lambda_t)$.
3. **Forward pass.** Set $\mu_0 = \mu^{ss}$. For $t = 0, \ldots, T-1$, evolve $\mu_{t+1} = \mathcal{T}_t[\mu_t]$.
4. **Update hazards.** Recompute $\lambda_t$ from $(V_t, \mu_t)$ at every $t$.
5. **CES update.** At each $t$, compute the new inflation $\pi_t^{\text{new}}$ from the CES condition above.
6. **Damped update.** $\pi_t \leftarrow (1-\omega)\pi_t + \omega\, \pi_t^{\text{new}}$. Return to step 1 until $\max_t |\pi_t^{\text{new}} - \pi_t| < \text{tol}$.

The terminal and initial conditions $V_T = V^{ss}$ and $\mu_0 = \mu^{ss}$ must be the stationary equilibrium of the economy without aggregate noise: at $\sigma_\eta = 0$ the aggregate state rests at $s^\star$, inflation equals $\bar{\pi}$ in every period, and the equilibrium is a fixed point of the deterministic operator $\mathcal{T}$, so that the no-shock path reproduces stationarity exactly.

The reported impulse response is the deviation of output from the contemporaneous no-shock transition path, $\psi_t = (y_t - y_t^0)/\Delta$, where $y_t^0$ is obtained by running the same operator $\mathcal{T}$ with $\Delta = 0$. The cumulative response is $\text{CIR}_t = \frac{1}{12}\sum_{s \le t} \psi_s$ in annualized units, reported at the one-year horizon in Table 6.

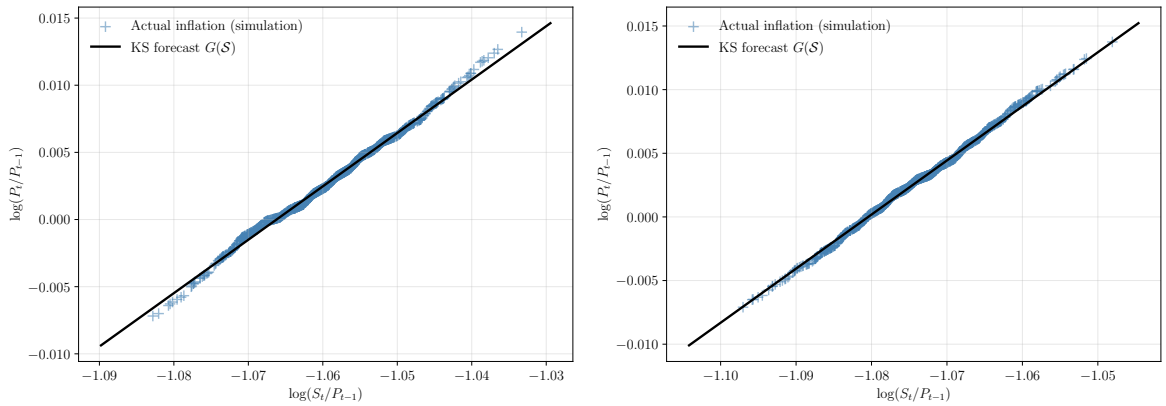
# D Accuracy and robustness of the forecasting rule

The stationary equilibrium relies on a parametric forecasting rule $\Gamma(\mathcal{S})$ for log inflation as a function of the aggregate state $\mathcal{S} = \log(S/P_{-1})$. In the main text I use a linear specification, $\Gamma(\mathcal{S}) = \alpha_0 + (1-\alpha_1)\mathcal{S}$ from (8), which is the standard Krusell and Smith (1998) choice and is fitted analytically by weighted least squares on the aggregate-state grid (Nakamura and Steinsson, 2010). This appendix assesses the accuracy of that specification and shows that the headline results are robust to a more flexible cubic alternative.

## D.1 Diagnostics under the linear rule

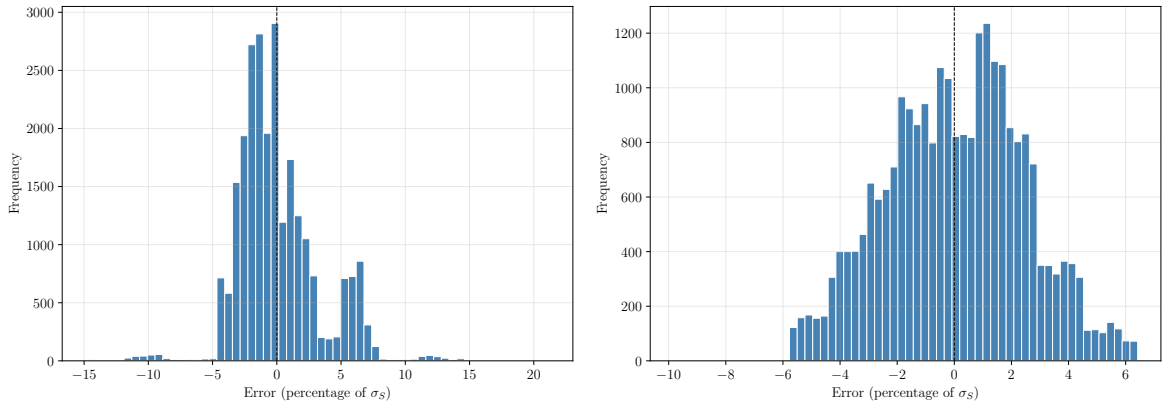
I evaluate the rule by simulating a long path of the aggregate state $\mathcal{S}_t$ under the converged equilibrium and recording realized inflation $\pi_t$ at each date. I then plot $(\mathcal{S}_t, \pi_t)$ pairs against the linear rule $\Gamma$ (Figure A.8) and the distribution of forecast errors $\pi_t - \Gamma(\mathcal{S}_t)$, expressed as a percentage of $\sigma_S$, the standard deviation of the nominal-demand innovation (Figure A.9).

**Figure A.8: Forecasting rule: perceived vs. realized inflation, linear rule.**




*Notes*: For each economy, the figure plots realized log inflation along the simulated path against the aggregate state $\log(S_t/P_{t-1})$, with the linear forecasting rule $\Gamma$ superimposed in black. The Benchmark (left) displays a visible bow-shape: at low values of the aggregate state realized inflation lies below the rule, and at high values it lies above. The No-selection economy (right) shows essentially no such pattern.

**Figure A.9: Distribution of one-step forecast errors, linear rule.**




*Notes*: For each economy, the histogram reports the distribution of $\pi_t - \Gamma(\mathcal{S}_t)$ as a percentage of $\sigma_S$. The Benchmark exhibits a few isolated tail masses around $\pm$10–15% of $\sigma_S$, corresponding to rare visits of the simulated aggregate state to its support boundaries; under No-selection, errors are smaller and more uniformly distributed.

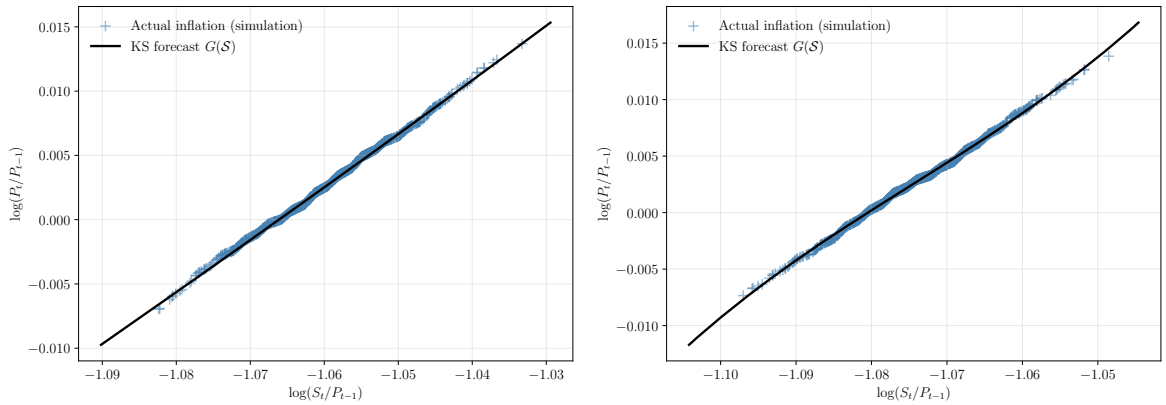
Under No-selection, model-implied inflation is essentially linear in $\mathcal{S}$ and the linear rule fits well throughout the support. Under the Benchmark, selective exit reshapes the cross-section in a state-dependent way, and the model-implied inflation is mildly nonlinear in $\mathcal{S}$. The mass-weighted least-squares fit absorbs that nonlinearity well in the center of the simulated support, where most of the stationary mass lies, but allows residuals to grow at the boundaries, producing the S-shape in the scatter and the tail humps in the histogram. Therefore, I fit a cubic forecasting rule next.

## D.2 Cubic forecasting rule

To assess whether these residuals matter quantitatively, I re-solve all three economies under a cubic specification, $\Gamma(\mathcal{S}) = \sum_{k=0}^{3} b_k \mathcal{S}^k$, fitted by the same weighted least-squares procedure on the aggregate-state grid. Under the cubic rule the diagnostics shown in Figures A.10 and A.11

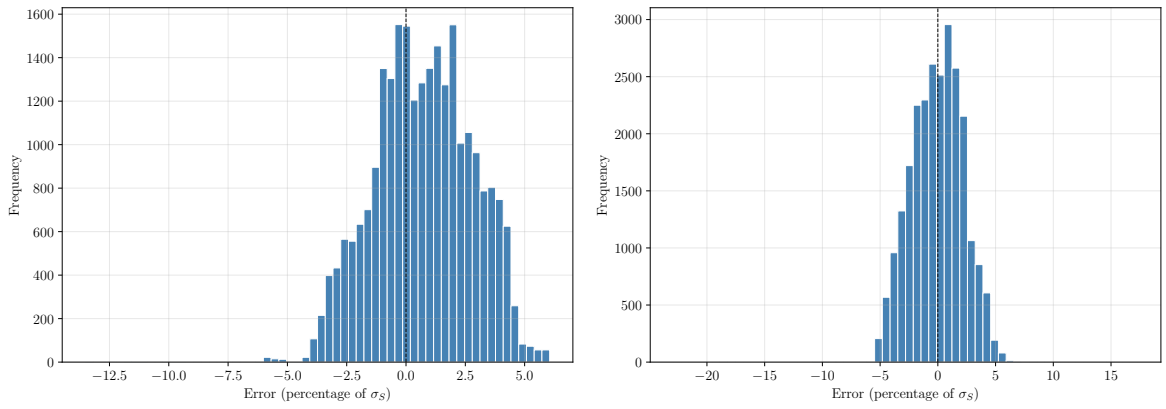
look cleaner: the tail humps are absorbed into a smooth, roughly bell-shaped error distribution, with the standard deviation of the forecast error dropping by approximately one-third in the Benchmark.

**Figure A.10: Forecasting rule: perceived vs. realized inflation, cubic rule.**




*Notes*: Same construction as Figure A.8, with the rule replaced by a cubic polynomial. The bow-shape that was present in the Benchmark scatter under the linear rule is no longer visible.

**Figure A.11: Distribution of one-step forecast errors, cubic rule.**




*Notes*: Same construction as Figure A.9. The Benchmark histogram is now smooth and unimodal; the isolated tail masses observed under the linear rule have disappeared.

I now report the stationary moments under the cubic rule (Table A.9).

**Table A.10: Moments in the Benchmark and No selection economies.**

| Moment | Data | Benchmark | No selection |
|---|---|---|---|
| Frequency of price changes | 0.09 | 0.09 | 0.09 |
| Monthly exit rate | 0.02 | 0.02 | 0.02 |
| Rel. size of exiting firms | 0.66 | 0.64 | 1.00 |
| Rel. size of entrants | 0.67 | 0.69 | 1.00 |
| Mean $\Delta \log p$ | 0.04 | 0.04 | 0.03 |
| Untargeted | | | |
| Mean $\lvert\Delta \log p\rvert$ | 0.10 | 0.21 | 0.21 |
| St. dev. $\Delta \log p$ | 0.11 | 0.21 | 0.21 |
| Fraction upward changes | 0.63 | 0.62 | 0.60 |

*Notes*: Moments of the stationary equilibrium. The first five rows are the calibration targets of Table 5. Relative sizes are average revenue of exiting products, or of entrants, over average revenue of incumbents.

**Table A.9: Targeted moments under linear vs. cubic forecasting rule.**

| | Benchmark | | No selection | | No churn | |
|---|---|---|---|---|---|---|
| Moment | Linear | Cubic | Linear | Cubic | Linear | Cubic |
| Frequency of price changes | 0.09 | 0.09 | 0.09 | 0.09 | 0.09 | 0.09 |
| Monthly exit rate | 0.02 | 0.02 | 0.02 | 0.02 | 0.00 | 0.00 |
| Rel. size of exiting firms | 0.64 | 0.64 | 1.00 | 1.00 | – | – |
| Rel. size of entrants | 0.69 | 0.69 | 1.00 | 1.00 | – | – |
| Mean $\Delta \log p$ | 0.04 | 0.04 | 0.03 | 0.03 | 0.03 | 0.03 |

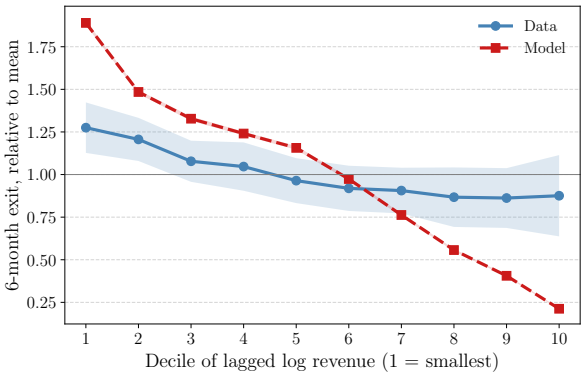
*Notes*: Targeted moments of the stationary equilibrium solved under the linear and the cubic forecasting rule. Frequency is the monthly frequency of price adjustment. Relative sizes are the average revenue of exiting firms and of entrants relative to incumbents. Mean $\Delta \log p$ is the average log price change conditional on a change. The No churn economy has no exit or entry.

The exercise confirms that the linear-rule approximation is adequate for the headline results: pricing moments are unchanged and the one-year cumulative impulse responses are essentially identical across rules.

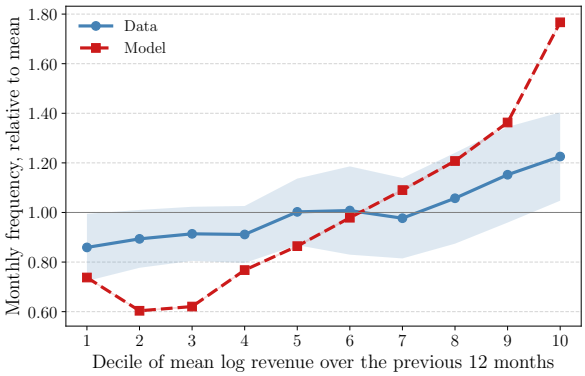
# E Moments in the No selection economy

**Figure A.12: Exit and price changes by revenue decile at short horizons.**

**(a)** Exit within six months




**(b)** Monthly frequency of price changes




*Notes*: Panel (a): for each decile of lagged log revenue, the share of product-months followed by exit within six months. Panel (b): for each decile of mean log revenue over the previous twelve months, the monthly frequency of price changes. Both profiles are divided by their sample mean. Data: Dutch scanner data. Model: simulated benchmark panel. Shaded: 95 percent intervals.

**Table A.11: Frequency of price changes by revenue decile, model and data.**

| | Monthly frequency | | Change within 12 months | |
|---|---|---|---|---|
| Decile | Data | Model | Data | Model |
| 1 | 0.069 | 0.064 | 0.565 | 0.558 |
| 2 | 0.071 | 0.052 | 0.620 | 0.516 |
| 3 | 0.073 | 0.053 | 0.611 | 0.558 |
| 4 | 0.073 | 0.066 | 0.615 | 0.605 |
| 5 | 0.080 | 0.074 | 0.617 | 0.639 |
| 6 | 0.080 | 0.084 | 0.640 | 0.671 |
| 7 | 0.078 | 0.094 | 0.653 | 0.700 |
| 8 | 0.084 | 0.104 | 0.668 | 0.729 |
| 9 | 0.092 | 0.117 | 0.727 | 0.774 |
| 10 | 0.098 | 0.152 | 0.780 | 0.844 |

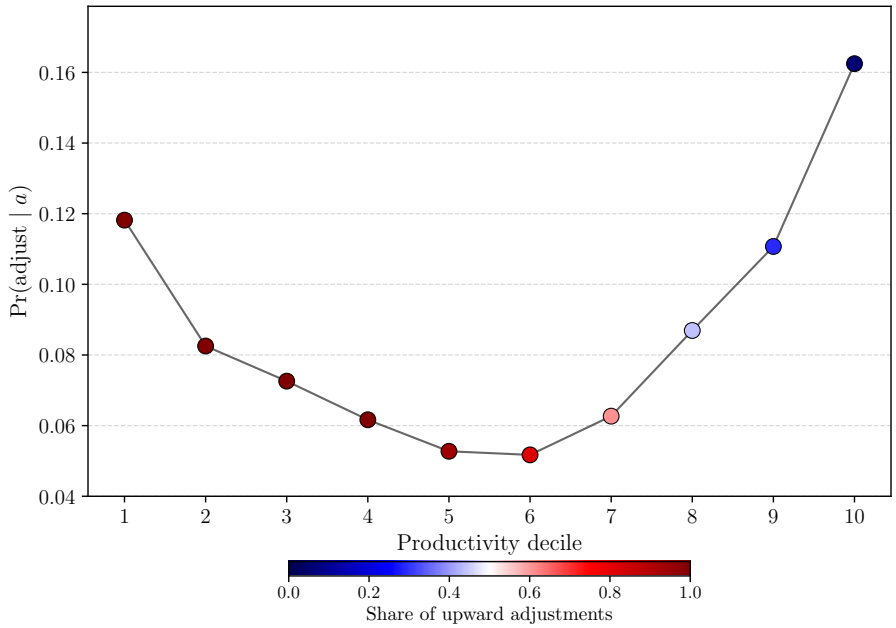
*Notes*: Deciles of mean log revenue over the previous twelve months. Monthly frequency: share of months with a price change. Change within 12 months: probability of at least one price change in the next twelve months. Data: Dutch scanner data. Model: simulated benchmark panel.

# F Additional results

## F.1 Validation at short horizons

The fit is less close at shorter horizons. Exit is more concentrated among low-revenue products over six months in the model than in the data, and the monthly repricing profile is steeper, particularly among the largest products (Figure A.12 and Table A.11). These differences indicate that the model captures the broader relationship between size, survival, and pricing more closely than the short-run timing of exit and adjustment. In particular, products can remain small for longer before exiting in the data, a margin that the calibration does not directly discipline.

**Figure A.13: Frequency of price adjustment by productivity decile.**




*Notes*: For each decile of productivity $z$, the figure reports the fraction of firms in that decile that adjust their price in a given month. Adjustment is concentrated in the tails: very low- and very high-productivity firms adjust relatively most often, while firms in the middle of the distribution adjust the least.

## F.2 Censoring and phantom adjustments

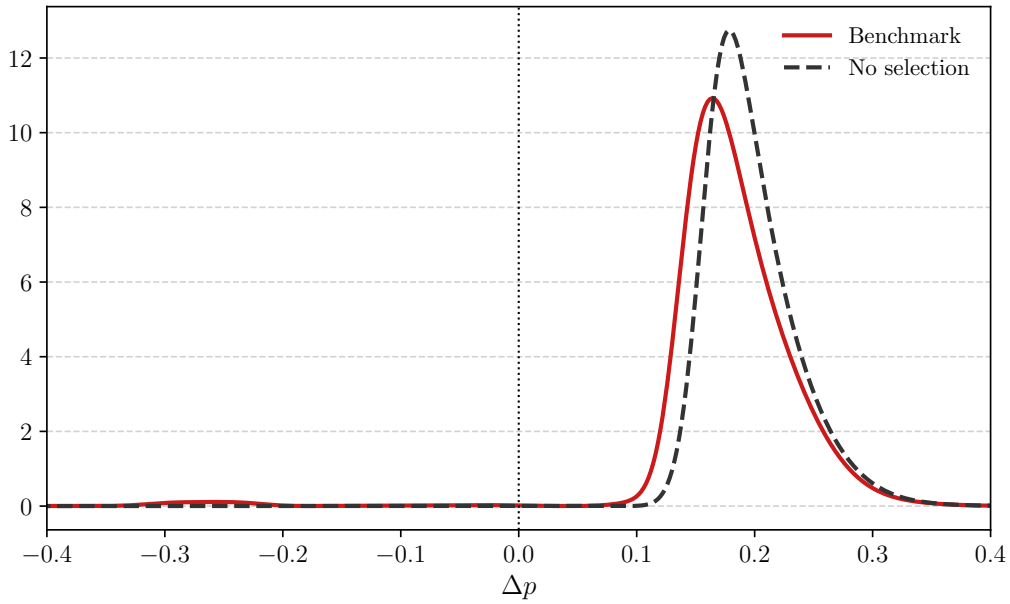
The second channel works through what would have happened to the firms that selective exit removes. Figure A.13 reports the conditional frequency of price changes by productivity decile in the No selection economy.

Adjustment is concentrated in the tails of the productivity distribution: low-$z$ firms reset upward (their inherited price is too low relative to a higher marginal cost), high-$z$ firms reset downward, and firms in the middle stay put.[25] The firms that selective exit removes, those at the bottom of the productivity distribution, therefore come from the high-adjustment tail. Once they exit, the price changes they would have made never appear in the data. Those changes are not random. A firm that survives a sharp negative productivity draw typically resets its price upward in the period of the shock and then gradually trims that price downward as productivity mean-reverts. Selective exit removes both legs of this trajectory: the firm is gone before either the upward reset or the subsequent downward corrections can take place. The two sides have different fates. The upward reset is replaced on the spot: the exiting firm's slot goes, within the period, to an entrant that sets its optimal price on arrival,[26] so the correction the incumbent was about to make happens anyway, recorded as an exit and an entry rather than as a price change. The downward corrections are not replaced, because the entrant draws

[25] This U-shape is a feature of menu-cost models with idiosyncratic shocks and is observed also in Cooper and Öztürk (2022).

[26] An entrant's price is optimal in its first period. From the next period it ages with productivity shocks like any incumbent's price.

**Figure A.14: Distribution of price changes within productivity quartile Q1.**




*Notes*: Kernel density of $\Delta \log p$ conditional on adjustment, for firms in the lowest productivity quartile (Q1), across the two economies. Quartile boundaries are defined using the Benchmark productivity distribution, so the same productivity range is compared across models.

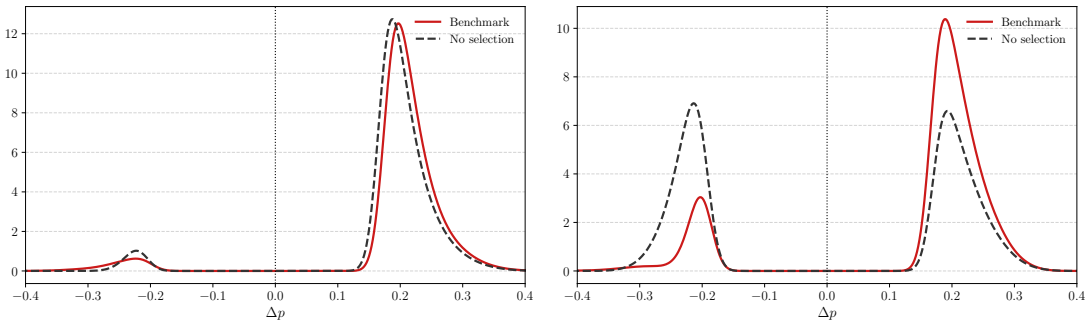
a fresh productivity and does not inherit the exiter's recovery path. The cross-section of price adjustments is therefore systematically different under selection, and this difference is visible directly in the conditional distribution of price changes by productivity quartile.

This is where entry and exit already matter. When selection removes firms from the left tail, it simply replaces one price change with another; this is exactly what selection does. Uniform exit, by contrast, removes more firms, on average, from the interior of the distribution. Firms in the middle are the ones that would not have changed their price (and so would have remained mispriced); uniform exit replaces them with new entrants, who tend to be more flexible. Uniform turnover therefore effectively subsidizes flexibility. Removing it, i.e. moving from uniform exit to selection, eliminates this subsidy and so amplifies the real effects of a monetary shock.

Figure A.14 shows the lowest quartile, Q1, which captures firms that have just received a sharp negative productivity draw. Their inherited price is too low relative to a higher marginal cost, and most of their adjustment activity is upward resets. Selective exit removes some of these firms, and the missing mass shows up directly in the right tail of the Q1 distribution.

Figure A.15 shows the next two quartiles. Relative to No selection, the Benchmark has visibly less mass in the moderate-downward-adjustment region. These are firms that, in the No-selection economy, survived an initial low-productivity episode and are now mean-reverting upward; as productivity recovers, their inherited price becomes too high relative to current conditions, and they generate a sequence of moderate downward corrections. Selective exit removes a fraction of these firms before they reach the migration stage. The missing mass therefore shows up not only at the moment of exit (Q1) but also one step later in the productiv-

**Figure A.15: Distribution of price changes within productivity quartiles Q2 and Q3.**




*Notes*: Kernel density of $\Delta \log p$ conditional on adjustment, for firms in productivity quartiles Q2 (left) and Q3 (right), across the two economies. Quartile boundaries are defined using the Benchmark mass.

ity trajectory (Q2 and Q3), where mean reversion has begun.

To quantify the price adjustments that go missing along with the missing mass, I propagate the cohort of selective exiters forward under the No-selection dynamics and compute the price changes they would have made. I call these missing partial-equilibrium corrections phantom adjustments. Let $\mu_b$ and $\lambda_b$ denote the stationary distribution and exit hazard of the Benchmark economy, and define the exit-source measure

$$de_0(s) = \lambda_b(s)\, d\mu_b(s),$$

which records the mass that exits under selection. I propagate this cohort forward under the No-selection transition kernel $Q_{ns}$ and survival rate $1-\delta$:

$$v_{t+1}(s') = (1-\delta) \int Q_{ns}(s', s)\, dv_t(s), \qquad v_0 := de_0, \tag{20}$$

and define the horizon-$T$ and lifetime cumulative phantom measures

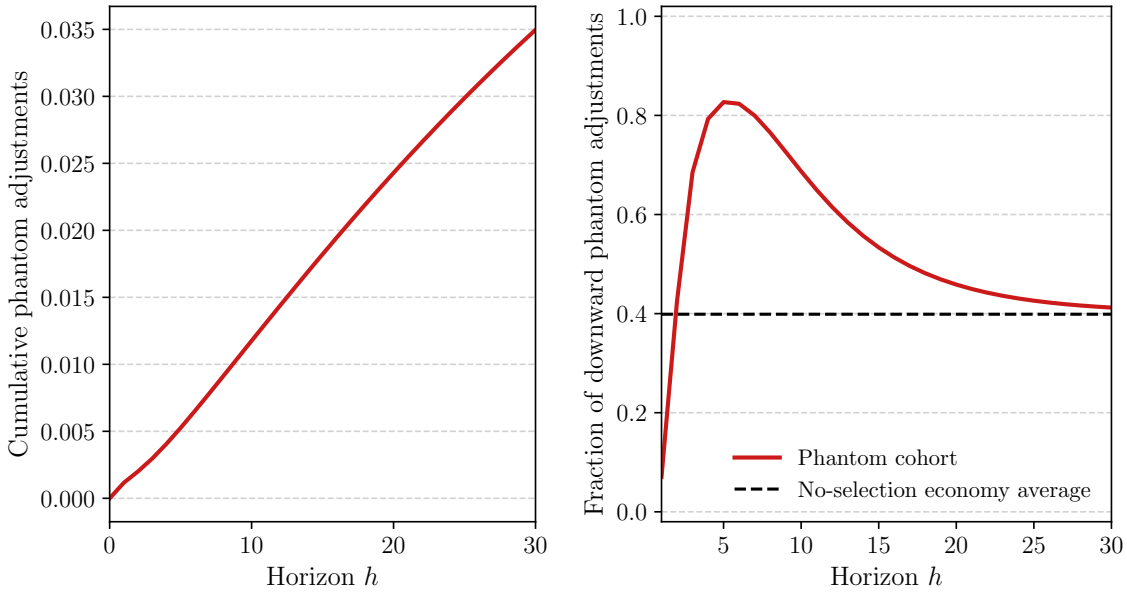
$$\Phi_T = \sum_{t=1}^{T} v_t, \qquad \Phi_\infty = \sum_{t=1}^{\infty} v_t.$$

At each horizon, I evaluate the No-selection price-adjustment policy $p_{ns}(s)$ against the phantom measure to compute the cumulative mass of adjustments and their sign composition. $\Phi_\infty$ measures the entire missing trajectory of price changes that the selective-exit cohort would have made over its remaining lifetime in the absence of selection.

Figure A.16 reports the cumulative mass of phantom adjustments (left panel) and the share of those adjustments that are downward (right panel). The downward share moves with the horizon: at short horizons the missing cohort generates mostly upward resets (the immediate response to a bad productivity draw), but as the cohort propagates forward and productivity mean-reverts, the share of downward corrections rises. Selective exit therefore censors a price-adjustment trajectory that is upward at impact and downward thereafter, and different from what a purely random exit would remove.

The phantom exercise delivers two facts. First, it shows that the removed cohort was about

**Figure A.16: Phantom price adjustments by horizon.**




*Notes*: The left panel reports the cumulative mass of phantom adjustments at different horizons, computed as the integral of the no-selection adjustment policy against the cumulative phantom measure $\Phi_T$. The right panel reports the share of these adjustments that are downward. At short horizons, the phantom cohort consists of immediate upward resetters from the low-productivity tail; the downward share rises with the horizon as productivity mean-reverts and the surviving cohort migrates into states that generate downward corrections.

to reprice. The removed cohort's counterfactual adjustments arrive almost entirely at short horizons, so selection removes firms whose correction was imminent. Second, it shows that the mispricing selection eliminates matters beyond impact as well: these firms would have kept adjusting their price for several periods, as their productivity mean-reverts only gradually. The phantom mass therefore records how much mispricing selection removes and how long that mispricing would otherwise have persisted: the time path of idiosyncratic states, beyond their cross-sectional distribution, shapes the transmission of monetary shocks.

### F.3 Selection on entry and on exit

Table A.12 separates selection on exit from selection on entry. The four economies share the same exit flow and differ only in $\xi$ and $\tau$. Selective exit on its own delivers an even larger amplification, while selective entry by itself does almost nothing, and added to selective exit it takes about four points off.

**Table A.12:** Selectivity routed through entry and through exit.

| Economy | $\xi$ | $\tau$ | CIR (1 year) | vs. No selection |
|---|---|---|---|---|
| No selection | 0 | 0 | 0.155 | – |
| Entry only | 0 | 5 | 0.156 | +0.3% |
| Exit only | 20 | 0 | 0.184 | +18.5% |
| Benchmark | 20 | 5 | 0.178 | +14.6% |

*Notes*: One-year cumulative output response to a one-time shock to nominal spending, computed along the perfect-foresight transition. All four economies carry the same exit flow $\delta = 0.02$. The exit elasticity is $\xi$ and the entry elasticity is $\tau$. Each economy is a stationary equilibrium with firms pricing under its own hazard; no counterfactual beliefs are involved.

The reason is the same as in Section 4.5: the amplification comes from the difference between the average firm removed by selective exit and the average firm removed by uniform exit. Selective exit removes a firm about to raise its price, while random exit removes the average firm, mid-spell with a stale price. A positive entry elasticity tones down this difference. Entrants start at low productivity, where exit concentrates, and they price on arrival, so part of the exit flow lands on freshly priced entrants instead of on incumbents whose prices have drifted far below target. Table A.13 shows it: in the exit-only economy 95 percent of exits fall at price gaps below $-0.10$, and with the entry tilt 87 percent do. Exit does not become less selective on productivity; what weakens is the link between low productivity and an imminent price correction. Shutting down selective entry widens the difference again, hence the larger amplification of the exit-only economy.

**Table A.13:** Where exit lands, with and without selective entry.

| | Share of the exit flow | | Share of the cross-section | |
|---|---|---|---|---|
| Price gap $x$ | Exit only | Benchmark | Exit only | Benchmark |
| $x < -0.25$ | 0.315 | 0.304 | 0.013 | 0.012 |
| $-0.25 \le x < -0.20$ | 0.284 | 0.258 | 0.023 | 0.022 |
| $-0.20 \le x < -0.15$ | 0.250 | 0.228 | 0.052 | 0.049 |
| $-0.15 \le x < -0.10$ | 0.103 | 0.078 | 0.098 | 0.095 |
| $-0.10 \le x < -0.05$ | 0.014 | 0.023 | 0.165 | 0.162 |
| $-0.05 \le x < 0$ | 0.013 | 0.044 | 0.206 | 0.202 |
| $x \ge 0$ | 0.020 | 0.065 | 0.443 | 0.458 |
| Mean price gap | $-0.215$ | $-0.198$ | $-0.014$ | $-0.010$ |
| Mean log productivity | $-0.296$ | $-0.316$ | 0.032 | 0.021 |

*Notes*: Stationary equilibria of the exit-only economy $(\xi, \tau) = (20, 0)$ and the Benchmark $(20, 5)$, which share the exit flow $\delta = 0.02$ and differ only in the entry tilt. The price gap $x$ is the log distance between a firm's posted price and the price it would set if adjustment were free; negative gaps are underpriced firms that would raise their price. The exit-flow columns weight each state by $\lambda(s)\, d\mu(s)$, the cross-section columns by $d\mu(s)$. Columns sum to one up to rounding.

# G Calvo pricing

This appendix replaces the menu cost with a Calvo lottery. The household, the productivity process, the exit hazard, and the entry block stay as in Section 3.

## G.1 Setup

The household block, the final-good sector, the productivity process, the exit hazard, and the entry block are unchanged. The only change is to the pricing margin. There is no menu cost in flow profits,

$$\pi(p_{it}, z_{it}, \mathcal{S}_t) = \frac{p_{it}}{P_t} y_{it} - \frac{W_t}{P_t} l_{it},$$

and in each period a firm receives an independent adjustment opportunity with probability $\alpha$. With probability $1 - \alpha$ it keeps its inherited price; with probability $\alpha$ it resets the price.

Let $V_N(p, z, \mathcal{S})$ denote the firm's value if it cannot adjust this period and stays at inherited price $p$, and let $V_A(z, \mathcal{S})$ denote its value if it can. The two satisfy

$$\begin{aligned} V_N(p, z, \mathcal{S}) &= \pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s'))((1 - \alpha)V_N(p, z', \mathcal{S}') + \alpha V_A(z', \mathcal{S}'))], \\ V_A(z, \mathcal{S}) &= \max_{p'} V_N(p', z, \mathcal{S}). \end{aligned}$$

The first equation says that a non-adjusting firm earns flow profit at its inherited price and then enters next period with the same $p$, where it either draws an adjustment opportunity (continuation value $V_A$, probability $\alpha$) or does not (continuation value $V_N$, probability $1 - \alpha$). The second equation says that an adjusting firm picks the price $p'$ that maximizes its continuation; the chosen $p'$ then becomes the inherited price going forward, and the value the firm attaches to that inherited price is $V_N(p', z, \mathcal{S})$. Iterating the $V_N$ recursion forward gives the discounted-sum form,

$$V_N(p, z, \mathcal{S}) = \mathbb{E}\left[\sum_{t \geq 0} (1 - \alpha)^t D_t \pi(p, z_t, \mathcal{S}_t)\right] + \alpha \mathbb{E}\left[\sum_{t \geq 0} (1 - \alpha)^t D_{t+1} V_A(z_{t+1}, \mathcal{S}_{t+1})\right],$$

where $D_t = \prod_{j=0}^{t-1} \Xi_{j,j+1}(1 - \lambda(s_{j+1}))$ is the cumulative discount-survival factor up to date $t$, with $D_0 = 1$. The first sum is the expected discounted profit over the random spell of non-adjustment, weighted by the probability $(1 - \alpha)^t$ that the firm has not yet drawn an adjustment opportunity by horizon $t$. The second is the present value once an adjustment opportunity arrives. The optimal reset price $p^\star(z, \mathcal{S}) = \arg\max_p V_N(p, z, \mathcal{S})$ is the maximizer of $V_N$ in its first argument.

For implementation, the two recursions collapse into the unconditional value the firm enters the period with, before the Calvo coin is flipped:

$$V(s) = \alpha V_A(s) + (1 - \alpha)V_N(s). \tag{21}$$

Adjustment timing is independent of the firm's state. In the timing of Section 3, step 3 is replaced by an exogenous Bernoulli draw on whether each surviving incumbent is allowed to adjust. The exit hazard (5) is unchanged and still depends on $V(s)$. I fix $\alpha = 0.09$ in every Calvo economy, matching the monthly adjustment frequency of the calibrated menu-cost model and mirroring the menu-cost exercise of Section 4 in which the menu cost was held at its calibrated value across Benchmark and No-selection. The underlying nominal friction is therefore held

constant within each comparison, so any change in the conditional distribution of $\Delta \log p$ reflects the response of the equilibrium to the exit margin rather than a change in the friction. All other parameters remain at their calibrated values.

### G.2 Selective exit and the output response

Selective turnover also amplifies nominal shocks under Calvo pricing, but less than under menu costs (Table A.14). A Calvo firm cannot postpone its adjustment opportunity when survival prospects deteriorate: the lottery determines when it can reprice. Exit risk can still affect its reset price, while selective replacement changes how long inherited prices remain in the economy.

To isolate this second effect, consider a simple approximation in which prices receive equal weight and every reset fully incorporates the nominal shock. The output response then follows the fraction of prices that have not adjusted or been replaced since the shock. With a constant firm-specific exit hazard $\lambda_i$ and a Calvo adjustment probability $\alpha$, an inherited price disappears each period with probability

$$p_i = \lambda_i + (1 - \lambda_i)\alpha.$$

Its expected remaining duration, excluding the impact period, is

$$L = \sum_{t \geq 1} \mathbb{E}[(1 - p_i)^t] = \mathbb{E}\left[\frac{1 - p_i}{p_i}\right].$$

Because $(1 - p)/p$ is convex, dispersion in permanent exit hazards raises $L$ relative to uniform exit with the same mean hazard. Vulnerable firms disappear sooner, leaving an increasingly safe group of firms whose inherited prices remain exposed mainly to the Calvo lottery. Turnover can therefore continue at the same aggregate rate while correcting fewer of the prices that still carry the shock.

Persistence is essential to this argument. With time-varying hazards, the probability that an inherited price survives for $t$ periods is

$$S_t = (1 - \alpha)^t \mathbb{E}\left[\prod_{s=1}^{t}(1 - \lambda_{i,s})\right].$$

If hazards are independently redrawn each period with mean $\delta$, this becomes $[(1 - \alpha)(1 - \delta)]^t$, as under random exit. Persistent differences in survival prospects can instead prolong the remaining duration of inherited prices. This provides a source of amplification without endogenous adjustment timing. Its quantitative contribution depends on the evolution of exit risk, reset prices and the weights of surviving products in the price index, so the duration calculation illustrates a mechanism rather than reproducing the full output response.[27]

[27] The reported Calvo transition holds the hazard normalization at its stationary value, rather than adjusting it each period to maintain a fixed aggregate exit rate. The stationary turnover rates are matched, but the transition comparison need not isolate composition alone.

**Table A.14:** Selective exit and the one-year output response, Calvo model.

| Economy | $\xi$ | $\tau$ | CIR (1 year) | vs. No selection |
|---|---|---|---|---|
| No selection | 0 | 0 | 0.552 | – |
| Benchmark | 20 | 5 | 0.565 | +2.2% |

*Notes*: One-year cumulative output response to a one-time shock to nominal spending, computed along the perfect-foresight transition of the Calvo model with $\alpha = 0.09$. Both economies carry the same exit flow $\delta = 0.02$; $\xi$ is the selectivity of exit and $\tau$ the entry tilt. Each economy starts from its own stationary point and firms price under its own hazard. Along the transition the hazard normalization stays at its stationary value, and the stationary residual of the price-index condition is subtracted from realized inflation.

# H Distribution of productivity shocks

The distribution of productivity shocks matters for both price adjustment and survival selection. Frequent small innovations allow productivity and desired prices to change gradually; infrequent large innovations can move firms directly into regions where adjustment or exit becomes attractive. Gertler and Leahy (2008) study infrequent idiosyncratic shocks with uniformly distributed realizations, while Midrigan (2011) uses infrequent Gaussian innovations alongside economies of scope in price adjustment. Rare large shocks can weaken selection into adjustment following a small monetary shock: many firms either remain far from adjusting or would adjust regardless of the monetary disturbance. Here, changing the shock distribution also changes how firms reach the low-value states exposed to exit. I examine whether the amplification from selective turnover survives these changes.

I retain the productivity process

$$\log z_{t+1} = \rho \log z_t + u_{t+1}, \qquad \rho = 0.85, \qquad \mathbb{E}[u_{t+1}] = 0, \qquad \mathrm{Var}(u_{t+1}) = \sigma_u^2,$$

with $\sigma_u = 0.08569$, and vary the distribution of $u$. Holding persistence and innovation variance fixed isolates changes in the shape of the innovation distribution. It also preserves the unconditional variance of the underlying AR(1) process, $\sigma_u^2/(1-\rho^2)$, although selection can change the productivity distribution among operating firms. The Gaussian reference has $u \sim N(0, \sigma_u^2)$. I solve this reference on the same wider productivity grid as the alternative specifications, so comparisons do not confound the shock distribution with a change in grid support.

The first alternative is a mixture of two normal distributions:

$$u \sim \begin{cases} N(0, s_H^2), & \text{with probability } q, \\ N(0, s_L^2), & \text{with probability } 1-q, \end{cases} \qquad q = 0.2, \quad s_L = 0.4 s_H, \quad s_H = \frac{\sigma_u}{\sqrt{q + (1-q)0.4^2}}.$$

Thus, every month brings an innovation, but one fifth of draws come from a distribution with a standard deviation two and a half times as large. The variance normalization gives $s_H \simeq 0.150$ and $s_L \simeq 0.060$, with innovation kurtosis of approximately 6.15.

The second alternative concentrates innovations in occasional jumps:

$$u = \begin{cases} 0, & \text{with probability } 1 - P, \\ \dfrac{\sigma_u}{\sqrt{P}}\eta, \quad \eta \sim N(0,1), & \text{with probability } P. \end{cases}$$

Reducing $P$ makes innovations less frequent and increases their conditional size while preserving their variance; their kurtosis is $3/P$. I consider $P = 0.25$ and $P = 0.15$ as intermediate sensitivity exercises, and $P = 0.079$, the arrival probability reported for the BLS calibration of Midrigan (2011). This is a discrete-time arrival process, labeled "Poisson" in Table A.15. In months without an innovation, productivity still reverts toward its mean because $\rho < 1$. The exercise therefore borrows the arrival structure without reproducing Midrigan's full specification or calibration.

Selective turnover continues to amplify nominal shocks under the mixture and the intermediate arrival probabilities. Table A.15 reports percentage amplification of 15.6 percent under Gaussian innovations, 12.6 percent under the mixture, and 11.6 percent at $P = 0.25$; the additional experiment at $P = 0.15$ gives 10.0 percent. These declines mainly reflect greater monetary non-neutrality in the economy with random turnover. The absolute difference between the selective and random CIRs remains close to 0.024 across these experiments. Fatter tails therefore reduce the proportional contribution of survival selection without materially reducing its absolute contribution in this range.

Rare large innovations provide a possible reason for amplification to weaken more sharply. When a jump moves a firm well beyond its adjustment boundary, a wider inaction region induced by poor survival prospects may no longer delay repricing. Adjustment timing then depends more on the arrival of the jump, leaving less scope for the anticipation channel, which would explain why the specification with $P = 0.079$ generates smaller amplification. The evidence therefore supports positive amplification under moderately leptokurtic innovations.

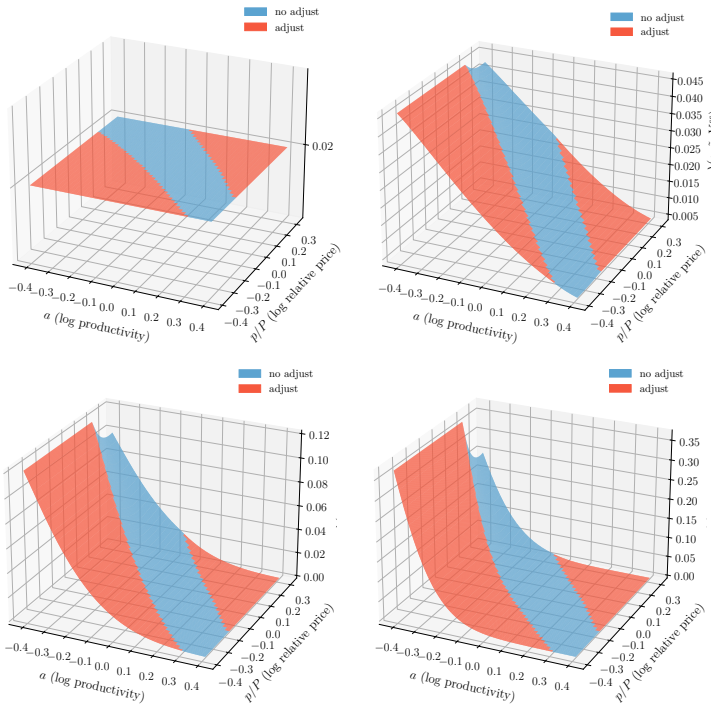
**Table A.15:** Calibration and amplification under four distributions of productivity shocks.

| | Data | Calibrated | Gaussian | Mixture | Poisson $P = 0.25$ | Poisson $P = 0.079$ |
|---|---|---|---|---|---|---|
| Menu cost $f$ | | 0.155 | 0.155 | 0.155 | 0.155 | 0.088 |
| Arrival probability $P$ | | 1 | 1 | 1 | 0.25 | 0.079 |
| Kurtosis of $\varepsilon$ | | 3.0 | 3.0 | 6.1 | 12.0 | 38.0 |
| Targeted moments | | | | | | |
| Frequency of price changes | 0.09 | 0.086 | 0.088 | 0.086 | 0.088 | 0.092 |
| Monthly exit rate | 0.02 | 0.020 | 0.020 | 0.020 | 0.020 | 0.020 |
| Rel. size of exiting firms | 0.66 | 0.643 | 0.620 | 0.637 | 0.656 | 0.681 |
| Rel. size of entrants | 0.67 | 0.688 | 0.666 | 0.657 | 0.637 | 0.607 |
| Mean $\Delta \log p$ | 0.04 | 0.042 | 0.045 | 0.047 | 0.052 | 0.084 |
| Untargeted moments | | | | | | |
| Mean $\lvert\Delta \log p\rvert$ | 0.10 | 0.209 | 0.212 | 0.216 | 0.222 | 0.229 |
| St. dev. $\Delta \log p$ | 0.11 | 0.210 | 0.212 | 0.221 | 0.232 | 0.254 |
| Fraction upward changes | 0.63 | 0.618 | 0.625 | 0.629 | 0.636 | 0.739 |
| One-year CIR, No selection | | 0.155 | 0.154 | 0.172 | 0.211 | 0.256 |
| One-year CIR, Benchmark | | 0.178 | 0.178 | 0.193 | 0.236 | 0.259 |
| Benchmark vs. No selection | | +14.6% | +15.6% | +12.6% | +11.6% | +1.2% |

*Notes*: Log productivity follows $\log z' = \rho \log z + \varepsilon$ with $\rho = 0.85$ and a standard deviation of $\varepsilon$ of 0.086 in every column; $\delta = 0.02$, $\xi = 20$ and $\tau = 5$ throughout. Calibrated is the economy of Table 5. Gaussian has the same shocks on the grid of the three columns after it, 40 nodes on $\pm 0.63$ in $\log z$, against $\pm 0.41$ in the calibrated economy. Mixture draws $\varepsilon$ from a normal with standard deviation 0.150 with probability 0.2 and 0.060 otherwise. In the Poisson columns a normal shock with standard deviation $0.086/\sqrt{P}$ arrives with probability $P$ each month and $\varepsilon = 0$ otherwise; $P = 0.079$ is the arrival rate of Midrigan (2011).

# I Additional figures

**Figure A.17: Exit hazard.**




*Notes*: The figure shows the exit hazard as a function of productivity and relative price, for values of $\xi = 0, 2, 4, 10$. The red region and blue region represent, respectively, the adjust and inaction regions of the value function. For positive selection ($\xi > 0$), firms can decrease their exit probability avoiding adjustment, i.e. saving on the menu cost.

# J Continuous time model

This appendix develops the reduced model of Section 6 and proves the results shown there. Section J.1 states the model. Section J.2 solves the option-value experiment, the band under exit risk. The remaining subsections study the censoring experiment at a fixed band. Section J.3 derives the three auxiliary objects that every proof uses: the expected value of a firm's remaining spell, the stationary density of gaps, and a stationarity identity linking the two. Section J.4 proves that the CIR equals the average slope of the spell value, Proposition 1, the closed form under a uniform hazard, and the placement results. Section J.5 proves the amplification theorem. Section J.6 proves the sufficient statistic of the literature under exit.[28] Firms are the intermediate producers of Section 3, viewed through a single state variable, the price gap. The results from Section J.3 on do not use the optimality of the band: they take the band as given. Fixing the band removes the option-value response of the pricing policy and isolates the censoring channel. Throughout, the exit hazard $\lambda$ is continuous on the closed band, hence bounded, with maximum $\max \lambda$ (Weierstrass theorem).

## J.1 The reduced model

A firm's state is its price gap $x$: the log difference between the price it charges and the price it would set if adjustment were free. Trend inflation erodes every gap downward (the target drifts up while the posted price stands still); idiosyncratic shocks move the target around.

Start from a discrete monthly version, matching the timing of the quantitative model. A firm ends last month with gap $x$, draws an innovation $\varepsilon \sim N(0, \sigma^2)$, and arrives at $y = x - \bar{\pi} + \varepsilon$, where $\bar{\pi}$ is the monthly erosion from trend inflation. On arrival: with probability $\lambda(y)$ the firm exits and is replaced by an entrant born at gap 0; a surviving firm adjusts, resetting to 0, if $y$ is outside the inaction band (in the Calvo variant, with constant probability $\alpha$ instead). Both events place a gap of 0 at that slot in the economy: the adjuster corrects its own price, the exiter hands the slot to a correctly priced entrant. The probability that something fires is

$$e(y) \;=\; \lambda(y) + (1 - \lambda(y))\, \mathbf{1}\{y \notin [b_\ell, b_u]\}. \tag{22}$$

Partition the month into short intervals of length $dt$: over one interval the gap moves by its share of the trend erosion, $-\bar{\pi}\, dt$, plus a Gaussian innovation with mean zero and variance $\sigma^2 dt$, independent across intervals. Denote the instantaneous change by $dx = -\bar{\pi}\, dt + \sigma\, dW$, then:

$$\mathbb{E}[dx] = -\bar{\pi}\, dt, \qquad \mathbb{E}[(dx)^2] = \sigma^2 dt + O(dt^2), \qquad \mathbb{E}[(dx)^k] = O(dt^2),\ k \geq 3 :$$

the squared increment is of order $dt$ and survives the limit, every higher power dies, which is

[28] The representation results extend formulas of Alvarez et al. (2016), Baley and Blanco (2021), and Alvarez et al. (2022); since the model here departs from that class through entry, exit, and an asymmetric gap-dependent hazard, I re-derive them in full, and the derivations double as the toolkit for the exit results, which are new.

why second derivatives appear in the equations below and higher ones never do. The band is $(-a, a)$; touching a boundary triggers adjustment and reset to 0; exit fires at hazard $\lambda(x)$, meaning probability $\lambda(x)\,dt$ per interval, with the entrant born at 0. $W$ is a standard Wiener process.

The monetary shock is a permanent increase in nominal money of size $\Delta$: every target price rises by $\Delta$, so every gap drops by $\Delta$ at $t = 0$. Output deviations are proportional to real balances, and the price level is the mean of prices, so the output deviation at $t$ is proportional to minus the deviation of the mean gap from its stationary value. The object of interest is the cumulated impulse response per unit of shock,

$$\text{CIR} \;=\; \lim_{\Delta \to 0} \frac{1}{\Delta} \int_0^\infty (\text{output deviation at } t)\, dt,$$

in months: the CIR divides the cumulated output deviation by the size of the shock, so a CIR of e.g. 5 means the shock generates the same cumulated output as output standing one full shock size above trend for five months. Amplification always means a higher CIR compared to some baseline.

The hazard of Section 3 is $\lambda = \delta\, e^{-\xi V} / \mathbb{E}[e^{-\xi V}]$, a function of the firm's value $V$; in the reduced model it is a function of the gap. Writing the hazard on the gap does not contradict the model's finding that exit is driven mainly by productivity and not by mispricing: the projection concerns what the hazard looks like along the dimension that matters for pricing. The gap is informative about who exits precisely because the driver is productivity: a firm with a large negative gap is typically a firm whose target price just jumped above its posted price after a bad (productivity) draw, that is, a low-value firm with a high hazard. The projection $\mathbb{E}[\lambda \mid \text{gap} = x]$ is simply a predictor of this feature of selection. The normalization in (5) fixes the average hazard over the cross-section at $\delta$ every period: the exit flow is always $\delta$, selection only decides who is in it. The no-selection benchmark is the uniform hazard $\lambda_u \equiv \delta$ (the $\xi = 0$ economy); the selective economy has some $\lambda_s(x)$ with the same flow.

Two remarks on the reduced model. First, a negative gap can have two sources in the full model: a bad productivity draw, which raises the firm's target price above its posted price at once, and trend inflation, which erodes every posted price slowly and can accumulate price gaps. The reduced model carries both, as the diffusion term and the drift term, and treats two firms with the same gap identically regardless of the source. For everything that happens on impact this is not an approximation: repricing incentives, and the event probabilities outside the inaction band, depend on the gap alone, however the firm got there. Yet, the zero-drift results of this section focus on the idiosyncratic sources of a price gap: at $\bar{\pi} = 0$ there is no erosion at all and the negative gaps are possible only because of a productivity draw, so erosion does not drive the amplification. Second, what the reduced model does give up is the persistence and mean-reversion of the shocks driving the price gap. In the full model productivity mean-reverts, so a firm whose gap came from a bad draw would correct its price now and then walk it back down as productivity recovers, while an erosion-driven gap carries no such trajectory. The gap process here has no mean reversion, so every statement in this section concerns the

impact aspect of the counterfactual; the delayed element belongs to the persistence margin the reduced model deliberately omits, and it is exactly what the phantom-adjustment exercise of Appendix F.2 measures in the full model.

The projection matches the exit flow out of every gap by construction, since it is a conditional expectation, but it collapses the dispersion of exit risk within a gap. A low-productivity firm that resets re-enters at gap zero and inherits the low hazard of the firms sitting there, while in the full model it keeps its productivity and its exit risk. The reduced model therefore lets a firm shed its exposure by adjusting, which the full model does not. This is again the continuation margin set aside above, and it leaves untouched where exit lands, which is what the results below price and what Figure 11 measures on the solved model.

## J.2 The option-value channel

This subsection proves Proposition 2: a higher exit hazard widens the firm's optimal inaction band. At zero trend inflation, the price gap follows $dx = \sigma\, dW$ between adjustments. The firm incurs a flow loss $Bx^2$, discounts at rate $r$, and pays a menu cost $\kappa$ to reset its gap to zero, with $B, \sigma, \kappa > 0$. Exit ends all subsequent losses and adjustment costs. An independent, constant exit hazard $\lambda$ therefore enters the pricing problem through the effective discount rate $\hat{\rho} = r + \lambda > 0$.

The hazard is exogenous to the firm's pricing decision and remains unchanged after adjustment. This differs from the cross-sectional projection $\mathbb{E}[\lambda \mid x]$, which reflects the association between productivity, price gaps and exit risk rather than the effect of repricing on one firm's survival. The experiment isolates an individual comparative static; the quantitative model determines its aggregate importance when survival prospects evolve with productivity.

Let $v(x)$ denote the minimum expected discounted sum of flow losses and future menu costs. The standard symmetric adjustment policy has an inaction band $(-\bar{y}, \bar{y})$ and resets the gap to zero (Stokey, 2009). Inside the band, the value function satisfies

$$\hat{\rho}\, v(x) = Bx^2 + \frac{\sigma^2}{2} v''(x),$$

with symmetry $v'(0) = 0$, value matching $v(\bar{y}) = v(0) + \kappa$, and smooth pasting $v'(\bar{y}) = 0$.

*Proof of Proposition 2.* Write $q = \sqrt{2\hat{\rho}}/\sigma$. The symmetric solution of the value-function equation is

$$v(x) = \frac{B}{\hat{\rho}} x^2 + \frac{B\sigma^2}{\hat{\rho}^2} + D \cosh(qx).$$

Smooth pasting gives $D = -2B\bar{y}/[\hat{\rho} q \sinh(q\bar{y})]$. Substituting into value matching yields

$$\frac{B}{\hat{\rho}} \left[ \bar{y}^2 - \frac{2\bar{y}}{q} \tanh\left( \frac{q\bar{y}}{2} \right) \right] = \kappa.$$

Define the rescaled boundary $y = q\bar{y}$. The boundary equation becomes

$$g(y) = \frac{2\kappa\hat{\rho}^2}{B\sigma^2}, \qquad g(y) := y^2 - 2y \tanh(y/2).$$

First, this equation has a unique positive solution. Let $T = \tanh(y/2)$ and $S = 1 - T^2$. Then

$$g'(y) = y - 2T + yT^2, \qquad g''(y) = 2T^2 + yTS > 0 \quad \text{for } y > 0.$$

Since $g(0) = g'(0) = 0$ and $g(y) \to \infty$ as $y \to \infty$, $g$ is strictly increasing from zero.

Next, implicit differentiation gives

$$\frac{dy}{d\hat{\rho}} = \frac{2g(y)}{\hat{\rho}g'(y)}.$$

Because the actual boundary is $\bar{y} = \sigma y/\sqrt{2\hat{\rho}}$, its derivative satisfies

$$\frac{d\log\bar{y}}{d\hat{\rho}} = \frac{1}{\hat{\rho}}\left[\frac{2g(y)}{yg'(y)} - \frac{1}{2}\right].$$

It is therefore positive if $4g(y) - yg'(y) > 0$. Write

$$4g(y) - yg'(y) = y\varphi(y), \qquad \varphi(y) := 3y - 6T - yT^2.$$

Here $\varphi(0) = 0$ and

$$\varphi'(y) = T(2T - yS) = TS(\sinh y - y) > 0 \quad \text{for } y > 0.$$

using $2T/S = \sinh y$.[29] Thus $4g(y) > yg'(y)$, implying $d\bar{y}/d\hat{\rho} > 0$. Since $\hat{\rho} = r + \lambda$, $d\bar{y}/d\lambda > 0$. ■

A higher exit hazard shortens the horizon over which the firm benefits from correcting its price, making it optimal to tolerate a larger gap before paying the menu cost. This result concerns an increase in one firm's exit risk. Redistributing a fixed aggregate exit flow raises some firms' hazards and lowers others', so the aggregate effect depends on which firms face each change.

### J.3 Spell values and the stationary density

The proofs use two objects: the mispricing a firm carries over its remaining spell and the stationary distribution of gaps. For a function $g$, write

$$\mathbb{E}_f[g] := \int_{-a}^{a} g(x)f(x)\,dx, \qquad \bar{x} := \mathbb{E}_f[x],$$

where $f$ is the stationary density. The expectation $\mathbb{E}_x$ instead follows one firm starting at gap $x$.

[29] See that, with $T = \tanh(y/2)$ and $S = 1 - T^2 = \operatorname{sech}^2(y/2)$, then:

$$2T/S = 2\tanh(y/2)/\operatorname{sech}^2(y/2) = 2 \cdot [\sinh(y/2)/\cosh(y/2)] \cdot \cosh^2(y/2) = 2\sinh(y/2)\cosh(y/2) = \sinh y$$

Let $\tau$ be the first reset, by adjustment or exit, and define

$$m(x) := \mathbb{E}_x[\tau], \qquad \bar{M}(x) := \mathbb{E}_x\left[\int_0^\tau (x_t - \bar{x})\, dt\right].$$

These are instances of the same construction. For a continuous function $g$, the expected accumulation

$$G[g](x) := \mathbb{E}_x\left[\int_0^\tau g(x_t)\, dt\right]$$

is the unique solution $w = G[g]$ of

$$\frac{\sigma^2}{2} w'' - \bar{\pi}\, w' - \lambda(x) w = -g, \qquad w(\pm a) = 0. \tag{23}$$

The equation follows from conditioning on a short initial interval; the boundary values are zero because reaching an edge ends the spell immediately. The solution is unique: the difference of two solutions solves the homogeneous problem with zero boundary values, and with $\lambda \geq 0$ the maximum principle leaves only the zero function. Thus $m = G[1]$ and $\bar{M} = G[x - \bar{x}]$, with

$$\frac{\sigma^2}{2} \bar{M}'' - \bar{\pi}\, \bar{M}' - \lambda(x) \bar{M} = -(x - \bar{x}), \qquad \bar{M}(\pm a) = 0. \tag{24}$$

The stationary density balances diffusion, exit, and reinjection at the reset point. The net rightward probability flux is

$$J(x) = -\bar{\pi}\, f(x) - \frac{\sigma^2}{2} f'(x).$$

Away from zero, stationarity requires $J' = -\lambda f$. Every adjustment and exit reinjects a firm at zero, so the density satisfies

$$\frac{\sigma^2}{2} f'' + \bar{\pi}\, f' - \lambda f + N\delta_0 = 0, \qquad f(\pm a) = 0, \qquad \int_{-a}^{a} f(x)\, dx = 1, \tag{25}$$

where $\delta_0$ is a unit point mass at zero and $N$ is the total reset rate. The density is continuous at zero, while its slope satisfies

$$\frac{\sigma^2}{2}[f'(0^+) - f'(0^-)] = -N.$$

The boundary adjustment rates and total reset rate are

$$N^+ = -\frac{\sigma^2}{2} f'(a), \qquad N^- = \frac{\sigma^2}{2} f'(-a), \qquad N = \mathbb{E}_f[\lambda] + N^+ + N^-.$$

The absorbing boundary conditions reflect immediate adjustment at the edges; the kink records reinjection at zero. Multiplying the stationary equation by a twice continuously differentiable function $g$ and integrating by parts on either side of zero gives

$$\mathbb{E}_f\left[\frac{\sigma^2}{2} g'' - \bar{\pi}\, g' - \lambda g\right] + N g(0) - N^+ g(a) - N^- g(-a) = 0. \tag{26}$$

This identity links individual gap dynamics to stationary moments. The first term records drift, diffusion, and exit within the band; the remaining terms account for reinjection and boundary adjustment.

### J.4 The CIR and gap dispersion

Two facts about spells come first.

**Lemma A.1** (A fresh spell carries no deviation)**.** $\bar{M}(0) = 0$ *at any drift and nonnegative hazard in the fixed band.*

*Proof.* Apply (26) first to $g = m$. Equation (23) and the zero boundary values give $Nm(0) = 1$, so $N > 0$. Applying the same identity to $g = \bar{M}$ gives

$$N\bar{M}(0) = \mathbb{E}_f[x - \bar{x}] = 0.$$

Hence $\bar{M}(0) = 0$. ■

**Lemma A.2** (Spells end geometrically fast, uniformly in the starting gap)**.** *There is* $p_0 > 0$*, independent of the starting gap, such that*

$$\mathbb{P}_x(\tau > t) \leq (1 - p_0)^{\lfloor t \rfloor} \quad \text{for every } t \geq 0, \qquad m(x) \leq \frac{1}{p_0}.$$

*Proof.* Survival without a reset for one unit of time requires the diffusion to remain inside the band, and therefore $|\sigma W_1 - \bar{\pi}| < 2a$. Set

$$p_0 := \mathbb{P}(|\sigma W_1 - \bar{\pi}| \geq 2a) > 0.$$

Then $\mathbb{P}_x(\tau > 1) \leq 1 - p_0$ for every starting gap. Iterating this bound with the Markov property proves the tail inequality. Integrating it gives

$$m(x) = \int_0^\infty \mathbb{P}_x(\tau > t)\, dt \leq \sum_{n=0}^{\infty} (1 - p_0)^n = \frac{1}{p_0}.$$

■

The CIR depends on how a small displacement of initial gaps changes their expected remaining mispricing.

**Proposition A.1** (CIR representation)**.**

$$\text{CIR} = \mathbb{E}_f[\bar{M}'(x)].$$

*Proof of Proposition A.1.* The monetary shock lowers every initial gap by $\Delta$. Write

$$\bar{x}_\Delta(t) := \int_{-a}^{a} \mathbb{E}_{x-\Delta}[x_t] f(x)\, dx.$$

Normalize output deviations to equal minus the deviation of the mean gap from its stationary value. Then

$$\text{CIR} = -\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \lim_{T \to \infty} \int_0^T (\bar{x}_\Delta(t) - \bar{x})\, dt. \tag{27}$$

A starting gap outside the band resets immediately to zero. Accordingly, extend $\bar{M}$ by zero outside $[-a, a]$.

For a smooth function satisfying $g(\pm a) = g(0)$, the reset process has generator

$$\mathcal{A}g = \frac{\sigma^2}{2} g'' - \bar{\pi}\, g' + \lambda(x)[g(0) - g(x)].$$

By Lemma A.1, $\bar{M}(\pm a) = \bar{M}(0) = 0$, so (24) implies $\mathcal{A}\bar{M} = -(x - \bar{x})$. Dynkin's formula, Itô's formula between resets (Øksendal, 2003, Theorem 4.1.2) with each hazard jump replaced by its compensator $\lambda(x_t)[\bar{M}(0) - \bar{M}(x_t)]\, dt$ (Brémaud, 1981, Chapter II, Theorem T8), gives the finite-horizon identity

$$\int_0^T (\mathbb{E}_x[x_t] - \bar{x})\, dt = \bar{M}(x) - \mathbb{E}_x[\bar{M}(x_T)]. \tag{28}$$

Started from $f$, the unshocked cross-section stays at $f$ at every later date, so

$$\int_{-a}^{a} \mathbb{E}_x[\bar{M}(x_T)] f(x)\, dx = \mathbb{E}_f[\bar{M}] \qquad \text{for every } T. \tag{29}$$

Averaging (28) over the shocked starting gaps $x - \Delta$, adding and subtracting $\mathbb{E}_f[\bar{M}]$, and using (29) yields

$$\int_0^T (\bar{x}_\Delta(t) - \bar{x})\, dt = \int_{-a}^{a} [\bar{M}(x - \Delta) - \bar{M}(x)] f(x)\, dx - \varepsilon_\Delta(T), \tag{30}$$

where

$$\varepsilon_\Delta(T) := \int_{-a}^{a} \{\mathbb{E}_{x-\Delta}[\bar{M}(x_T)] - \mathbb{E}_x[\bar{M}(x_T)]\}\, f(x)\, dx.$$

The terminal difference vanishes as $T \to \infty$. After the first reset, the process consists of independent spells launched from zero. A spell ends at a Brownian hitting time or at the first point of a clock with intensity $\lambda(x_t)$, so its length has a density and its distribution is non-lattice; its mean is positive and at most $1/p_0$ by Lemma A.2. For the bounded continuous reward $\bar{M}$, the expected reward before the first reset is continuous in $T$ and bounded in absolute value by $\|\bar{M}\|_\infty (1 - p_0)^{\lfloor T \rfloor}$. The key renewal theorem (Feller, 1971, Volume II, Chapter XI) therefore gives a limit $\psi_\infty$ with

$$\mathbb{E}_x[\bar{M}(x_T)] \longrightarrow \psi_\infty \qquad \text{for every starting gap } x,$$

the same for every $x$ because every path is a fresh spell from zero after its first reset. Averaging over $f$, legitimate because $\bar{M}$ is bounded, and using (29) identifies $\psi_\infty = \mathbb{E}_f[\bar{M}]$. The same averaging over the shocked starting gaps $x - \Delta$ gives $\varepsilon_\Delta(T) \to 0$ for each fixed $\Delta$.

Substituting into (27),

$$\mathrm{CIR} = \lim_{\Delta \downarrow 0} \int_{-a}^{a} \frac{\bar{M}(x) - \bar{M}(x-\Delta)}{\Delta} f(x)\, dx.$$

The zero extension of $\bar{M}$ is Lipschitz because $\bar{M}$ vanishes at the boundaries and has a bounded derivative inside the band. The difference quotient is therefore uniformly bounded and converges to $\bar{M}'(x)$ at every interior point. Dominated convergence gives $\mathrm{CIR} = \mathbb{E}_f[\bar{M}']$. ■

Proposition 1 is the zero-drift case of the identity

$$\mathrm{CIR} = \frac{\mathrm{Var}_f(x) + \bar{\pi}\, \mathbb{E}_f[\bar{M}]}{\sigma^2}. \tag{31}$$

*Proof of Proposition 1.* Apply (26) to $g(x) = x\bar{M}(x)$. All reset terms vanish because $g(0) = g(\pm a) = 0$. Using (24) gives

$$0 = \mathbb{E}_f\left[\sigma^2 \bar{M}' - \bar{\pi}\, \bar{M} - x(x - \bar{x})\right],$$

and hence

$$\sigma^2 \mathbb{E}_f[\bar{M}'] = \mathrm{Var}_f(x) + \bar{\pi}\, \mathbb{E}_f[\bar{M}].$$

Proposition A.1 identifies the left side as $\sigma^2$ CIR, which is (31). At zero drift, $\mathrm{CIR} = \mathrm{Var}_f(x)/\sigma^2$. ■

Under uniform exit, the stationary density and the CIR have closed forms.

**Proposition A.2** (Uniform exit)**.** *At zero trend inflation, with a uniform hazard $\lambda > 0$ and band $\pm a$, write $y = a\sqrt{2\lambda}/\sigma$. Then*

$$\mathrm{CIR}(\lambda, a) = \frac{1}{\lambda}\left(1 - \frac{y^2}{2(\cosh y - 1)}\right) = \frac{a^2}{6\sigma^2} - \frac{a^4}{60\sigma^4}\lambda + O(\lambda^2).$$

*The CIR is strictly decreasing in $\lambda$, from the no-exit value $a^2/(6\sigma^2)$ toward zero, and strictly increasing in $a$.*

*Proof of Proposition A.2.* Let $q = \sqrt{2\lambda}/\sigma$, so $y = qa$. Solving (25) on either side of zero, imposing the boundary conditions and normalizing gives

$$f(x) = \frac{q\, \sinh(q(a - |x|))}{2(\cosh y - 1)}.$$

The density is symmetric, so $\bar{x} = 0$, and its second moment is

$$\mathbb{E}_f[x^2] = \frac{q}{\cosh y - 1} \int_0^a x^2 \sinh(q(a - x))\, dx = \frac{2}{q^2} - \frac{a^2}{\cosh y - 1}.$$

Proposition 1 with $\bar{x} = 0$ gives $\mathrm{CIR} = \mathbb{E}_f[x^2]/\sigma^2$, which is the closed form. The expansion

follows from

$$\frac{y^2}{2(\cosh y - 1)} = 1 - \frac{y^2}{12} + \frac{y^4}{240} + O(y^6).$$

To establish monotonicity in $\lambda$, write

$$\text{CIR} = \frac{2a^2}{\sigma^2} g(y), \qquad g(y) = \frac{1}{y^2} - \frac{1}{2(\cosh y - 1)}.$$

Since $y$ increases with $\lambda$, it suffices to show $g'(y) < 0$. With $z = y/2$, this inequality is equivalent to

$$z^3 \cosh z < \sinh^3 z.$$

Define

$$F(z) := 3 \log\left(\frac{\sinh z}{z}\right) - \log\cosh z, \qquad H(z) := z(3 + 2\sinh^2 z) - 3\sinh z \cosh z.$$

Then

$$F'(z) = \frac{H(z)}{z \sinh z \cosh z}, \qquad H'(z) = 4\sinh z(z\cosh z - \sinh z) > 0 \quad (z > 0).$$

The last inequality holds because $z\cosh z - \sinh z$ vanishes at zero and has derivative $z \sinh z > 0$. Since $H(0) = 0$ and $F(z) \to 0$ as $z \downarrow 0$, $F(z) > 0$ for $z > 0$, the required inequality. The limit as $\lambda \to \infty$ follows from the closed form.

Finally, at fixed $\lambda$,

$$\text{CIR} = \frac{1}{\lambda}\left[1 - \left(\frac{z}{\sinh z}\right)^2\right], \qquad z = \frac{qa}{2}.$$

The ratio $z/\sinh z$ strictly decreases with $z > 0$, since its derivative has numerator $\sinh z - z\cosh z < 0$. Thus the CIR increases with $a$. The same conclusion holds at $\lambda = 0$ from the limiting formula. ■

Uniform replacement supplies fresh prices and shortens the life of standing mispricing; the limit in which every firm is replaced each period and money is neutral appears in Bilbiie (2021). A wider band works in the opposite direction and raises the CIR at a given uniform exit rate.

**Proposition A.3** (Censoring is exactly neutral)**.** *In the monthly reduced model, holding the band, gap dynamics, and hazard inside the band fixed, the aggregate response is invariant to exit probabilities on arrival states outside the band. Raising exit there replaces incumbent adjustments with entrant resets one for one.*

*Proof of Proposition A.3.* At an arrival state $y$ outside the band, (22) gives

$$e(y) = \lambda(y) + (1 - \lambda(y)) = 1.$$

Whether the incumbent exits or adjusts, the slot returns to gap zero. Changing the exit probability there therefore changes the recorded event but not the law of motion of gaps. ■

The placement kernel extends this comparison to exit inside the band. Throughout the remainder of this subsection, $\bar{\pi} = 0$. Proposition 3 follows from two identities for the operator $G$.

**Lemma A.3** (Two exact zeros, at any baseline hazard and band)**.** $\omega(0) = 0$ *and* $\omega(\pm a) = 0$.

**Lemma A.4** (Swap identity)**.** *For continuous* $u, v$,

$$\int_{-a}^{a} u\, G[v]\, dx = \int_{-a}^{a} G[u]\, v\, dx.$$

*Proof.* Let $w_1 = G[u]$ and $w_2 = G[v]$. Their zero boundary values and integration by parts give

$$\int_{-a}^{a} u\, w_2\, dx = \int_{-a}^{a} \left(\lambda w_1 w_2 + \frac{\sigma^2}{2} w_1' w_2'\right) dx = \int_{-a}^{a} w_1\, v\, dx.$$

■

**Lemma A.5** (Duality)**.** *For continuous* $w$,

$$\mathbb{E}_f[w] = N\, G[w](0), \qquad N = \frac{1}{m(0)}.$$

*Proof.* Apply (26) to $g = G[w]$. Its differential equation and zero boundary values give

$$-\mathbb{E}_f[w] + N\, G[w](0) = 0.$$

With $w = 1$, $Nm(0) = 1$. ■

Duality expresses a stationary average as the expected accumulation over a spell divided by expected spell length.

*Proof of Proposition 3 and Lemma A.3.* Consider $\lambda_\theta = \lambda + \theta h$, for perturbations preserving non-negative hazards. A dot denotes a derivative at $\theta = 0$.

The stationary solution and spell values depend smoothly on $\theta$. The differential equations have coefficients affine in $\theta$, and their boundary, matching, and normalization conditions form an invertible linear system. Uniqueness of the Dirichlet problem and the positive normalization ensure invertibility. Standard parameter dependence therefore gives locally analytic solutions (Coddington and Levinson, 1955, Chapter 1). For bounded measurable profiles, the same argument uses the integral form of the differential equations, which then hold almost everywhere.

Differentiating the stationary equation gives

$$\frac{\sigma^2}{2} \dot{f}'' - \lambda \dot{f} = hf - \dot{N}\delta_0, \qquad \dot{f}(\pm a) = 0. \tag{32}$$

Using (25) and the definition of $G$, its solution is

$$\dot{f} = -G[hf] + \frac{\dot{N}}{N} f.$$

The first term records the density removed by the additional hazard; the second accounts for reinjection. Normalization requires $\int \dot{f} = 0$, so the swap identity gives

$$\frac{\dot{N}}{N} = \int_{-a}^{a} G[hf]\,dx = \int_{-a}^{a} hfm\,dx = \mathbb{E}_f[hm].$$

Write $\mathrm{Var}_f = \mathrm{Var}_f(x)$. Since $\mathbb{E}_f[x - \bar{x}] = 0$, the derivative of the mean contributes zero to the variance derivative:

$$\frac{d\mathrm{Var}_f}{d\theta}\Big|_{\theta=0} = \int_{-a}^{a} (x - \bar{x})^2 \dot{f}(x)\,dx.$$

Substituting $\dot{f}$ and applying the swap identity once more,

$$\begin{aligned}\frac{d\mathrm{Var}_f}{d\theta}\Big|_{\theta=0} &= -\int_{-a}^{a} (x - \bar{x})^2 G[hf](x)\,dx + \mathbb{E}_f[hm]\mathrm{Var}_f \\ &= \int_{-a}^{a} h(x)f(x)\left\{m(x)\mathrm{Var}_f - G[(\cdot - \bar{x})^2](x)\right\}dx.\end{aligned}$$

Proposition 1 therefore implies

$$\frac{d\,\mathrm{CIR}}{d\theta}\Big|_{\theta=0} = \int_{-a}^{a} h(x)\omega(x)\,dx, \qquad \omega(x) = \frac{f(x)}{\sigma^2}\left\{m(x)\mathrm{Var}_f - G[(\cdot - \bar{x})^2](x)\right\}.$$

By the definition of $G$, this is exactly

$$\omega(x) = \frac{f(x)}{\sigma^2}\mathbb{E}_x\left[\int_0^{\tau} (\mathrm{Var}_f - (x_t - \bar{x})^2)\,dt\right],$$

as stated in Proposition 3.

The zeros follow from duality and the boundary conditions. At the reset point,

$$G[(\cdot - \bar{x})^2](0) = \frac{\mathrm{Var}_f}{N} = m(0)\mathrm{Var}_f,$$

so $\omega(0) = 0$. At either boundary, $f$, $m$, and the $G$-image vanish, giving $\omega(\pm a) = 0$.

At the no-exit baseline, the required objects are

$$f_0(x) = \frac{a - |x|}{a^2}, \qquad m_0(x) = \frac{a^2 - x^2}{\sigma^2}, \qquad G_0[x^2](x) = \frac{a^4 - x^4}{6\sigma^2}, \qquad \mathrm{Var}_{f_0}(x) = \frac{a^2}{6}.$$

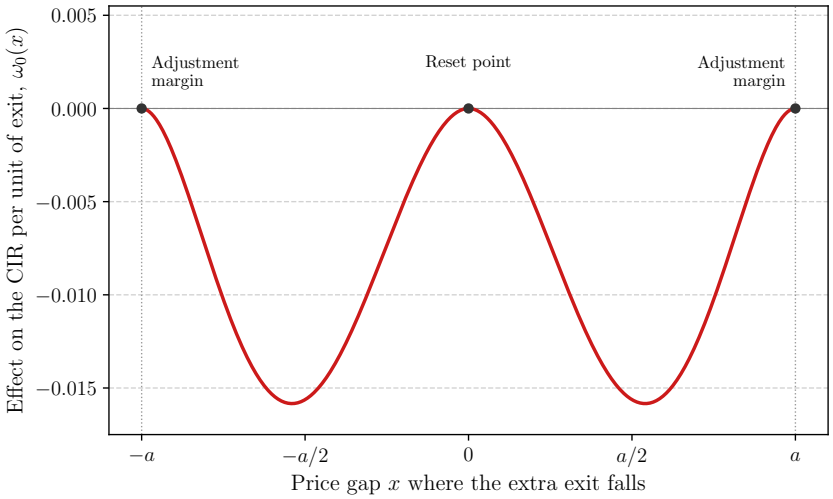
These expressions satisfy their respective equations and boundary conditions. Substitution gives

$$\omega_0(x) = -\frac{x^2(a - |x|)^2(a + |x|)}{6\sigma^4 a^2} \le 0, \qquad \int_{-a}^{a} \omega_0(x)\,dx = -\frac{a^4}{60\sigma^4}.$$

The integral agrees with the first-order effect of uniform exit in Proposition A.2. ■

Figure A.18 displays the no-exit kernel. Additional exit lowers the CIR at every interior gap except zero. Replacement at zero leaves the gap and its continuation unchanged; replacement at an adjustment margin brings forward no correction. Between these points, replacement

**Figure A.18: The price of exit at each point of the cross-section.**




*Notes*: The placement kernel $\omega_0$ at the no-exit baseline, with $a = \sigma = 1$. The kernel weights an additional hazard at each gap in the derivative of the CIR. The dots mark its zeros at the reset point and the adjustment margins.

removes mispricing that would otherwise persist.

## J.5 The amplification theorem

Theorem 1 compares selective and uniform exit near the no-exit economy. The comparison matches exit flows to first order, so its leading term measures the effect of placement.

*Proof of Theorem 1.* Let $c = \mathbb{E}_{f_0}[h]$. The smooth dependence established in the proof of Proposition 3 implies $f_{\theta h} = f_0 + O(\theta)$. The selective exit flow is therefore

$$\int_{-a}^{a} \theta h(x) f_{\theta h}(x)\, dx = \theta c + O(\theta^2),$$

whereas the uniform hazard $\theta c$ generates a flow of exactly $\theta c$.

Applying Proposition 3 at the no-exit baseline and expanding in $\theta$ gives

$$\begin{aligned} \mathrm{CIR}(\theta h) &= \mathrm{CIR}(0) + \theta \int_{-a}^{a} \omega_0(x) h(x)\, dx + O(\theta^2), \\ \mathrm{CIR}(\theta c) &= \mathrm{CIR}(0) + \theta c \int_{-a}^{a} \omega_0(x)\, dx + O(\theta^2). \end{aligned}$$

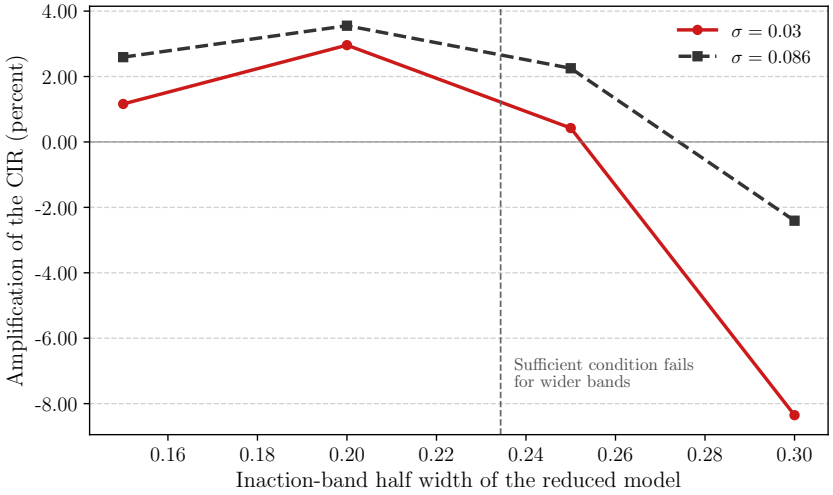
The quadratic remainders follow from the same local smoothness. Subtracting,

$$\mathrm{CIR}_s - \mathrm{CIR}_u = \theta A(h) + O(\theta^2), \qquad A(h) := \int_{-a}^{a} \omega_0(x)(h(x) - c)\, dx. \tag{33}$$

If $A(h) > 0$, the positive first-order term dominates the remainder for all sufficiently small $\theta > 0$. ∎

The condition compares two placements of the same first-order exit flow. Exit added where

**Figure A.19: Exit placement and amplification across inaction bands.**




*Notes*: Percentage difference between selective- and uniform-exit CIRs in the monthly reduced model, for two innovation volatilities. The selective hazard is the projected profile of Figure 11, renormalized each period so that its average over the arrival distribution equals $\delta = 0.02$. The dashed line reports the zero of the auxiliary diagnostic $\int_{-a}^{a} \omega_0(x)(\widehat{\lambda}(x) - \delta)\, dx$, where $\widehat{\lambda}$ is the unscaled projected hazard. This diagnostic differs from $A(h)$ in Theorem 1, which centers the hazard using its mean under the no-exit density.

the kernel is close to zero removes little persistent mispricing; exit withdrawn where the kernel is more negative preserves more, and $A(h)$ nets the two. If $A(h) < 0$, the same expansion instead gives dampening for sufficiently small positive exit intensities. When $A(h) = 0$, higher-order terms determine the comparison.

Figure A.19 runs the comparison numerically in the monthly reduced model, with the projected hazard of Figure 11 renormalized to the exit flow $\delta$. At band half widths from 0.15 to 0.25 the selective CIR exceeds the uniform one by 0.4 to 3.6 percent; at 0.30 it falls short, by 8 percent at the lower volatility and 2 percent at the higher. Widening the band while holding the hazard fixed moves the high-hazard gaps away from the adjustment margin, so their replacement removes prices whose correction was not imminent. The diagnostic $\int \omega_0(\widehat{\lambda} - \delta)\, dx$ of the figure changes sign at a half width of 0.23, the dashed line. The comparison is at a finite exit flow, so the local theorem does not fix the location of the reversal.

## J.6 The sufficient statistic under exit

Counting every replacement as a price change recovers the kurtosis-over-frequency statistic of Alvarez et al. (2016, 2022), with a correction when the stationary gap distribution has a nonzero mean.

At zero drift, count every reset as a pooled price change. An incumbent adjusting from gap $x$ changes its price by $-x$; replacement by an entrant at gap zero produces the same correction. Let $N$ be the pooled reset frequency and $\Delta p$ the distribution of these changes. Its moments satisfy

$$N\mathbb{E}[\Delta p^k] = \int_{-a}^{a} (-x)^k \lambda(x) f(x)\, dx + N^+(-a)^k + N^- a^k.$$

**Proposition A.4** (Sufficient statistic with entry and exit)**.** *At zero drift,*

$$\mathbb{E}[\Delta p] = 0, \qquad N\mathbb{E}[\Delta p^2] = \sigma^2, \qquad N\mathbb{E}[\Delta p^4] = 6\sigma^2\mathbb{E}_f[x^2].$$

*Consequently,*

$$\text{CIR} = \frac{\text{Kurt}(\Delta p)}{6N} - \left(\frac{\bar{x}}{\sigma}\right)^2.$$

*Proof of Proposition A.4.* Apply (26) at zero drift to $g = x$, $x^2$, and $x^4$, respectively:

$$\begin{aligned} 0 &= \mathbb{E}_f[\lambda x] + N^+ a - N^- a = -N\mathbb{E}[\Delta p], \\ \sigma^2 &= \mathbb{E}_f[\lambda x^2] + (N^+ + N^-)a^2 = N\mathbb{E}[\Delta p^2], \\ 6\sigma^2\mathbb{E}_f[x^2] &= \mathbb{E}_f[\lambda x^4] + (N^+ + N^-)a^4 = N\mathbb{E}[\Delta p^4]. \end{aligned}$$

Since $\mathbb{E}[\Delta p] = 0$, kurtosis is the ratio of the fourth raw moment to the squared second moment. Therefore

$$\frac{\text{Kurt}(\Delta p)}{6N} = \frac{N\mathbb{E}[\Delta p^4]}{6(N\mathbb{E}[\Delta p^2])^2} = \frac{\mathbb{E}_f[x^2]}{\sigma^2} = \text{CIR} + \frac{\bar{x}^2}{\sigma^2},$$

where the last equality uses Proposition 1. ■